# MoCHII: Physics, Atomic Data, Algorithms, and Validation

**Abstract**

MoCHII (MOnte Carlo for H II regions) adds a photoionization gas layer to the validated octree-AMR dust radiative-transfer engine of MoCafe v2.00. This memo documents the ionizing frequency band and its zero-variance transport, the complete atomic-data inventory with sources and the analytic fitting functions evaluated at run time, the iteration and equilibrium algorithms (ionization, thermal balance, trace-metal cascade, explicit diffuse field), the $n$-level emissivity solver, the ported nebular continuum and Storey & Hummer recombination lines, and the quantitative validation gates: rate integrals vs. analytic attenuation (bias +0.004%); the Strömgren sphere vs. an exact 1D reference (+1.05% in the ionized-volume radius at level 6, +0.73% at level 7; AMR ≡ uniform); the thermal H/He nebula (median 0.12% in $T_e$); the MOCASSIN Lexington benchmarks ($T_e$ = 8180 K at HII40 and 6940 K at HII20 with the density-suppressed cooling default, run to the drift stopping test, matching a current CLOUDY c25.00 run to −0.37% and +0.44%, with the HII40 He I front 0.033 pc inside CLOUDY's; the zero-density limit gives 7877 and 6282 K); and the line-flux comparison (H$\beta$ within a few percent of the published values, the optical collisional lines within the published spread). Also documented: the solution-driven I-front re-refinement (re-refined vs. native high-resolution octree +0.006% in the ionized-volume radius with 4.5× fewer cells) and an Illustris-TNG post-processing demonstration; the completed dust coupling (ionizing-band dust absorption with the dusty Strömgren reduction 0.724 reproduced to +0.34%; EUV/FUV dust *scattering* with forced-first interaction sampling, validated by bracketing; live ionization-dependent dust with a PAH-specific survival factor; and the equilibrium dust temperature + IR spectrum consuming the transported grain heating, energy closure 1.000); and the incremental metal additions (argon, magnesium, iron, and silicon/chlorine/calcium) as pure data operations.

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# 1. Scope and design rules

MoCHII copies validated modules from MoCafe v2.00 (octree, event-driven traversal, MPI shared memory, HDF5/FITS I/O) and adds gas microphysics under three design rules: (1) *atomic data are fitted offline, evaluated online* — the Python fitting code under `tools/fitting/` is the auditable source of every coefficient in `data/atomic/`, and Fortran only evaluates closed forms; (2) *adding an ion is a data operation*, realized by the species registry (Section 4.6); (3) *validation by quantitative gates*, each capability compared against an analytic or external reference.

The capabilities documented here: the ionizing band + rate integrals; ionization equilibrium + opacity feedback (case B, fixed $T_e$); thermal balance + trace-metal frame; the explicit diffuse field (case A); solution-driven I-front re-refinement + snapshot post-processing; the dust coupling (absorption competition, scattering, live PAH split, grain heating/temperature, and the SEDust stochastic-emission coupling); the incremental ions (argon, magnesium, iron, silicon, chlorine, calcium); plus the output-time diagnostic layer: $n$-level emissivities, Storey & Hummer recombination lines, and the nebular continuum.

# 2. The ionizing band and transport

## 2.1 Continuous energy and diagnostic bins

The ionizing band covers photon energies $[E_{\min}, E_{\max}] = [13.598, 100]$ eV. By default (`par%ion_energy_mode = 'continuous'`), a packet energy is sampled from the normalized source-energy CDF, rather than from a frequency bin. For a Planck source the CDF is built adaptively from the Planck energy spectrum; for a tabulated source it is the cumulative integral of the input $L_E$. H I, He I, He II, and active metal thresholds are retained as knots in the adaptive Planck CDF. With $u \in [0, 1)$, the draw is

$$F_E(E) = \frac{\int_{E_{\min}}^{E} L_{E'}\,\mathrm{d}E'}{\int_{E_{\min}}^{E_{\max}} L_{E'}\,\mathrm{d}E'}, \qquad E_{\rm pkt} = F_E^{-1}(u), \qquad L_{\rm pkt} = \frac{L_{\rm band}}{N_{\rm pkt}}. \tag{1}$$

Here $L_{\rm band}$ is the luminosity integrated over the whole ionizing band, not the luminosity of one energy interval. Each packet is therefore an equal-luminosity sample of the source spectrum.

The packet retains $E_{\rm pkt}$ throughout transport. Gas opacity, photoionization cross sections, and the excess energy used for heating are evaluated at this exact energy. The $n_\nu$ bins (`par%nnu_ion`; gates use 32) receive only a diagnostic $J_E$ tally: the sampled energy is located by a binary search of the bin-edge array and assigned an output index. Consequently the number or placement of those bins cannot alter the continuous-mode physical solution; it changes only the spectral

resolution of saved tallies and rate diagnostics. This is important at thresholds: a photon exactly on a stored edge is assigned to the bin above it, but its interaction physics still uses its own energy.
Every source configuration is sampled this way: one or several internal point sources, an external isotropic field, plane-parallel slab illumination, and the analytic ISRF presets, each drawing from its own spectrum through its own inverted CDF.

**Diagnostic bin edges.** `par%ion_align_edges` (default on) pins the ionizing-band bin edges onto the ionization thresholds — H I (13.6 eV), He I (24.6 eV), He II (54.4 eV), plus the active metal photoionization thresholds gathered from the registry — so that no bin straddles a threshold. The sub-segments between consecutive thresholds are log-uniform, and bins are distributed among the segments by log width (largest-remainder, floor 1). When the thresholds outnumber the bins the lowest-priority (metal) edges are dropped gracefully — the H/He edges are always kept — so the option is safe at any `nnu_ion`. Setting it `.false.` uses a single log grid. Because the transported energies are continuous, this choice changes the resolution of the saved $J_E$ tally and of the rate diagnostics only; it cannot move the physical solution.

**FUV extension.** With `par%add_fuv` the band grows a second, lower segment: `par%nnu_fuv` log bins on $[E_{\rm FUV}, E_{\rm min}]$ (`par%efuv_min`, default 6 eV) *below* the unchanged ionizing bins, so widening the band does not coarsen the ionizing resolution (the earlier single-segment caveat). `par%luminosity` keeps its meaning as the ionizing $[E_{\rm min}, E_{\rm max}]$ luminosity; the FUV part of the same source spectrum rides on top ($L_{\rm FUV}/L_{\rm ion} = 1.097$ for a $4\times10^4$ K Planck source at 6 eV), so the ionizing photon budget $Q_{\rm H}$ is preserved without manual band-ratio rescaling. How the packets are shared between the two segments is `par%fuv_sampling` (Section 4.2): by default each segment is a stratum with its own packet count, and a packet carries its stratum's luminosity over that count; `'proportional'` draws every packet from one CDF over the whole band, with $L_{\rm pkt} = (L_{\rm ion} + L_{\rm FUV})/N$. An input that leaves `add_fuv` and `efuv_min` out gets the extension from 4.78 eV whenever the He I $2^3$S population is used, since its photoionization rate is scored from this band (Section 3.13). H and He cross sections vanish below their thresholds, so the FUV bins interact only with dust and with the low-threshold metal stages (next subsection) — they penetrate the neutral gas beyond the I-front and photoionize Mg I (7.65 eV), C I (11.26 eV), S I (10.36 eV), and Fe I (7.90 eV) there, producing the PDR-side Mg II and C II that the ionizing-only band cannot.

## 2.2 Source components: multiple point sources and an external field

The band is fed by any number of *source components*: `par%nsource` internal point sources (0–16; positions `par%src_x/y/z`, ionizing luminosities `par%src_lum` — all set, or none for an equal split of `par%luminosity`, which becomes the sum) plus, independently, one isotropic external radiation field, on when `par%ext_intensity` $> 0$. Each packet draws a component in proportion to its band-total luminosity, is emitted from it, and draws its frequency bin from that component's own spectrum CDF; every packet carries $L_{\rm pkt} = \sum_c L_c/N$. A single-component run takes the legacy code path (the RNG stream is unchanged, so earlier results reproduce). The spectrum of internal source $i$ resolves as `par%src_spectrum_file` column $i >$ `src_tstar`($i$) Planck $>$ the global `ion_spectrum` file $>$ the global `tstar`; `src_spectrum_file` is one multi-column text file — column 1 the photon energy $E$ [eV] ascending, then one

$L_E$ column for each source in arbitrary units, each independently normalized so source $i$'s ionizing segment carries `src_lum`($i$).

The external field is an isotropic band-integrated mean intensity $J$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] entering through an illuminating surface (`par%ext_geometry`): `'rec'`, the box faces (face-area-weighted — correct for a non-cubic box), or `'sph'`, the bounding sphere of radius `par%rmax` about the box center. The inward flux through the surface is the hemisphere integral of $J\cos\theta$, so the entering band power is

$$F_{\rm in} = \int_{2\pi} J\cos\theta\, d\Omega = \pi J, \qquad L = \pi J A_{\rm surf}. \tag{2}$$

Entry directions are cosine-weighted (Lambert) into the inward hemisphere, $\mu = \sqrt{\xi}$, which reproduces an isotropic interior field with energy density $u = 4\pi J/c$: an optically thin box tallies an interior mean intensity equal to $J$ (the gate of Section 5.8). Its spectrum resolves as `ext_spectrum` (file) > `ext_tstar` Planck > the global spectrum.

**Physical-unit spectra.** Every file slot is read in the column units set by `par%spectrum_type` (default `'shape_ev'`, arbitrary units renormalized to the scale — the legacy behavior). The physical types `'per_ev'`, `'per_hz'`, `'per_ang'`, `'per_um'` give the first column as $E$ [eV], $\nu$ [Hz], or $\lambda$ [Å or $\mu$m] and the value columns as a luminosity density $L_X$ (internal sources) or an intensity density $J_X$ (external field) in the matching spectral unit; wavelength tables are converted to energy and sorted. The `'shape_ev'` value columns are read as the density per eV $L_E$ on the $E$ [eV] grid; a shape tabulated in wavelength or frequency must use the matching `per_*` type, whose conversion applies the $\mathrm{d}\lambda/\mathrm{d}E$ (or $\mathrm{d}\nu/\mathrm{d}E$) Jacobian, so a raw re-grid onto $E$ under `'shape_ev'` would be wrong by $E^{-2}$. A physical spectrum is *absolute*: its bin luminosities follow the tabulated integral directly. If the scale (`luminosity` / `src_lum`($i$) / `ext_intensity`) is unset it is derived from the file and reported; if set, the spectrum is rescaled to it and acts as a shape. With `add_fuv` the same table fills the FUV bins with no separate normalization; a spectrum with no luminosity inside the band aborts.

**ISRF presets.** `par%ext_spectrum` may instead name one of three analytic interstellar radiation fields: `'draine'` (Draine 1978), `'habing'` (the Draine shape at Habing 1968 normalization, $/1.71$), and `'mathis'` (Mathis, Mezger & Panagia 1983), the shape of the SEDust radiation-field module. They are FUV-only (5–13.6 eV, exactly zero in the ionizing bins), so they require `add_fuv`, and `par%ext_scale` multiplies them. The Draine field is the energy-weighted photon polynomial $F_{\rm ph}(E) = 1.658\times10^{6}E^{2} - 2.152\times10^{5}E^{3} + 6.919\times10^{3}E^{4}$ with $J_E = e\,F_{\rm ph}$ ($e = 1.602\times10^{-12}$ erg eV$^{-1}$; $F_{\rm ph} = E\,N(E)$ is energy-weighted — a common pitfall). The presets carry the canonical FUV energy densities $u_{\rm FUV} = 8.94$, $5.23$, $6.01\times10^{-14}$ erg cm$^{-3}$ (Draine / Habing / Mathis; $G_0 = 1.69$, $0.99$, $1.14$ in Habing units). When `ext_intensity` is set the preset or physical file is rescaled to that band-integrated $J$ (a note is printed; `ext_scale` is then ignored).

## 2.3 Opacity and the zero-variance edge walk

Each leaf carries the band absorption coefficient per code length

$$\kappa_b \;=\; n_{\rm H}\,[x_{\rm HI}\,\sigma_{\rm HI}(E_b) + y_{\rm He}\,(x_{\rm HeI}\sigma_{\rm HeI} + x_{\rm HeII}\sigma_{\rm HeII})(E_b)]\,d_{\rm cm}\;(+\,\kappa_{\rm dust}), \tag{3}$$

stored as the full $(n_\nu, n_{\rm leaf})$ array `kap_ion` — the grey $s_{\rm ext}$ rescaling of the dust SED band cannot apply because the gas opacity varies with both leaf and bin. With `par%ion_metal_abs` (default on when `use_metals`) the registry metals join

the absorption, $\Delta\kappa_b = n_{\rm H}\sum_{\rm el}A_{\rm el}\sum_i f_i\,\sigma_i(E_b)\,d_{\rm cm}$, with $\sigma_i$ the cross section of §3.1 (Verner's `phfit2` prescription, inner shells included), the stage fractions $f_i$ from the same product chain as the cooling, iterated by the opacity feedback exactly like $x_{\rm HI}$. Above 13.6 eV this is a $\sim 10^{-22}$ cm$^2$ per H perturbation next to H/He and dust; in the FUV bins it is the only *gas* opacity. The absorbed share returns to the metals through their own $\Gamma$ integrals, so the bookkeeping stays consistent, and the photoelectron energy it carries enters the thermal balance under `par%metal_heat` (on by default) through the shell-resolved heating coefficient of §3.1.

Where the band has no scattering and the diffuse field is emitted explicitly, every packet's contribution to the mean-intensity tally is computed *analytically* along its ray (the forced-first-flight estimator of MoCafe, made exact):

$$\Delta J_b(\text{leaf}) \;=\; w\,L_{\rm pkt}\,\frac{e^{-\tau_{\rm in}}-e^{-\tau_{\rm out}}}{\kappa_b}, \tag{4}$$

accumulated leaf by leaf until the domain edge (`raytrace_ion_to_edge_amr`). No interaction sampling is needed; the only Monte Carlo variance is in direction and bin sampling. The rate-integral gate verifies this estimator to a mean bias of +0.004%.

### 2.4 Rate integrals

With $V$ the leaf volume and $J_b$ the reduced tally, the code forms for each absorber $i$

$$\Gamma_i = \sum_b \frac{[4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)}{h\nu_b}, \qquad \mathcal{H}_i = \sum_b [4\pi J\,\Delta\nu]_b\,\sigma_i(E_b)\left(1-\frac{E_{{\rm th},i}}{E_b}\right), \tag{5}$$

the photoionization rate [s$^{-1}$] and photoheating [erg s$^{-1}$] for each particle of species $i$. For a metal ion at energies where more than one subshell absorbs, the heating factor $\sigma_i(1-E_{{\rm th},i}/E)$ becomes the sum over the absorbing subshells $\sum_s \sigma_s\,(1-E_{{\rm th},s}/E)$, since each subshell releases its photoelectron with $E-E_{{\rm th},s}$ (§3.1).

## 3. Atomic data

Every coefficient file in `data/atomic/` carries a provenance header (source, version, fit range, maximum fit error). The generators live in `tools/fitting/`.

### 3.1 Photoionization cross sections

The one-electron ions H I ($Z$=1) and He II ($Z$=2) use the *exact* ground-state hydrogenic cross section (Osterbrock & Ferland 2006, Eq. 2.4; `sigma_hydrogenic`, the same form as EXHALE):

$$\varepsilon = \sqrt{E/E_{\rm th}-1}, \qquad \sigma(E) = \frac{\sigma_0}{Z^2}\left(\frac{E_{\rm th}}{E}\right)^4 \frac{\exp\!\left(4-4\arctan\varepsilon/\varepsilon\right)}{1-e^{-2\pi/\varepsilon}}, \tag{6}$$

with $\sigma_0 = 6.30\times10^{-18}\,$cm$^2$ the threshold value and $E_{\rm th}$ the true ionization potential (13.598 and 54.416 eV), tied to the physical edge (not $13.6Z^2$) so it stays aligned with the band grid. He I (two electrons) uses the full VFKY96 single-shell analytic form (Verner, Ferland, Korista & Yakovlev 1996, ApJ 465, 487, Eq. 1):

$$x = \frac{E}{E_0}-y_0, \quad z=\sqrt{x^2+y_1^2}, \quad Q = 5.5-\tfrac{1}{2}P, \quad \sigma(E) = \sigma_0\left[(x-1)^2+y_w^2\right]z^{-Q}\left(1+\sqrt{z/y_a}\right)^{-P}, \tag{7}$$

zero below threshold, $\sigma_0$ in Mb, with the parameters parsed from the authors' `phfit2.f` `PH2` (outer-shell) table, verified against the EXHALE implementation, and compiled in `photo_xsec.f90`:

| ion | $E_{\rm th}$[eV] | $E_0$ | $\sigma_0$[Mb] | $y_a$ | $P$ | $y_w$ | $y_0$ | $y_1$ |
|---|---|---|---|---|---|---|---|---|
| He I | 24.587 | 13.61 | 949.2 | 1.469 | 3.188 | 2.039 | 0.4434 | 2.136 |

The exact H I and He II cross sections give the threshold values $\sigma_0/Z^2 = 6.30$ and 1.575 Mb; the VFKY96 fits they replace for H/He agreed with the exact curve to 0.73% (H I) and 0.43% (He II), and $\sigma_{\rm HeI} = 7.43$ Mb (VFKY96).

**Metal ions: Verner's `phfit2` prescription, inner shells included.** The metals follow what `phfit2.f` itself evaluates, which is not one fit at all energies. `phfit2` uses the VFKY96 outer-shell fit [Eq. (7), `PH2`] only below the first inner-shell threshold $E_{\rm inn}$ — its `einn`, the `PH1` threshold of shell `NINN`($N_e$) for an ion with $N_e$ electrons — and at and above $E_{\rm inn}$ it sums the Verner & Yakovlev (1995, A&AS 109, 125) `PH1` fits of every occupied subshell $s$ with $E \geq E_{{\rm th},s}$:

$$\sigma(E) = \sum_{s:\,E\geq E_{{\rm th},s}} \sigma_{0,s}\left[(y-1)^2 + y_{w,s}^2\right] y^{-(5.5+l_s-P_s/2)} \left(1+\sqrt{y/y_{a,s}}\right)^{-P_s}, \qquad y = E/E_{0,s}, \tag{8}$$

with $l_s$ the orbital quantum number of the subshell. `phfit2` sets $E_{\rm inn} = 0$ for $Z = 15$, 17, 19 and for $Z > 20$ except iron — the elements absent from the VFKY96 outer-shell table, Opacity-Project gaps — so for chlorine the subshell sum is used at every energy, and $E_{\rm inn} = 10^{30}$ eV (never reached) for ions with fewer than three electrons. Both branches are zero below the NIST ionization potential carried in the element file (`ETH`). The element files carry the three pieces as `PHOTO` (the `PH2` fit, when $E_{\rm inn} > 0$), `PHINNER` ($E_{\rm inn}$) and one `PHSHELL` row for each occupied subshell ($l$, $E_{\rm th}$, $E_0$, $\sigma_0$, $y_a$, $P$, $y_w$), all parsed directly from `phfit2.f` by `tools/fitting/make_element_data.py`; the evaluator is `species_mod::species_cross_section`.

Seven registry ions have their first inner-shell edge inside the transported 13.6–100 eV band (below), and the four chlorine ions, summed over subshells at every energy, add their $3s$ subshell at 25.3, 36.9, 50.2 and 64.7 eV (Cl I–IV); until 2026-09-24 chlorine carried only the outer $3p$ subshell fit:

| ion | subshell | $E_{\rm inn}$ [eV] | $\sigma$ just above the edge [Mb] | |
|---|---|---|---|---|
| | | | outer-shell fit alone | `phfit2` (subshell sum) |
| Ca I | $3p$ | 34.4 | 0.090 (50 eV) | 1.43 |
| Ca II | $3p$ | 40.9 | | |
| Mg I | $2p$ | 54.9 | 0.087 (70 eV) | 6.03 |
| Mg II | $2p$ | 65.7 | 0.050 (85 eV) | 6.68 |
| Fe I | $3p$ | 66.0 | 8.88 (100 eV) | 5.29 |
| Fe II | $3p$ | 76.2 | | |
| Fe III | $3p$ | 87.1 | | |

Until 2026-09-24 MoCHII evaluated the outer-shell fit alone up to 100 eV for every ion, which is physically wrong above $E_{\rm inn}$: the `PH2` fit describes the outer subshell only, and above the edge the $2p$ (Mg) or $3p$ (Ca, Fe) subshell absorbs as well. For Mg and Ca the omission was a factor 16–130 in the cross section above the edge; for iron the two forms differ by up to $\pm$20% across the band.

*Photoheating.* Each subshell releases its photoelectron with $E - E_{\mathrm{th},s}$, so the heating coefficient is $\sum_s \sigma_s(1 - E_{\mathrm{th},s}/E)$ over the absorbing subshells, accumulated with the cross section at each packet's own energy (`heat_metal`, `species_rate_path_score`). The excitation $E_{\mathrm{th},s} - E_{\mathrm{th}}$ left in the ion by an inner-shell ionization is assumed to leave the gas as fluorescence (not transported), and the ion moves up *one* stage: Auger emission, which would release a second electron and move the ion up a second stage, is not carried. **That assumption does not hold for five of the seven ions above.** The Kaastra & Mewe (1993, A&AS 97, 443) electron yields, as tabulated in CLOUDY c25.00 `data/mewe_nelectron.dat`, give two electrons for a vacancy in Mg I $2p$ with probability 0.997, Ca I $3p$ 0.995, and Fe I, Fe II and Fe III $3p$ 0.9999 each; only Mg II $2p$, Ca II $3p$ and the chlorine $3s$ vacancies give one electron. The energetics agree: the $Mg^+$ ($2p^{-1}$) state left by a $2p$ ionization of Mg I lies 54.9 eV above the Mg I ground, above the $Mg^{2+}$ ground at $7.646 + 15.035 = 22.68$ eV, so it autoionizes. For those five ions an inner-shell absorption above $E_{\mathrm{inn}}$ should therefore raise the stage by two, with the Auger electron's kinetic energy going to heat rather than to fluorescence; the adjacent-stage chain of §4.6 cannot represent that, which is the stage matrix of the hard-spectrum plan (HS6). The cross section and the opacity are unaffected; the stage bookkeeping and the heating of Mg, Ca and Fe above their inner edges are not yet right (flagged 2026-09-24; see §6.).

*Gate.* `tests/photo_shells/` (`check_photo_shells.f90`, `run_check.sh`) compares $\sigma$ and the heating coefficient of every registry ion against `phfit2.f` itself on a logarithmic 5–1000 eV grid, 9112 points, excluding the points within 1% of a threshold: worst relative difference $2.313 \times 10^{-6}$, PASS. Measured end to end (pre-change binary against the current one, same input): on the 11-element smoke run in the HII20 geometry (20 kK blackbody) the Mg, Fe and Ca stage fractions do not move at the $10^{-4}$ level — the in-band inner edges sit above 34 eV, where that spectrum carries almost no photons; on the 40 kK PDR run of §4.18 the $Mg^+$ and $Fe^+$ fractions rise by 0.0013 absolute ($0.2369 \to 0.2382$, $0.2155 \to 0.2168$), the change folded in with the other items of that update.

## 3.2 H/He recombination

Recombination coefficients in the ionization balance are chosen by `par%recomb_model`. In every case the default `'detailed_balance'` obtains the ground-level ($n$=1) direct-capture rate $\alpha_1$ from the Milne relation applied to the *same* photoionization cross sections the transport absorbs with [Eq. (46)]. For He II $\to$ He I it takes the total (case A) radiative recombination $\alpha_A$ from the same Badnell (2023) RR fit used for the metals [Eq. (13)] and forms case B by removing the ground term, $\alpha_B = \alpha_A - \alpha_1$. For the two hydrogenic ions, H II and He III, it takes $\alpha_B$ from an $nl$-resolved radiative cascade in the zero-density limit on Storey & Hummer's (1991) hydrogenic A-values and cross sections (`hydrogenic_recombination_mod`; the cascade and its checks are described with the two-photon continuum, Section 4.20) and forms case A by adding the same Milne term, $\alpha_A = \alpha_B + \alpha_1$. Until 2026-09-26 the hydrogenic ions took $\alpha_A$ from the Badnell fit as well. That fit is 0.6% above the Storey & Hummer (1995) total for H II at $10^4$ K, and subtracting an exact $\alpha_1$ from it carried the error into $\alpha_B$ amplified: +1.13% at $10^4$ K, +0.7 to +1.2% over 5000–$2\times10^4$ K (+0.35% for He III). The cascade's $\alpha_B$ is within 0.1% of the SH95 alpha-tot at $10^4$ K for both ions. Toward low temperature both zero-density determinations fall below SH95's value at $n_e = 100\,\mathrm{cm}^{-3}$ (–3.5% at 500 K), and there the Badnell total and the cascade agree with each other to 0.1%: that departure is the collisional transitions SH95 include, not an error of either. At run time $\alpha_1$ is evaluated from the seven-coefficient functional form of Mao & Kaastra (2016,

A&A 587, A84) ($T_{\rm eV} = k_B T$; $a_0$ in $10^{-10}\,\mathrm{cm^3\,s^{-1}}$),

$$\alpha_1(T) = a_0\,10^{-10}\,T_{\rm eV}^{-b_0-c_0\ln T_{\rm eV}}\,\frac{1+a_2\,T_{\rm eV}^{-b_2}}{1+a_1\,T_{\rm eV}^{-b_1}}, \tag{9}$$

but the coefficients are fitted to that Milne integral instead of being taken from Mao & Kaastra; §4.10 lists them with their fit residuals and gives the measurement that forced the replacement. The routine name `rr_mao` in `recomb_mod` therefore records the functional form, not the origin of the numbers, while the namelist value names what the default does for all three ions — the ground level by detailed balance on the transport cross section — and not where the totals come from, which is the best determination for each ion (it was `'badnell_milne'` until 2026-09-26, when the hydrogenic ions stopped using the Badnell total; the old name is refused). Because $\alpha_1$ and the ground-continuum emissivity of §4.10 are now one integral over one cross section, the case A/B split and the diffuse packet count cannot drift apart. He II→He I adds the Badnell three-term dielectronic sum to $\alpha_A$ (negligible below $\sim 5\times10^4$ K); H II and He III are bare nuclei with no DR. The coefficients at $10^4$ K ($10^{-13}\,\mathrm{cm^3\,s^{-1}}$), with the independently level-resolved Cloudy c25.00 values in parentheses, are

| reaction | $\alpha_A$ | $\alpha_1$ | $\alpha_B$ |
|---|---|---|---|
| H II→H I | 4.16 (4.16) | 1.579 (1.584) | 2.58 |
| He II→He I | 4.37 (4.36) | 1.609 (1.603) | 2.76 |
| He III→He II | 21.8 (21.8) | 6.512 (6.515) | 15.3 |

Setting `par%recomb_model = 'hui_gnedin'` restores the earlier Hui & Gnedin (1997, MNRAS 292, 27) case-A/B fits with $\lambda = 2T_{\rm TR}/T$ ($T_{\rm TR}$ = 157807, 285335, 631515 K for H I, He I, He II):

$$\alpha_A^{\rm HII} = 1.269\times10^{-13}\,\lambda^{1.503}\big/\left[1+(\lambda/0.522)^{0.470}\right]^{1.923}, \qquad \alpha_B^{\rm HII} = 2.753\times10^{-14}\,\lambda^{1.500}\big/\left[1+(\lambda/2.740)^{0.407}\right]^{2.242},$$
$$\alpha_A^{\rm HeII} = 3.000\times10^{-14}\,\lambda^{0.654}, \qquad \alpha_B^{\rm HeII} = 1.260\times10^{-14}\,\lambda^{0.750}, \tag{10}$$

He III hydrogenic ($Z$=2). These are retained so the recorded gates reproduce; they agree with the default to $< 0.6\%$ for H II and He III, but give He II→He I 5% lower ($\alpha_B = 2.62$ vs $2.76\times10^{-13}\,\mathrm{cm^3\,s^{-1}}$). Recombination *cooling*, the kinetic energy the captured electrons remove, is evaluated from the *same* $\alpha$ the balance uses, for either `recomb_model` and either case: for a Maxwellian at $T$ and any capture cross section $\sigma(E)$ — every level, radiative and dielectronic — differentiating $\alpha = \langle\sigma v\rangle$ with respect to $T$ gives exactly

$$\beta(T) = kT\left[\tfrac{3}{2}\,\alpha(T) + \frac{d\alpha}{d\ln T}\right]\ \mathrm{erg\,cm^3\,s^{-1}}, \tag{11}$$

evaluated with a centered difference in $\ln T$ (step $10^{-4}$; `maxwellian_recombination_cooling` in `cooling_mod`), for H II, He II and He III. This replaced (2026-09-24) the Hui & Gnedin companion fits for H II and He III, which fit a different $\alpha$ and sat 2.3% below Eq. (11) on MoCHII's own case-B $\alpha$ at $10^4$ K, and $k_B T\,\alpha^{\rm HeII}$ for He II, which is the mean kinetic energy only for $\alpha \propto T^{-1/2}$: at $10^4$ K the mean energy removed is $0.674\,kT$ (H II) and $0.700\,kT$ (He II), so $k_B T\alpha$ overstated the He II term 1.43×. The metal cascade carries the same identity on its own $\alpha_{\rm rec}$, together with the collisional-ionization energy loss, as the loss side of `par%metal_heat` (`metal_ionization_cooling`).

## 3.3 H/He microphysics: comparison with other codes

The H/He cross sections and recombination can be placed next to three reference codes. For the photoionization cross section, MoCHII and EXHALE evaluate the exact hydrogenic formula [Eq. (6)] for H I/He II; MOCASSIN fits them with VFKY96 (agreeing with the exact curve to $< 1\%$), and Cloudy resolves them from its iso-electronic H-like/He-like model atoms. For the recombination that closes the ionization balance (Table 1), MoCHII's default takes $\alpha_A$ of He I from the modern Strathclyde (Badnell) data and $\alpha_B$ of the hydrogenic ions from a level-resolved Storey & Hummer cascade, with $\alpha_1$ from the Milne relation in both, which is how Cloudy resolves its H-like and He-like recombination level by level; MOCASSIN uses the older Verner & Ferland (1996) total; EXHALE, a planetary-atmosphere code, adopted the Badnell RR + Mao & Kaastra $\alpha_1$ construction as its default on 2026-07-17, before the $\alpha_1$ correction of §4.10 (Hui & Gnedin remains a legacy option). MOCASSIN and Cloudy always work in case A and transport the diffuse ground-recombination field explicitly, whereas MoCHII defaults to case B (on-the-spot) with an optional explicit diffuse field, and EXHALE is case B only.

Table 1: H/He recombination in the ionization balance: source and $\alpha$ at $10^4$ K ($10^{-13}\,\mathrm{cm^3\,s^{-1}}$; case A / case B where both apply).

| code | source | H II→I | He II→I | He III→II |
|---|---|---|---|---|
| MoCHII (default) | Badnell RR + Milne $\alpha_1$ | 4.19 / 2.61 | 4.37 / 2.76 | 21.9 / 15.4 |
| MoCHII (`hui_gnedin`) | Hui & Gnedin 1997 | 4.30 / 2.59 | 4.23 / 2.62 | 22.3 / 15.5 |
| EXHALE (default) | Badnell RR + Mao & Kaastra $\alpha_1$ | – / 2.61 | – / 2.81 | – / 15.4 |
| EXHALE (`legacy_hhe_rates`) | Hui & Gnedin 1997 (case B) | – / 2.59 | – / 2.62 | – / 15.5 |
| MOCASSIN | Verner & Ferland 1996 (case A) | 4.19 | 4.84 | 21.9 |
| Cloudy c25.00 | level-resolved Milne, own $\sigma_{\mathrm{pi}}$ (case A) | 4.16 | 4.36 | 21.8 |

The H II and He III rates agree across all four codes to within a few percent. The one substantive difference is He II→He I: MoCHII's Badnell total (4.37) and Cloudy's explicit sum over levels (4.36) agree to 0.3%, MOCASSIN's Verner & Ferland value (4.84) is the high outlier, and the earlier Hui & Gnedin (4.23) was low. Moving MoCHII onto the modern Badnell value therefore raises its case-B He recombination by $\sim 5\%$, in the direction that reduces the He over-ionization seen relative to converged MOCASSIN in the HII40 benchmark (§5. discusses that residual). Because that 11% spread in a single coefficient is the largest code-to-code difference anywhere in the H/He data, §3.4 takes it apart.

## 3.4 The helium ionization network, code by code

*Record (2026-09-25).* The front shifts and bounds in this part were measured on runs stopped by the earlier convergence rule (§5.4), which ended the HII40 benchmark 0.17 pc short of its converged He I front; they document the search for a residual that was mostly that early stop, and are not current bounds. The converged residual is 0.034 pc.

Neutral helium is the one species in the H/He inventory on which the photoionization codes disagree at more than the percent level. Reading each code's own data rather than its published description gives, for $\alpha_A$(He I) at $10^4$ K,

| code | how $\alpha_A$(He I) is obtained | $10^{-13}\,\mathrm{cm^3\,s^{-1}}$ | vs. Cloudy |
|---|---|---|---|
| Cloudy c25.00 | level-by-level Milne from its own $\sigma_{\mathrm{pi}}$, total cached | 4.3583 | — |
| MoCHII | Badnell RR total + DR | 4.3719 | +0.3% |
| Wood `ionize` / CMacIonize | sum of the four BSS99 *effective* coefficients | 4.2630 | –2.2% |
| MOCASSIN | Verner & Ferland 1996 total fit | 4.8394 | +11.0% |

The MOCASSIN entry is decoded from `data/radrec.dat`, replicating the read order of `update_mod.f90` lines 1707–1735; the decode is validated on the two channels whose answer is known independently, giving H I 4.1923 against Cloudy's $4.1642\times10^{-13}$ (+0.7%) and He II 2.1880 against $2.1817\times10^{-12}$ (+0.3%).

**The spread is not the extent of the cascade sum.** A natural first reading of an 11% spread is that the codes stop the recombination cascade at different levels. None of them does. Three treatments are in play and all three are complete: Cloudy sums explicitly over resolved levels; MoCHII and MOCASSIN use totals over all $n$ (Badnell, VF96); Wood and CMacIonize use *effective* coefficients, in which the cascades from higher levels are already folded into the $n = 2$ terms. MoCHII's Badnell total agrees with Cloudy's explicit level sum to 0.3%, which is what a truncation on either side would break. (The 20-level list in Cloudy's `source/sanity_check.cpp` is a unit test of threshold cross sections, not the extent of its recombination sum.)
The control that settles it is the electron count:

| species | electrons | code-to-code spread in $\alpha_A$ |
|---|---|---|
| H I | 1 (hydrogenic) | 0.7% |
| He II | 1 (hydrogenic) | 0.3% |
| He I | **2** | **11–13%** |

The one-electron species have an exact analytic solution, so there is nothing for two calculations to disagree about, and they do not; the two-electron one has no closed form and carries screening and the singlet/triplet structure, and it spreads by 11–13%. The disagreement is therefore in the atomic data for a two-electron atom, not in how far anybody sums.

**Three of the four cluster.** Within He I the values are not scattered but split three-to-one: Cloudy's level sum 4.3583, Badnell 4.3719 and BSS99 $4.2630\times10^{-13}\,\mathrm{cm^3\,s^{-1}}$ all lie within 2.5% of each other, and Verner & Ferland (1996) alone sits at $4.8394\times10^{-13}$. MOCASSIN applies its `radRecFit` uniformly with no special case for H or He (`source/update_mod.f90:1636--1639`, whose own comment names "H-like, He-like, Li-like, Na-like — Verner & Ferland, 1996, ApJS, 103, 467"), and adds dielectronic recombination only for $Z_{\mathrm{elem}} \geq 3$, so its He I rate is the VF96 radiative value alone. Offered as reasoning rather than as a verified claim: VF96's He-like fits lean on hydrogenic scaling for the excited levels, and $Z = 2$ is where that approximation is worst, since neutral helium is exactly where screening and the singlet/triplet splitting matter most.

**A caution for the MOCASSIN regression suite.** MOCASSIN's 11-case suite is used here as a three-dimensional validation reference. Every helium comparison against it inherits a known +11% in $\alpha_A$(He I), and that difference has been traced to VF96 by the first-principles Cloudy determination above, so it should be subtracted before a disagreement

in a helium ionization front or a helium line intensity is attributed to transport or geometry. The same comparison across the metal registry, where no such independent determination is available, is in §3.7.

**Excited He I levels are not a hidden channel.** What Cloudy's resolved He I model atom has and an effective treatment does not is a population in the $n = 2$ terms that can itself be ionized. MoCHII cannot photoionize those levels at all: the band floor is `par%eion_min` = 13.598 eV while every excited He I level has an ionization potential below it ($2^3$S 4.767, $2^1$S 3.971, $2^3$P 3.623, $2^1$P 3.369 eV). The size of what is missed is bounded by the metastable population, since $2^1$S decays by two-photon emission at $51.3\,\mathrm{s}^{-1}$ and is left $10^7$ times less populated than $2^3$S: with $n(2^3\mathrm{S})/n(\mathrm{He}^0) = 6.2\times10^{-5}$ at $r = 3$ pc, $Q(>4.767\ \mathrm{eV})/Q(>24.587\ \mathrm{eV}) = 33.8$ for a 40-kK blackbody and Cloudy's own $2^3$S threshold cross section of $5.48\times10^{-18}\,\mathrm{cm}^2$, the extra ionization is 0.2–0.6%, and collisional ionization out of $2^3$S adds a comparable $\sim$0.6%. Taken together with the front sensitivity measured in §4.10 (0.80 pc per unit fractional change in the He I recombination rate), the 11–13% code spread is worth 0.09 pc — more than the converged HII40 front residual against Cloudy, 0.034 pc — but MoCHII itself lies within 0.3% of Cloudy, so that lever is not available to MoCHII: it measures how much of a helium front comparison between two arbitrary codes can be recombination data alone.

The measurements of this subsection and of the HII40 He I front residual predate the helium–hydrogen charge exchange added on 2026-09-24 (§3.8), which CLOUDY c25 carries and MoCHII did not; it acts where $\mathrm{He}^+$ meets atomic hydrogen, i.e. at and beyond the fronts. With the 2026-09-24 atomic-data update (He–H charge exchange on, recombination cooling from Eq. 11) the production HII40 run puts $r(\mathrm{He}^0 = 0.5)$ at 4.12931 pc against 4.12776 pc before, so the residual to CLOUDY's 4.30531 pc is 0.17600 pc (was 0.17755; `tests/continuous_energy/analyze_hii40_residuals.py` on `results/continuous_energy/m2_atomic/hii40_atomic_rates.h5`). *Those residuals were measured on runs stopped by the earlier convergence rule* (§5.4); run to the drift test, the front is at 4.27107 pc and the residual is 0.034 pc (`results/continuous_energy/m5_converged`).

## 3.5 Collisional ionization (Voronov 1997)

$$k(T) = A\,\frac{1+P\sqrt{U}}{X+U}\,U^K e^{-U}, \qquad U = \frac{\Delta E}{kT}, \tag{12}$$

with parameters $(\Delta E\,[\mathrm{eV}], P, A, X, K)$ parsed from the author's `cfit.f` table:

| ion | $\Delta E$ | $P$ | $A\,[\mathrm{cm}^3\,\mathrm{s}^{-1}]$ | $X$ | $K$ |
|---|---|---|---|---|---|
| H I | 13.6 | 0 | $2.91\times10^{-8}$ | 0.232 | 0.39 |
| He I | 24.6 | 0 | $1.75\times10^{-8}$ | 0.180 | 0.35 |
| He II | 54.4 | 1 | $2.05\times10^{-9}$ | 0.265 | 0.25 |

plus the 16 registry rows in the element files. An optional `par%ci_model = 'dere_hybrid'` multiplies each Voronov rate by the constant Dere (2007)/Voronov ratio per (element, stage) taken from Cloudy c23.01's `DereRatio` table; the default `'voronov'` leaves the rate unchanged. In photoionized gas at $\sim10^4$ K the metal collisional ionization is Boltzmann-suppressed far below photoionization, so the hybrid shifts the ionization structure only in shielded, low-ionization zones where collisions are the sole metal-ionization channel.

## 3.6 Metal recombination: Badnell RR + DR

For the registry cascade, total recombination of the recombining ion is Badnell (2006, ApJS 167, 334) radiative recombination

$$\alpha_{\rm RR}(T) = A\left[\sqrt{T/T_0}\left(1+\sqrt{T/T_0}\right)^{1-b}\left(1+\sqrt{T/T_1}\right)^{1+b}\right]^{-1}, \qquad b = B + Ce^{-T_2/T}, \tag{13}$$

plus the Badnell-group dielectronic fits $\alpha_{\rm DR}(T) = T^{-3/2}\sum_i c_i e^{-E_i/T}$. Coefficients are read from the Badnell/Strathclyde AMDPP TAMOC tables (the `clist_K` files, 2023 release, copied into `data/atomic/badnell_{rr,dr}.dat` and verified md5-identical to the copies Cloudy c23.01 redistributes), using the $M$=1 ground-level fit of each recombining ion. Most ions are numerically identical to the CHIANTI v11.0.2 `.rrparams`/`.drparams` (which *are* Badnell data), the notable exception being Si II: CHIANTI 11 still carries the Abdel-Naby et al. (2012) DR, whose $\sim 10^4$-K resonance is stronger, so the 2023 refit lowers the total $\mathrm{Si}^+ + e \rightarrow \mathrm{Si}^0$ recombination to 0.48× the CHIANTI value at $10^4$ K. Ions Badnell does not tabulate (the low Ar/Fe/Cl/Ca stages, confirmed absent at every $M$) are taken from the CHIANTI files instead. Where CHIANTI carries no Badnell DR (Ar II, Ar III, Cl II, Ca II, Ca IV, Ca V), the `.drparams` type-2 entries are the Shull & Van Steenberg (1982) form $\alpha_{\rm DR} = A\,T^{-3/2}e^{-T_0/T}(1+B\,e^{-T_1/T})$, written as `DR2` lines and supported by the same reader (negligible at nebular temperatures; kept for completeness). One further CHIANTI RR type appears with iron and the higher calcium stages: the Fe II–IV, Ca IV and Ca V `.rrparams` are type-3 power laws, written as `RR2` lines and evaluated as $\alpha_{\rm RR} = A\,(T/10^4\,\mathrm{K})^{-\eta}$. Section 3.7 audits all of this entry by entry.

## 3.7 Recombination data across the registry: an audit

The $\alpha_1$ source-mixing defect of §4.10 was found in the H/He channels, and it prompted a sweep of every recombination coefficient the code carries — 32 registry entries over 11 elements, plus the hardcoded H/He channels. Three questions were put to them: is each coefficient the one its provenance header claims, is the functional form the best available for that ion, and how far does the answer depend on which data set is used.

**Transcription: no errors.** Of the 32 registry entries, 22 take their coefficients from the Badnell tables, and all 22 reproduce `data/atomic/badnell_{rr,dr}.dat` at $10^4$ K to a ratio of 1.0000. The key is fixed by the registry convention: `TRANSITION` $i$ of element $Z$ is the recombination of the ion of charge $i$, so the Badnell row is ($Z$, $N = Z-i$, $M = 1$). MoCHII's hardcoded H I coefficients ($8.318\times10^{-11}$, 0.7472, 2.965, $7.001\times10^5$) are exactly the $Z = 1$, $N = 0$ row of the same table. There is no transcription error anywhere in the registry, and no coefficient traceable to the wrong ion.

**The older forms cannot be migrated, and should not be.** Badnell's tabulation covers $N \leq 15$ (through the Al-like sequence) plus $N = 18$ (Ar-like) for $Z \geq 20$. Every use of `RR2` or `DR2` in the registry falls at a $(Z,N)$ outside that coverage. Where Badnell has nothing, `tools/fitting/make_element_data.py` takes CHIANTI v11.0.2 and writes *exactly the functional form CHIANTI declares*: `rrparams` types 1 and 2 are the Badnell form and become `RR`, type 3 is the power law and becomes `RR2`; `drparams` type 1 becomes `DR`, and type 2, the Shull & Van Steenberg (1982) form, becomes `DR2`. Checked against the CHIANTI file of the recombining ion, entry by entry:

| registry entry | recombining ion | $N$ | Badnell | CHIANTI `rr/dr` type → form |
|---|---|---|---|---|
| Ar 1 | Ar II | 17 | absent | 2 / 2 → RR + DR2 |
| Ar 2 | Ar III | 16 | absent | 1 / 2 → RR + DR2 |
| Ar 3 | Ar IV | 15 | covered | 2 / 1 → RR + DR |
| Cl 1 | Cl II | 16 | absent | 1 / 2 → RR + DR2 |
| Ca 1 | Ca II | 19 | absent | 1 / 2 → RR + DR2 |
| Ca 2 | Ca III | 18 | covered | 2 / 1 → RR + DR |
| Ca 3 | Ca IV | 17 | absent | 3 / 2 → RR2 + DR2 |
| Ca 4 | Ca V | 16 | absent | 3 / 2 → RR2 + DR2 |
| Fe 1, 2, 3 | Fe II–IV | 25, 24, 23 | absent | 3 / 1 → RR2 + DR |

Ar IV and Ca III are the controls: Badnell covers $N = 15$, and $N = 18$ for $Z \geq 20$, and the registry duly uses RR + DR at both. So the Shull & Van Steenberg form is not a stale choice made in preference to Badnell — it is what CHIANTI itself still ships for those ions. There is nothing to migrate, and the registry already uses the best source available at every element and stage. A suggestion recorded earlier, to move the RR2 and DR2 entries onto Badnell, is therefore withdrawn on both counts: Badnell has no row for them, and CHIANTI's current data for them is in those forms.

**Code to code: variance between data sets, not an error bar on MoCHII.** What transcription cannot settle is which data set is right. Comparing radiative recombination only — dielectronic recombination is excluded on both sides, because MOCASSIN adds it separately and keeping it on one side alone would compare unlike quantities — MoCHII's Badnell/CHIANTI values against the Verner & Ferland (1996) fits MOCASSIN uses, at $10^4$ K:

| recombination forming | MoCHII / VF96 |
|---|---|
| C II, C III, N II, Ne II | 0.999–1.016 |
| N I | 0.971 |
| Si IV (Na-like) | 0.932 |
| Ne I | 0.898 |
| C I | 0.861 |
| O I | 0.837 |
| Mg II (Na-like) | 0.791 |

Eleven ions: seven within 10%, ten within 20%, median 0.971. The spread is concentrated in the rates that form *neutral atoms*, which is where He I sits as well, at 11% (§3.4).

What that table is has to be stated precisely. It is the variance between two data sets, *not* a measured error in MoCHII, and for the metals there is no third arbiter available here: Cloudy ships `data/badnell_rr.dat` and `badnell_dr.dat` md5-identical to MoCHII's copies and evaluates them with `RR_Badnell_rate_coef` and `DR_Badnell_rate_coef` in `source/ion_recomb.cpp`, so Cloudy and MoCHII agree on metal recombination by construction and Cloudy cannot adjudicate. With only two determinations in play, which one is right for C I, O I, Ne I or Mg II is undetermined.

He I is the one exception, and there MoCHII is demonstrably right. Cloudy does *not* use Badnell for the He-like and H-like sequences; it derives them level by level from its own photoionization cross sections through the Milne relation, which is an independent first-principles determination. Against it, Badnell (MoCHII) is +0.3% and VF96 (MOCASSIN)

is +11.0% (§3.4), so for He I the spread is attributable to VF96. One may reason by analogy that the metal neutrals behave the same way — Badnell (2006) is the later and more careful calculation, VF96 (1996) leans on simplified treatments of many-electron systems, and He I is the case where that was actually demonstrated — but that is inference, not measurement. Settling it needs an independent determination for each ion: R-matrix values for C I, O I and Ne I recombination, or CHIANTI totals that do not derive from Badnell.

An earlier phrasing, "a 10–20% uncertainty in MoCHII's neutral-forming recombination rates", is withdrawn as imprecise: it reads a spread between two data sets as an error bar on one of them. The two accurate statements are that MoCHII's He I rate agrees with an independent first-principles calculation to 0.3% while the VF96 set MOCASSIN uses is 11% high (measured), and that for the metal neutrals Badnell and VF96 differ by 10–16% with no third source in this configuration able to say which is right (a code-to-code spread).

**What the audit establishes, and what it does not.** It found no defect in MoCHII's recombination data: no transcription error, no form that a better source could replace, and no coefficient attached to the wrong ion. That is not the same statement as "the data are accurate". For H and He an independent determination exists and MoCHII matches it to 0.3–0.7%; for the metals no arbiter is available in this configuration, so the verdict there is the absence of a defect found, not accuracy established.

## 3.8 Charge exchange with hydrogen

First-stage reactions use Huang et al. (2023, ApJ 951, 123) Table 4 (sources Stancil et al. 1998/1999, Lin et al. 2005, Zhao et al. 2005, Kingdon & Ferland 1996), e.g. for oxygen

$$k(\mathrm{O}+\mathrm{H}^+) = \left(1.26\times10^{-9}\,T_4^{0.517} + 4.25\times10^{-10}\,T_4^{0.00669}\right) e^{-227/T},$$
$$k(\mathrm{O}^+ + \mathrm{H}) = 2.08\times10^{-9}\,T_4^{0.405} + 1.11\times10^{-11}\,T_4^{-0.458}. \tag{14}$$

The oxygen rows required a correction. Huang's Table 4 prints the two oxygen reactions with their reactant labels interchanged, placing the barrier factor $e^{-227/T}$ on the exothermic recombination $\mathrm{O}^+ + \mathrm{H}^0$, where no barrier belongs, and leaving the endothermic ionization $\mathrm{O}^0 + \mathrm{H}^+$ — endothermic by 0.0196 eV (IP(O) − IP(H)), i.e. 227.7 K — without one. Taken as printed the pair violates detailed balance by 1.47× at 8 000 K; with the labels restored to the assignment above it satisfies it to 0.910, and the two directions then match the Stancil et al. (1999) fit that CLOUDY c25.00 uses to 1–3% over 4 000–20 000 K — the same determination, since Huang's Table 4 footnote for oxygen is Stancil et al. (1999). This check is now the gate `tests/charge_exchange/check_detailed_balance.f90`.

*Nitrogen (2026-09-24) does not use Huang's rows.* Table 4 prints for $\mathrm{N}+\mathrm{H}^+$ and $\mathrm{N}^+ + \mathrm{H}$ two polynomials in $\ln T$ (citing Lin et al. 2005) under which the endothermic direction *falls* with temperature — $1.6\times10^{-15}$, $8.9\times10^{-20}$ and $3.3\times10^{-23}\,\mathrm{cm^3\,s^{-1}}$ at $10^3$, $10^4$ and $3\times10^4$ K — and the exothermic one is 70–86× below the Kingdon & Ferland (1996) value. MoCHII carried them from the first transcription until 2026-09-24; the detailed-balance gate did not see it, because it screened out every pair whose smaller rate was below $10^{-15}\,\mathrm{cm^3\,s^{-1}}$ at $10^4$ K as unable to affect the balance, and the defect itself had made the rate that small. The pair is now KF96, read from the published scan: Table 1 $\mathrm{N}^+ + \mathrm{H}$, $a = 1.01\times10^{-3}$, $b = -0.29$, $c = -0.92$, $d = -8.38$ ($10^2$–$5\times10^4$ K; Butler & Dalgarno 1979), and Table 3 $\mathrm{N} + \mathrm{H}^+$ with the same $b, c, d$, $a = 4.55\times10^{-3}$ and $\Delta E/k = 1.086\times10^4$ K. The two $a$ differ by 4.505, the ground-term weight ratio

$g(N^+)g(H)/[g(N)g(H^+)] = 9 \cdot 2/4 = 4.5$: one calculation and its detailed-balance reverse. At $10^4$ K the pair is $1.54\times10^{-12}$ / $1.01\times10^{-12}$ cm$^3$ s$^{-1}$; CLOUDY c25.00's transcription of Lin et al. (2005) gives $1.68\times10^{-12}$ / $1.26\times10^{-12}$. Nitrogen is near-resonant with hydrogen ($\Delta E = 0.936$ eV), so like oxygen it is locked to hydrogen's ionization in partly neutral gas. The gate now has no rate floor: every pair with data in both directions is tested from 3 000 to 20 000 K, and the one exemption is by name and for a physical reason — carbon, whose exothermic $C + H^+$ is dominated by radiative charge transfer (Stancil et al. 1998) with no collisional reverse, is tested one-sided: the endothermic rate may not exceed the detailed-balance reverse of the whole forward rate. Higher stages ($X^{i+} + H^0 \rightarrow X^{(i-1)+}$, recombination direction) use the Kingdon & Ferland (1996) fits $k = a\,T_4^b\,(1 + c\,e^{d\,T_4})$, transcribed from the MOCASSIN `update_mod.f90` `chex` table and evaluated at $T$ clamped into the fit validity (∼5 000–50 000 K). *Stage mapping (corrected 2026-07-13).* MOCASSIN's `chex`$(Z, \mathrm{ion})$ is the reaction $X^{\mathrm{ion}+} + H^0 \rightarrow X^{(\mathrm{ion}-1)+}$, and its inline comment labels the *product* (lower) ion. The recombining ion of transition $i$ is $X^{i+}$, so the entry is `chex`$(Z, i)$. An earlier version read the product-labeled comment as the reactant and used `chex`$(Z, i{+}1)$, placing every higher-stage rate one ionization stage too low; the corrected assignment (verified against MOCASSIN's own ratio loop and CMacIonize) is below.

| reaction | $a$ [$10^{-9}$cm$^3$ s$^{-1}$] | $b$ | $c$ | $d$ |
|---|---|---|---|---|
| $C^{2+}$ + H | $1.67\times10^{-4}$ | 2.79 | 304.72 | −4.07 |
| $C^{3+}$ + H | 3.25 | 0.21 | 0.19 | −3.29 |
| $N^{2+}$ + H | tabulated, Barragan et al. (2006); see below | | | |
| $O^{2+}$ + H | tabulated, Barragan et al. (2006); see below | | | |
| $Ne^{2+}$ + H | $10^{-5}$ | 0 | 0 | 0 |
| $S^{2+}$ + H | $10^{-5}$ | 0 | 0 | 0 |
| $S^{3+}$ + H | 2.29 | 0.0402 | 1.59 | −6.06 |
| $Ar^{2+}$ + H | $10^{-5}$ | 0 | 0 | 0 |
| $Ar^{3+}$ + H | 4.57 | 0.27 | −0.18 | −1.57 |
| $Si^{2+}$ + H | 1.23 | 0.24 | 3.17 | $4.18\times10^{-3}$ |
| $Si^{3+}$ + H | 0.49 | −0.0874 | −0.36 | −0.79 |
| $Si^{4+}$ + H | 7.58 | 0.37 | 1.06 | −4.09 |
| $Mg^{2+}$ + H | $8.58\times10^{-5}$ | $2.49\times10^{-3}$ | 0.0293 | −4.33 |
| $Fe^{2+}$ + H | 1.26 | 0.0772 | −0.41 | −7.31 |
| $Fe^{3+}$ + H | 3.42 | 0.51 | −2.06 | −8.99 |

*$N^{2+}$ and $O^{2+}$ (2026-09-24).* These two now use the Barragan et al. (2006, ApJ 636, 544) rate coefficients directly: Table 1 $k_1$ for $O^{2+}(^3P) + H \rightarrow O^+ + H^+$ and Table 2 $k_4$ for $N^{2+}(^2P^\circ) + H \rightarrow N^+ + H^+$, tabulated over $10^2$–$10^5$ K and carried as the 26-point `CXRTAB` row of the element file (`CXR 8`), interpolated log–log in $T$ and held at the end values outside the table (no extrapolation). CLOUDY c25 and CHIANTI v11 use the same calculation. Against the KF96 fits they replace (the old rows $a = 1.04$, $b = 0.27$, $c = 2.02$, $d = -5.92$ for $O^{2+}$ and 0.305, 0.60, 2.65, −0.93 for $N^{2+}$), KF96 was 1.56× higher for $O^{2+}$ at 5000 K and 1.22× at $10^4$ K, and 0.57–0.64× for $N^{2+}$ over the same range. The pairwise charge conservation, the root-solve gate G-M1f and the detailed-balance gate of `tests/charge_exchange/` pass with the new rows (the latter, with nitrogen and 3 000 K now included, at worst 0.303 against tolerance 0.35).

Mg and Fe were already correct: their Huang Table 4 reactant labels happen to land on `chex`$(Z, i)$. Silicon's first

stage uses the Huang A9/A10 fits ($Si + H^+$ and $Si^+ + H^0$, the latter $2.371\times10^{-12}$cm$^3$ s$^{-1}$ at $10^4$ K); Cl and Ca have no KF96/Huang transcription. The Mg and Fe first-stage reactions follow Huang Table 4: Mg + $H^+$ is the KF96 form with an added reverse barrier $e^{-6.91/T_4}$ (endothermic; the fitted factor is clamped into its validity range but the barrier factor takes the actual $T$ — until 2026-09-24 it was clamped too, which below 5000 K froze $e^{-\Delta E/kT}$ at its 5000 K value); Fe + $H^+$ is a fast constant $4.0\times10^{-9}$cm$^3$ s$^{-1}$ with reverse $a\,(T/300)^b\,e^{-6.61/T_4}$. The correction raises the HII40 doubly/triply-ionized fractions (Section 5.4); the earlier good agreement of some metal fractions with the underconverged MOCASSIN baseline was partly fortuitous.

With the shipped abundances 29 of these reactions are active — 7 in the $H^+ \rightarrow H^0$ direction (C, N, O, Mg, Si, S, Fe, all at the neutral stage) and 22 in the reverse — and every one of them enters hydrogen's balance as well as the metal's (§4.4, `par%charge_exchange`).

**Helium–hydrogen charge exchange (`par%helium_charge_exchange`, default on, 2026-09-24).** Four reactions couple the H and He balances (`ion_balance_mod`: `cx_HeII_H`, `cx_HeI_Hp`, `cx_HeIII_H`, `cx_HeII_Hp`):

- $He^+ + H \rightarrow He + H^+$, the sum of two channels that the underlying calculation (Zygelman, Dalgarno, Kimura & Lane 1989, PRA 40, 2340) keeps apart. The *radiative* channel, through the $A\,^1\Sigma^+ \rightarrow X\,^1\Sigma^+$ transition, is $1.25\times10^{-15}(T/300)^{0.25}$ cm$^3$ s$^{-1}$ (Glover & Jappsen 2007, ApJ 666, 1, Table 1 R26, citing Zygelman et al.); that fit is one quarter of Zygelman's Table I "Direct" column (5.36, 4.21, 4.41, 4.50, 4.83, $5.99\times10^{-15}$ at 1, 10, 100, 200, 400, 1000 K), the quarter being the probability of approach in the singlet, $(2S+1)(2-\delta_{0\Lambda})/[(2L_a+1)(2S_a+1)(2L_b+1)(2S_b+1)] = 1/4$ for $He^+(^2S)+H(^2S)$ (Courtney et al. 2021, ApJ 919, 70, Eqs. 4–6). Zygelman tabulates only to 1000 K, so above that the $T^{0.25}$ is an extrapolation. The *non-radiative* channel is the Kingdon & Ferland (1996, ApJS 106, 205) Table 1 $He^+$ row, their Eq. 7 with $a = 7.47\times10^{-6}$, $b = 2.06$, $c = 9.93$, $d = -3.89$, fitted over $6\times10^3$–$10^5$ K. Below 6000 K the fit is evaluated as written: it falls as $t_4^{2.06}$, as a curve-crossing channel must at low collision energy, whereas holding the 6000 K value would overstate it tenfold at 1000 K; above $10^5$ K $T$ is held at $10^5$ K. At 8000 K the two channels give $2.84\times10^{-15}$ and $6.8\times10^{-15}$ cm$^3$ s$^{-1}$.
- $He^{2+} + H \rightarrow He^+ + H^+$: the KF96 $He^{2+}$ row, a constant $10^{-14}$ cm$^3$ s$^{-1}$ ($10^3$–$10^7$ K; West, Lane & Cohen 1982).
- $He + H^+ \rightarrow He^+ + H$ (endothermic by 10.989 eV): the detailed-balance reverse of the non-radiative channel, $4\,e^{-10.989\,\mathrm{eV}/kT}\,k_{\mathrm{nr}}$ (statistical weights $g(He^+)g(H)/[g(He)g(H^+)] = 4$; the reduced-mass factor is $1+6\times10^{-4}$ and is dropped). The radiative channel has no collisional reverse. The direct calculation of this direction by Kimura, Lane, Dalgarno & Dixson (1993, ApJ 405, 801; Glover & Jappsen 2007 R27) is not used. Read from the published paper, its Table 1 gives charge-transfer cross sections from 20 eV only, above the 10.99 eV threshold, and its Table 3 rates imply an apparent activation energy of 4.33 eV between 8000 and $10^4$ K (15.2 eV between 6000 and 8000 K), below the $E_{\mathrm{th}} - \frac{3}{2}kT \approx 9.7$ eV that the Tolman relation allows an endothermic rate there; the table is not reliable at nebular temperatures. (It was adopted for a few hours on 2026-09-24, at 35× this rate at $10^4$ K and 0.16× at $3\times10^4$ K; the Lexington benchmarks moved by 0.02 K or less either way.)
- $He^+ + H^+ \rightarrow He^{2+} + H$: the detailed-balance reverse of the $He^{2+}$ reaction, $e^{-40.819\,\mathrm{eV}/kT}\times10^{-14}$ cm$^3$ s$^{-1}$ (weight ratio 1).

Each reaction enters both the hydrogen and the helium balance (§4.4), so the charge it moves is conserved in the pair. The default is on because the physics is real and the reference codes carry it: CLOUDY c25, MOCASSIN and EXHALE all include $He^+ + H$, and without it $He^+$ is overestimated wherever it meets atomic hydrogen. CLOUDY c25 uses 1.25× the KF96 value for the non-radiative channel: its `ctrecombdata.dat` row is KF96 scaled by 0.25, and `atmdat_char_tran.cpp`:93 adds the full $7.47\times10^{-15}$ expression on top. `.false.` drops all four reactions and reproduces runs made before 2026-09-24. Gate `tests/charge_exchange/check_helium_charge_exchange.f90` (in `run_check.sh`): the rate forms and detailed balance to $5.0\times10^{-7}$; on four partially neutral cells the three balances plus charge neutrality close to $2.6\times10^{-11}$; and the exchange lowers $He^+$ and raises $H^+$ (on the cell with $x_{\rm HII} = 0.041$, $x_{\rm HII}$ rises 3.1× and $x_{\rm HeII}$ falls to 0.968×). The heat these reactions exchange with the gas is the next paragraph.

**Charge-exchange heating (`par%charge_exchange_heating`, default on, 2026-09-25).** A charge-exchange reaction moves one electron between two binding energies. In this code the ionization energy is not thermal energy (photoionization deposits $h\nu - \mathrm{IP}$, recombination removes the captured electron's kinetic energy), so what a reaction does to the gas is the kinetic energy of its products:

$$\Delta E_{\rm kin} = \left[\mathrm{IP(acceptor's\ product\ stage)} - \mathrm{IP(donor)}\right] - E_{\rm rad}, \tag{15}$$

with $E_{\rm rad}$ the excitation of the product state that leaves as radiation — an allowed or intercombination decay, a forbidden line below its critical density, or the photon of a radiative charge transfer. An endothermic reaction has $\Delta E_{\rm kin} < 0$ and takes it from the gas (Kingdon & Ferland 1996, §2 and Table 4, write the reaction $X^{q+} + H \rightarrow X^{(q-1)+} + H^+ + \Delta E$ with the same sign convention, and note that the recombined ion "can and usually will be in an excited state"). The thermal balance (§4.5) adds $\sum k\, n_{\rm reactant}\, n_{\rm partner}\, \Delta E_{\rm kin}$ over the carried reactions: `species_mod::metal_charge_exchange_heating` (under `use_metals`) with the cascade's stage fractions and rate coefficients, and `ion_balance_mod::helium_charge_exchange_heating` (under `helium_charge_exchange`). The defects are data: each transition of `data/atomic/element_<el>.txt` carries a `CXHEAT` row, $\Delta E_{\rm ion}$, $\Delta E_{\rm rec}$ and $g_{\rm split}$, written by `tools/fitting/make_element_data.py`, which states the product state and its source for every reaction and refuses a carried reaction without one; the loader refuses a file without the rows while the switch is on. Term energies are CHIANTI observed energies, $(2J+1)$-weighted over the term. KF96 print no defect for their Table 1 fits; the electronic version of their tables that CLOUDY c25.00 distributes (`data/ctrecombdata.dat` column 7, `ctiondata.dat` column 8; row mapping verified against KF96 Table 1 and the MoCHII rows) carries one, and where it equals ΔIP minus a level energy it identifies the product ($O^+$ 4P, $N^+$ 3D, $Mg^+$ 3p, $Si^+$ 4P, $Si^{3+}$ 4s, all to 0.01 eV). Where it does not, it is used as distributed and marked unidentified.

Three product assignments deserve their reasoning. For $O^{2+} + H$, Barragan et al. (2006) name no product but place the avoided crossing of the quartet subsystem at $R_0 = 4.57$ a.u.; by KF96 Eqs. (3)–(4) that is an asymptotic defect near 6.5 eV, the $O^+$ $2s2p^4\ ^4P$ channel (6.65 eV), whose 833 Å decay lies above 13.6 eV (a photon that would ionize hydrogen, not transported). $S + H^+$ is weighted over Zhao et al.'s Table III channels at $10^4$ K ($^2$P 93%, $^2$D 7%); the $^2$P share runs from 0.978 at 4000 K to 0.888 at $2\times10^4$ K, which moves $E_{\rm rad}$ by less than 0.05 eV. For $Fe + H^+$ Rutherford & Vroom (1972) state that "the state of excitation of the products could not be determined" and attribute the large cross sections to near resonance; the $z^4F/z^4D$ terms 0.15 eV below the defect are the near-resonant ones, and their decay chain ends in

Table 2: Kinetic energy $\Delta E_{\rm kin}$ [eV] one event gives the gas (Eq. 15); $k$ at $10^4$ K. "ion": $X^i + H^+ \rightarrow X^{i+1} + H^0$; "rec": the reverse direction.

| reaction | dir | $\Delta$IP | $\Delta E_{\rm kin}$ | product; source |
|---|---|---|---|---|
| $C+H^+$ | ion | 2.338 | 2.338 | $C^+$ $^2P$, non-radiative part only (below) |
| $C^+ + H$ | rec | −2.338 | −2.338 | C $^3P$, ground |
| $C^{2+} + H$ | rec | 10.785 | 4.010 | not identified; KF96 electronic |
| $C^{3+} + H$ | rec | 34.290 | 5.730 | not identified; KF96 electronic, $2.86(q-1)$ |
| $N+H^+$ | ion | −0.936 | −0.936 | $N^+$ $^3P$, ground by energetics |
| $N^+ + H$ | rec | 0.936 | 0.936 | N $^4S$, ground by energetics |
| $N^{2+} + H$ | rec | 16.003 | 4.566 | $N^+$ $2s2p^3$ $^3D$; Barragan et al. 2006 |
| $O+H^+$ | ion | −0.020 | −0.020 | $O^+$ $^4S$, ground |
| $O^+ + H$ | rec | 0.020 | 0.020 | O $^3P$, ground |
| $O^{2+} + H$ | rec | 21.523 | 6.653 | $O^+$ $2s2p^4$ $^4P$; Barragan et al. 2006, KF96 electronic |
| $Ne^{2+}$, $S^{2+}$, $Ar^{2+}+H$ | rec | | 0 | radiative ($10^{-14}$ cm$^3$ s$^{-1}$ rows, KF96 §3.1) |
| $S+H^+$ | ion | 3.238 | 0.276 | $S^+$ $^2P/^2D/^4S$; Zhao et al. 2005 Table III |
| $S^+ + H$ | rec | −3.238 | −3.238 | S $^3P$, ground |
| $S^{3+}$, $Ar^{3+}+H$ | rec | 21.2, 27.1 | 5.730 | not identified; KF96 electronic |
| $Mg+H^+$ | ion | 5.952 | −1.258 | $Mg^+$ 3p (+1.522) or 4s (−2.703), branched by Allan et al. (1988) Table 1; value at $10^4$ K (+0.61 |
| $Mg^+ + H$ | rec | −5.952 | −5.952 | Mg $^1S$, ground |
| $Mg^{2+} + H$ | rec | 1.437 | 1.437 | $Mg^+$ 3s, ground by energetics |
| $Fe+H^+$ | ion | 5.696 | 0.151 | $Fe^+$ $z^4F/z^4D$, near resonance (derived) |
| $Fe^+ + H$ | rec | −5.696 | −5.696 | Fe $a^5D$, ground |
| $Fe^{2+} + H$ | rec | 2.601 | 2.601 | $Fe^+$ $a^6D$, ground; KF96 electronic 2.56 |
| $Fe^{3+} + H$ | rec | 17.053 | 6.300 | not identified; KF96 electronic |
| $Si+H^+$ | ion | 5.446 | 0.115 | $Si^+$ $3s3p^2$ $^4P$; KF96 electronic |
| $Si^+ + H$ | rec | −5.446 | −5.446 | Si $^3P$, ground |
| $Si^{2+} + H$ | rec | 2.748 | 2.748 | $Si^+$ $^2P$, ground by energetics |
| $Si^{3+} + H$ | rec | 19.895 | 4.230 | not identified; KF96 electronic |
| $Si^{4+} + H$ | rec | 31.544 | 7.493 | $Si^{3+}$ 4s $^2S$; KF96 electronic |
| $He^+ + H$, non-rad. | rec | 10.989 | 10.989 | He $1^1S$; Zygelman et al. 1989 |
| $He^+ + H$, radiative | rec | 10.989 | 0 | photon |
| $He+H^+$ | ion | −10.989 | −10.989 | reverse of the non-radiative channel |
| $He^{2+} + H$ | rec | 40.819 | 0 | radiative (KF96 $10^{-14}$ row) or $He^+(n=2)$ |
| $He^+ + H^+$ | ion | −40.819 | −40.819 | H 1s, ground |

[Fe II] lines, so $E_{\rm rad}$ is their mean — derived, not a determination of the product. Counting a metastable product ($S^+$ $^2$D, $^2$P) as radiated holds below its critical density ($\sim 10^3$ cm$^{-3}$ for $^2$D).

Detailed balance fixes the defects of a pair that is one channel and its reverse: the two carry opposite $\Delta E_{\rm kin}$ and deposit nothing where they balance. That holds for N, O and the non-radiative $He^+ + H$ pair. $C + H^+$ is mostly radiative charge transfer (KF96 §3.1 assume the reverse of $C^+ + H$ to be radiative), so only its non-radiative part, the detailed-balance reverse of $C^+ + H$, $k_{\rm nr} = \min[k_{\rm ion},\, g_{\rm split}\, k_{\rm rec}\, e^{\Delta E/kT}]$ with $g_{\rm split} = g(C^+)g(H)/[g(C)g(H^+)] = 4/3$, heats. The S, Mg, Si and Fe pairs are *not* one channel: the exothermic direction ends in a radiating term, whereas the endothermic direction starts from whatever term the ion is in. Huang et al. (2023, sec. 2.4.2) "estimate the reverse rates assuming microscopic balance" from the TOTAL forward rate, which is the reverse with the ion's excited terms in LTE; in nebular gas the ion sits in its ground term, and that overstates the reverse (the reason EXHALE drops $Mg^+ + H$).

*Sulfur is resolved by product term (2026-09-26, charge-exchange form 9).* Zhao et al. (2005, PRA 71, 062713; read) Table III gives the forward rate into each $S^+$ term, 25 temperatures from 20 K to $10^6$ K; at $10^4$ K $^4$S $8.90\times10^{-16}$, $^2$D $1.19\times10^{-12}$, $^2$P $1.63\times10^{-11}$ cm$^3$ s$^{-1}$. The element file carries that table (`CXSTATE` rows), the forward rate is its sum, and the reverse is each channel's detailed balance weighted by the term's actual population,

$$k_{\rm rev} = \sum_j p_j(T, n_e)\, k_j(T)\, \frac{g(\mathrm{S})}{g_j\, g(\mathrm{H})}\, e^{-(\mathrm{IP(H)}-\mathrm{IP(S)}-E_j)/kT}, \tag{16}$$

$p_j$ from the [S II] $n$-level atom, tabulated at setup on a (log $T$, log $n_e$) grid and interpolated in $\ln p_j$. At $10^4$ K it is $2.34\times10^{-17}$ at $n_e \to 0$ (the $^4$S channel alone) and $6.0\times10^{-16}$ at $n_e = 100$, where the slightly populated $^2$D and $^2$P terms already dominate, against the $3.7\times10^{-13}$ of the LTE reverse, which it recovers at $n_e = 10^{12}$ to 0.2% (gate `check_detailed_balance`). The heat is resolved the same way. Measured on the converged states (restarts from `m8_defaults`): the Lexington models move by $\leq 0.023$ K, $L(\mathrm{H}\beta)$ by $2\times10^{-6}$ and no line by more than 0.13% (the largest is [Ne III] at HII20, the run-to-run level; the sulfur lines move less), and the PDR test's neutral-zone $x(S^0)$ by −23% ($1.59 \to 1.23\times10^{-5}$), $T_e$ by −1.4 K, lines by $\leq 0.1\%$ — small because the FUV band photoionizes $S^0$ and the endothermic reverse is slow at the neutral zone's 700 K. *Mg, Si and Fe* keep the total-balance reverse (Mg: Huang et al. from the Kingdon & Ferland forward; Si, Fe: the same form in Huang et al.'s Table 4), marked in the element files as an upper bound on the reverse from the ground term: no forward rate into that term is available (Allan et al. 1988 give $Mg^+$ 3p and 4s only). Likewise $He^+ + H^+ \to He^{2+} + H$ is the reverse of a ground channel that the $10^{-14}$ forward rate does not describe; at $10^4$ K it is $\sim 10^{-35}$ cm$^3$ s$^{-1}$.

Gate `tests/charge_exchange/check_charge_exchange_heating.f90` (in `run_check.sh`): all 29 metal reactions have a defect and nothing else does; every defect equals $\Delta\mathrm{IP} - E_{\rm rad}$ with $E_{\rm rad}$ transcribed independently into the gate, to $10^{-6}$ eV; the N, O, C (non-radiative) and He pairs cancel where they balance to $1.2\times10^{-16}$ relative (tolerance $10^{-12}$) at 3000, $10^4$ and $2\times10^4$ K; and `metal_charge_exchange_heating` equals the sum of the reactions formed in the gate to $3.3\times10^{-14}$.

Measured on the converged Lexington states (§5.4; `results/continuous_energy/m5_converged`, with the code's own functions applied to the written leaf states, not a rerun): the charge-exchange heat is $3.55\times10^{-4}$ of the volume-integrated photoheating at HII40 and $3.0\times10^{-6}$ at HII20. At HII40 it is $O^{2+} + H$ ($3.0\times10^{-4}$) and $N^{2+} + H$ ($4.3\times10^{-5}$); the oxygen pair $O^0 \leftrightarrow O^+$, individually $\pm 1.4\times10^{-4}$ (HII40) and $\pm 2.8\times10^{-4}$ (HII20), cancels to $1.7\times10^{-6}$ and $2.2\times10^{-6}$ because charge exchange locks oxygen's ionization to hydrogen's; He–H adds $3.0\times10^{-6}$ and $5\times10^{-8}$. The local share rises

outward with $x_{\rm HI}$ while $O^{2+}$ survives, to a maximum of $7.9\times10^{-4}$ at $r = 3.98$ pc (HII40, 0.7 pc inside the H front, where $O^{2+}$ is still abundant), and falls to $10^{-5}$ in the front itself, where $O^{2+}$ is gone; at HII20 the maximum is $7.1\times10^{-5}$ at the front. `.false.` restores the thermal balance of runs made before 2026-09-25 exactly.

## 3.9 Cooling fits

Collisional line cooling for the thermal loop uses the EXHALE multi-exponential form fitted to direct CHIANTI v11.0.2 evaluations (`fit_cooling.py`; reader `chianti_cooling.py` parses `.elvlc`/`.scups` with Burgess & Tully 1992 descaling):

$$\Lambda_{\rm ion}(T) = T^{-1/2}\sum_{i=1}^{N} A_i\, e^{-T_i/T} \qquad [\mathrm{erg\,cm^3\,s^{-1}\ per\ } (n_e n_{\rm ion})], \tag{17}$$

evaluated in the optically thin $n_e \to 0$ limit of statistical equilibrium: every ion sits in its lowest level, excitation happens out of that level only, and the cascade radiates the full excitation energy, so $\Lambda_{\rm ion}$ is the excitation power out of the ground level and equals what the $n$-level solve (Section 3.10) returns as $n_e \to 0$. Transition energies are the observed `.elvlc` level differences in both the Boltzmann factor and the radiated photon (the `.scups` de is the Burgess–Tully scaling energy, and detailed balance ties the excitation rate to the level gap that appears in the equilibrium matrix). Fit range $10^3$–$10^5$ K; Levenberg–Marquardt on $\log\Lambda$ with the $T_i$ ladder derived from the excitation temperatures $\Delta E_{1u}/k$ of each ion.

| ion | $N$ | max err | ion | $N$ | max err |
|---|---|---|---|---|---|
| H I | 4 | 0.52% | O I | 6 | 0.20% |
| C I | 6 | 0.27% | O II | 4 | 0.27% |
| C II | 5 | 0.35% | O III | 5 | 0.32% |
| C III | 6 | 0.50%[a] | S II | 3 | 2.23% |
| N I | 5 | 0.58% | S III | 5 | 0.23% |
| N II | 5 | 0.66% | S IV | 3 | 0.61% |
| N III | 5 | 0.70% | Ne II | 5 | 0.40% |
| | | | Ne III | 5 | 0.34% |
| | | | Ar III | 2 | 0.22%[b] |
| | | | Ar IV | 4 | 0.27% |

[a] in the relevant range ($\Lambda > 10^{-3}$ max); 5.5% at the negligible low-$T$ tail. [b] at $10^4$ K; 3.9% at the $10^5$-K endpoint.

By construction Eq. (17) carries no collisional de-excitation, so it is exact for lines whose critical density is far above the local $n_e$ and increasingly too large for those below it — the low-$n_{\rm crit}$ lines [C II] 158 $\mu$m, [O III] 52/88 $\mu$m and the [O II]/[S II] optical doublets, whose electron critical densities are $10^2$–$10^{3.5}\,\mathrm{cm^{-3}}$, so at the $n_e \sim 10^2\,\mathrm{cm^{-3}}$ of an H II region they overcool the line budget by $\sim$12% (measured against a CLOUDY c25.00 cooling decomposition on a fixed state). The thermal loop therefore treats Eq. (17) as the $n_e \to 0$ *carrier* and suppresses it to the local density. A $(T, n_e)$ table, built once at setup (`nlevel_cooling_mod`) by solving the same $n$-level atom over a grid ($\log_{10} T$ = 2–6, $\log_{10} n_e$ = –6–10), stores for each ion the suppression ratio

$$S_{\rm sup}(T,n_e) = \frac{\Lambda_{\rm SE}(T,n_e)}{\Lambda_{\rm SE}(T,n_e\to 0)} \in (0,1], \qquad \Lambda_{\rm SE}(T,n_e) = \frac{1}{n_e}\sum_k n_u^{(k)} A_k\, \Delta E_k, \tag{18}$$

and the low-density reference $\Lambda_{\rm SE}(T, n_e \to 0)$; the runtime cooling coefficient combines the two as

$$\Lambda(T, n_e) = \Lambda_{\rm fit}(T)\left[ f_{\rm cov}(T)\, S_{\rm sup}(T, n_e) + (1 - f_{\rm cov}(T)) \right], \qquad f_{\rm cov} = \min\!\left(1,\ \frac{\Lambda_{\rm SE}(T, n_e \to 0)}{\Lambda_{\rm fit}(T)}\right). \tag{19}$$

Here $f_{\rm cov}$ is the fraction of the low-density cooling the (often truncated) $n$-level model resolves; the remainder $1 - f_{\rm cov}$ — the high-excitation resonance lines, of far higher critical density — is left in its low-density limit, so a truncated model cannot bias the total. As $n_e \to 0$, $S_{\rm sup} \to 1$ and $\Lambda \to \Lambda_{\rm fit}$ *exactly*, and an ion with no $n$-level model would keep the plain fit — there is none left, since C I and N I, the last two gaps, now carry models of their own (their far-infrared fine-structure lines have $n_{\rm crit} \sim 2$–$10\,{\rm cm^{-3}}$, so the suppression is severe: $S_{\rm sup} = 0.43$ at $10^4$ K and $n_e = 10^4\,{\rm cm^{-3}}$). This density-suppressed cooling (`par%cooling_model = local_ne`, the default) is built from the very solve that produces the diagnostic lines (Section 3.10), so the thermal balance and the emergent line list are one physics; the pure zero-density fit is retained as the `low_density` option, exact in the diffuse gas the limit is written for. The density-dependent emissivities themselves come from the $n$-level data.

## 3.10 $n$-level (diagnostic) data

For output-time diagnostics, `fit_nlevel.py` exports for each ion the lowest $n$ levels (energies, statistical weights), the radiative $A$-values (`.wgfa`), and the effective collision strengths $\Upsilon(T)$ as low-order Chebyshev fits in Burgess–Tully scaled space: with $e_t = kT/\Delta E$,

$$s = \begin{cases} 1 - \ln C / \ln(e_t + C) & \text{types 1, 4} \\ e_t/(e_t + C) & \text{types 2, 3, 5, 6} \end{cases}, \qquad u = \frac{2(s - s_{\rm lo})}{s_{\rm hi} - s_{\rm lo}} - 1, \qquad y = \sum_k c_k T_k(u), \tag{20}$$

descaled by type ($y\ln(e_t{+}e)$; $y$; $y/(e_t{+}1)$; $y\ln(e_t{+}C)$; $y/e_t$; $10^y$), with a log flag storing $\log_{10} y$ for weak peaked intercombination transitions. The $n$-level export now covers the 27 line-emitting ions of the eleven-element registry (Table 3); the worst transition fit error is 1.7% (O III $^1{\rm S}_0$–$^5{\rm S}_2$, $\Upsilon \sim 10^{-4}$), typically $\lesssim 0.5\%$. Ar II has no level/collision data in CHIANTI v11 (only RR/DR), so that stage is tracked without cooling or lines ([Ar II] 6.98 $\mu$m omitted). Verification is layered: the fitted files reproduce direct CHIANTI spline evaluation of the classic ratios to 0.05–0.5%; the PyNeb cross-check gives [O III] 5007/4363 within 1.2% and [N II] 6584/5755 within 0.3% (the density diagnostics differ by 6–10% at $10^4\,{\rm cm^{-3}}$ from PyNeb's independent collision data — database spread); and the Fortran solver (Section 4.19) matches the Python reference implementation to four significant digits on 26 lines. Three atoms, C I, C II and O I, also carry atomic-hydrogen impact rates for their ground-term fine structure in a sibling file `hcoll_<el>_<stage>.txt`, read by `nlevel_load` and solved in the same statistical equilibrium as the electron rates (§4.18).

Table 3 lists the principal collisional diagnostic lines each metal block produces (the complete transition list for each ion is written to `<base>_lines.txt`); only active-registry ions with an `nlevel_<el>_<stage>` file get a block. The H I/He II/He I recombination-line set is documented separately in the recombination-lines subsection below.

Table 3: Metal collisional diagnostic lines from the $n$-level solve (block `emis_<el>_<stage>` in `<base>_emis`; principal lines, Å unless $\mu$m noted). C, N, O, Ne, S are active by default; Ar, Mg, Fe, Si, Cl, Ca are activated by `par%abund_<El>`.

| ion (block) | principal lines |
|---|---|
| `c_1` [C I] | 4623, 8729, 9827, 9853, 370 $\mu$m, 609 $\mu$m |
| `c_2` [C II] | 2325 (UV multiplet), 158 $\mu$m |
| `c_3` C III] | 977, 1907, 1909, 178 $\mu$m |
| `n_1` [N I] | 3467/3468, 5199, 5202, 10400/10410 (multiplet) |
| `n_2` [N II] | 3064/3071, 5756, 6550, 6585, 122 $\mu$m, 205 $\mu$m |
| `n_3` [N III] | 1750 (UV multiplet), 57 $\mu$m |
| `o_1` [O I] | 5579, 6302, 6366, 63 $\mu$m, 145 $\mu$m |
| `o_2` [O II] | 2471, 3727, 3730, 7321/7331 |
| `o_3` [O III] | 1661/1666, 2322/2332, 4364, 4960, 5008, 52 $\mu$m, 88 $\mu$m |
| `ne_2` [Ne II] | 12.8 $\mu$m |
| `ne_3` [Ne III] | 3343, 3870, 3969, 15.55 $\mu$m, 36 $\mu$m |
| `s_2` [S II] | 4070/4078, 6718, 6733, 1.03 $\mu$m, 315 $\mu$m |
| `s_3` [S III] | 3722, 6314, 8831, 9070, 9532, 12 $\mu$m, 18.7 $\mu$m, 33.5 $\mu$m |
| `s_4` S IV] | 1398–1424 (UV multiplet), 10.5 $\mu$m |
| *Optional — activate with* `par%abund_<El>` | |
| `ar_3` [Ar III] | 3006/3110, 5193, 7138, 7753, 8.99 $\mu$m, 21.8 $\mu$m |
| `ar_4` [Ar IV] | 2854/2869, 4713, 4741, 7170 (multiplet), 56 $\mu$m, 78 $\mu$m |
| `mg_2` Mg II | 2796, 2804 |
| `fe_2` [Fe II] | 1.26/1.32 $\mu$m, 5.34 $\mu$m, 17.9 $\mu$m, 26 $\mu$m (40 lines total) |
| `fe_3` [Fe III] | 4659, 4703, 4881, 5011, 22.9 $\mu$m (38 lines; 16/34-level with `fe_levels_full`) |
| `si_2` [Si II] | 2335 (multiplet), 34.8 $\mu$m |
| `si_3` Si III] | 1206, 1883, 1892, 38 $\mu$m |
| `si_4` Si IV | 1394, 1403 |
| `cl_2` [Cl II] | 3587/3679, 6164, 8581, 9126, 14.4 $\mu$m, 33 $\mu$m |
| `cl_3` [Cl III] | 3344/3354, 5519, 5539, 8480 (multiplet) |
| `cl_4` [Cl IV] | 3120/3205, 5325, 7263, 7533, 11.8 $\mu$m, 20.3 $\mu$m |
| `ca_2` Ca II | 3935, 3970, 7293/7326, 8500/8544/8665 |
| `ca_5` [Ca V] | 3999, 5311, 6088, 3.05 $\mu$m, 4.16 $\mu$m, 11.5 $\mu$m |

Some registry ions are tracked in the ionization balance but carry no line block because CHIANTI v11 has no level/collision data for them: Ar II, Cl V, Ca III, Ca IV.

## 3.11 Free-free Gaunt factors and continuum data

`gaunt.f90` is the verbatim MOCASSIN copy of the Hummer (1988) two-dimensional Chebyshev fit of the Karzas & Latter (1961) free-free Gaunt factor ($\leq 0.7\%$; interface `getGauntFF(z, log10Te, xlf, g)`). The thermal loop uses the thermally averaged $\bar{g}(T,Z) = \int g_{\rm ff}(u)\, e^{-u}\, du$ tabulated at setup ($\bar{g}(10^4\,\mathrm{K}, Z{=}1) = 1.2626$), now over net charge $Z = 1$–4 so the trace-metal free-free (`par%metal_freefree`) has its Gaunt factor. The van Hoof et al. (2014) table is now available as a drop-in (`par%gaunt_vh14`; `data/gauntff_vh14.dat`, an $81 \times 146$ grid in $\log_{10}\gamma^2 = \log_{10}(Z^2\,\mathrm{Ry}/kT)$ and $\log_{10} u = \log_{10}(h\nu/kT)$, selected at both call sites — the cooling average and the continuum $\gamma_{\rm ff}$). The two compilations agree: switching moves $T_e$ by a median $5\times10^{-5}$, at most 0.5%, consistent with the sub-percent Hummer accuracy van Hoof et al. report. Hummer remains the default so the recorded gates reproduce.

The nebular continuum tables `data/gamma{HI,HeI,HeII}.dat` (MOCASSIN inheritance; 21 temperatures $\times$ paired-edge frequency points) were *numerically verified* (2026-07-12) against a hydrogenic Kramers Milne sum: they contain free-bound *plus* free-free combined coefficients (at $\nu = 0.0069$ Ryd, $10^4$ K the table value 9.47 decomposes into ff $\simeq 8.9$ + fb $\simeq 0.6$ in $10^{-40}$ units; near the Balmer edge the residual is $\sim$10%, the size of the neglected bound-free Gaunt factor). Consequently the port never adds free-free inside the table range. The Ercolano & Storey (2006) item is resolved as *provenance*: the MOCASSIN 2.02 tables ship in the ES06 compact paired-edge format, so no swap is needed — the verification above is the record. The tables leave three ranges out, and until 2026-09-26 the code returned zero emission there. (i) Each edge pair runs from just above one edge to just below the next, and the nodes, rounded to $10^{-4}$ Ryd, leave a gap of about $2\times10^{-4}$ Ryd around every edge (10 in the $H^+$ table, 7 in $He^+$, 1 in $He^{2+}$); five of the written-continuum frequencies fell in one. The edge is now placed at the gap's midpoint, within the $10^{-4}$ Ryd precision of the nodes, and each side extends its own pair. (ii) From the last node to the ground edge the last pair is extended ($He^+$: 1.7999–1.8080 Ryd). (iii) Below the first node. The $He^+$ table starts at 0.0911 Ryd (1.0 $\mu$m), so $He^+$ contributed neither free-free nor free-bound at longer wavelengths — the same omission as in MOCASSIN's `fb_ff`, whose arrays are zero there. The levels whose edges lie below that frequency ($n \geq 4$) are nearly hydrogenic, and a capture into them weighs $g_n/g_{\rm ion} = 4n^2/2$, the $2n^2$ of $H^+$, so the $He^+$ coefficient per ion is the $H^+$ one; the two tables agree at the join to 0.4% ($10^4$ K) and 2.2% (5000 K). Below the $H^+$ (0.0069 Ryd) and $He^{2+}$ (0.0278 Ryd) tables the coefficient is free-free plus the captures into every level with an edge below $h\nu$, summed in the Kramers approximation, $\sum_{n\geq n_1} \gamma_{{\rm fb},n} = [e^{\chi_{n_1}/kT} - 1]\,\gamma_{\rm ff}/g_{\rm ff}$, which lands −0.3% ($H^+$) and +3.6% ($He^{2+}$) off the first table node at $10^4$ K where free-free alone would be −6.6% and −30%. Two-photon continua: the Nussbaumer & Schmutz (1984) spectral shape for H I and He II, fed at $\alpha_B f_{2s}$ with $f_{2s}$ from a zero-density $nl$-resolved cascade in the case of the recombination lines (Section 4.20), and the collisional $2s \to 2p$ suppression for H I; the Drake, Victor & Dalgarno (1969) shape (`data/HeI2phot.dat`) for He I, fed at the $2^1$S decay rate of the He I cascade exits.

## 3.12 H I and He II recombination lines (Storey & Hummer 1995)

`make_sh95_lines.py` parses the SH95 case-B data (`e1bx.d`) with the reference reader indexing $k = (2n_{\rm top} - n_{ll} - n_l + 1)(n_l - n_{ll})/2 + n_{\rm top} - n_u + 1$ and exports H$\alpha$, H$\beta$, H$\gamma$, H$\delta$, P$\alpha$, P$\beta$, Br$\gamma$ on the full $(T, n_e)$ grid ($11 \times 13$) to `data/atomic/sh95_hi_caseB.txt`; Fortran interpolates bilinearly in $(\log T, \log n_e)$. Checks: $4\pi j_{\mathrm{H}\beta}/(n_e n_p) = 1.235\times10^{-25}$ at $(10^4\,\mathrm{K}, 10^2\,\mathrm{cm}^{-3})$ and H$\alpha$/H$\beta$ = 2.864 (canonical values).

He II ($Z = 2$) comes from the *original* CDS file `e2b.d`, whose format differs from the repackaged `e1bx.d` (six-token block headers; index $k = (n_{\rm cut} - n_u)(n_{\rm cut} + n_u - 1)/2 + n_l$ with $n_{\rm cut} = 25$, from the SH95 `intrat.f` reader): six lines (1640, 4687, 3204, 10126, 5413, 4543 Å) on a $12 \times 13$ grid in `data/atomic/sh95_heii_caseB.txt`, weighted in the line table by $n_e n_{\rm HeIII}$. Checks: $4\pi j_{4687}/(n_e n_{\rm HeIII}) = 1.495 \times 10^{-24}$ at $(10^4\,{\rm K}, 10^4\,{\rm cm^{-3}}) = \alpha^{\rm eff}_{4687}(3.57 \times 10^{-13})\,h\nu$, and $1640/4687 = 6.56$ there.

He I lines use the same interpolator (a third table in `sh95_mod`) but a different source: the Porter et al. (2012, 2013) case-B emissivities, parsed by `make_hei_lines.py` from Cloudy c23.01's `he1_case_b.dat` (copied into `data/atomic`) onto a $21 \times 14$ $(T, n_e)$ grid. Unlike the H I and He II tables, these are case-B collisional-radiative TOTALS, not pure recombination cascades: the tabulated $4\pi j/(n_e n_{\rm He^+})$ is density-dependent (the 10833 Å coefficient rises 6.6x from $n_e = 10$ to $10^4$ cm$^{-3}$ at $10^4$ K) because it already includes collisional excitation out of the $2^3$S metastable. Eight H II-region diagnostics are exported (3890, 4027, 4473, 5877, 6680, 7067, 7283, 10833 Å, vacuum), weighted in the line table by $n_e n_{\rm HeII}$ ($He^+$ recombining). Emergent ratios are physical: $5877/4473 = 2.88$, $6680/4473 = 0.82$, $7067/4473 = 0.49$ at $\sim$7 kK.

The H I and He II tables follow `par%lyman_lines`, the fate of the Lyman-line photons, not the continuum treatment: case-A tables (`sh95_{hi,heii}_caseA.txt`, parsed from the original CDS files `e1a.d`/`e2a.d` onto the $10 \times 9/12 \times 9$ grids case A tabulates) are loaded when the Lyman lines escape (`'escaping'`); case B (H$\alpha$/H$\beta = 2.94$) is the default. The He I Porter set is case B only. An optional collisional H I Balmer component (`par%h_coll_effects`, off by default) solves a 25-level H I atom (from CHIANTI `h_1` through the $n$-level fit and solve) with collisional excitation from the ground level, groups the $l$-resolved emission into H$\alpha$/H$\beta$/H$\gamma$, weights it by $n_{\rm HI}$, and writes it as separate `hc` rows — a first-cut estimate (optically-thin cascade) for the front and hot-cell bias, kept apart from the recombination lines.

## 3.13 He I $2^3$S metastable population and its coupling

The Porter case-B 10833 Å table above is a collisional-radiative *total* emissivity, correct in the dense, optically-thin, recombination-dominated H II regime it was built for. It does *not* give the $2^3$S population $n_3$, its line-of-sight column, or the 10833 Å *absorption* opacity — the observables of exoplanet transit, chromospheric, and PDR 10833 work, where the line is seen against a background. `hei_metastable_mod` computes $n_3$; the output diagnostic `par%hei_metastable` writes it, and it does not add or modify any emission row — in particular *nothing is summed onto* the Porter 10833 line (which already carries the metastable-collisional term). Since 2026-09-25 the same population is also *coupled* (`par%hei_metastable_coupling`, default on) into the ionization balance, the thermal balance and the He I diffuse channel, below.

**The $2^3$S balance.** Because $n_3/n_{\rm He} \sim 10^{-7}$ the balance is linear in $n_3$ — a closed-form quotient in each leaf, no iteration (`hei_triplet_population`, the one $n_3$ every user reads):

$$n_3 = \frac{n_e\,[\alpha_3\, n_{\rm HeII} + (q_{13} + q_{\rm feed})\, n_{\rm HeI}]}{D}, \qquad D = A_{31} + \Phi_3 + n_e\,(q_{31g} + q_{31a} + q_{31b} + q_{31s} + C_3) + n_{\rm HI}\,Q_{31}. \tag{21}$$

Sources: recombination into the triplets, $\alpha_3 n_e n_{\rm HeII}$; electron impact $1^1{\rm S} \to 2^3{\rm S}$ ($q_{13}$) and $1^1{\rm S} \to n^3L$ for every other triplet term $n \le 5$ ($q_{\rm feed}$: each such atom decays within the triplets to $2^3$S). Sinks: the radiative decay $A_{31} = 1.272 \times 10^{-4}$ s$^{-1}$ (19.82 eV; the CHIANTI `he_1.wgfa` $1.73 \times 10^{-4}$ is $\sim$36% high and is *not* used), photoionization $\Phi_3$, electron impact to

the ground ($q_{31g}$), to $2^1$S ($q_{31a}$), $2^1$P ($q_{31b}$) and to $n^1L$, $n \geq 3$ ($q_{31s}$), electron-impact ionization ($C_3$), and Penning plus associative ionization by atomic hydrogen ($n_{\rm HI}Q_{31}$). The excitations $2^3{\rm S} \rightarrow n^3L$ ($2^3$P and above, rate $q_{33}$) return to $2^3$S by radiative cascade (10830 Å for $2^3$P): they change no population and are an energy loss only. Two limits anchor the gates: the low-density radiative limit $n_3/n_{\rm He^+} = \alpha_3 n_e/A_{31}$ and the collisional plateau $\alpha_3/(q_{31g}+q_{31a}+q_{31b}+q_{31s}+C_3)$, crossing at $n_{\rm crit} = A_{31}/(q_{31g}+q_{31a}+q_{31b}+q_{31s}+C_3) = 3502$ cm$^{-3}$ at $10^4$ K (4131 at 8000 K, 2921 at $1.5\times10^4$ K).

*Recombination.* $\alpha_3(T) = f_{3S}(T)\,\alpha_B({\rm He\,II}, T)$, the triplet share of the case-B cascade (Eq. 49, `hei_cascade_mod`) times the case-B coefficient the ionization balance itself uses (`recomb_mod`). Every triplet recombination ends in $2^3$S and ground captures feed no triplet, so it is $\alpha_B$ whatever `recombination_continuum` is. Until 2026-09-25 the module carried its own fit, $2.10\times10^{-13}T_4^{-0.778}$ (Oklopcic & Hirata 2018, OH18): a second normalization of the same recombinations supplied beside the one the balance fixes — the defect of lesson (4). It sits 1.5% below $f_{3S}\alpha_B$ at $10^4$ K, 0.6% below at 8000 K and 8% above at 3000 K.

*Electron impact.* All rates are $q = \beta\,\Upsilon(T)\,e^{-\Delta E/kT}/(g_l\sqrt{T})$ with the exact Maxwellian constant $\beta = h^2/[(2\pi m_e)^{3/2}k^{1/2}] = 8.629132\times10^{-6}$ (Bray et al.'s eq. 1 is the same expression), $g_l = 1$ or 3, and the effective collision strengths $\Upsilon$ of Bray, Burgess, Fursa & Tully (2000, A&AS 146, 481), Table 2 (from $1^1$S, the triplet columns) and Table 3 (from $2^3$S), all terms $n \leq 5$: 41 transitions, each a row of `data/atomic/hei_metastable.txt` tagged with its role (`Q13`, `FEED`, `Q31A`, `Q31B`, `SINGLET`, `TRIPLET`). $\Delta E$ are the $g$-weighted observed term energies of CHIANTI `he_1.elvlc`. Bray et al. tabulate $\Upsilon$ at $\log T = 3.75$–$5.75$ and stop there because of the pseudo-resonances of their close-coupling cross sections near threshold; they also state that for a neutral target $\Omega \propto E_j^{1/2}$ near threshold, so $\Upsilon$ must fall as $T \rightarrow 0$ and holding the $10^{3.75}$ K value would contradict their own threshold law. Below the table the file therefore carries, at $\log T = 3.00$, 3.25, 3.50, the temperature dependence of the CHIANTI v11.0.2 `he_1` strength of the same term (its $J$ levels summed, Burgess–Tully descaled) anchored to the Bray value at $10^{3.75}$ K; the CHIANTI file states its strengths (Sawey & Berrington 1993, Bray et al. 2000) are used over $3.0 < \log T < 5.75$. The one term CHIANTI lacks from $2^3$S ($4^3$S) holds its $10^{3.75}$ K value. The Fortran interpolates $\log\Upsilon$ linearly in $\log T$ and holds the end values outside $[10^3, 10^{5.75}]$ K; the Boltzmann factor is always at the actual $T$ (below $10^3$ K every excitation here is suppressed by $e^{-\Delta E/kT} < e^{-9.2}$). $q_{31g}$ is the detailed-balance inverse of the `Q13` row, $q_{31g} = q_{13}(g_1/g_3)e^{E_{13}/kT}$, evaluated with the Boltzmann factor removed. At $10^4$ K:

| rate [cm$^3$ s$^{-1}$] | MoCHII | EXHALE (cross-check) |
|---|---|---|
| $q_{13}$ | $5.720\times10^{-19}$ | $5.720\times10^{-19}$ |
| $q_{\rm feed}$ | $6.055\times10^{-20}$ | $6.055\times10^{-20}$ |
| $q_{31g}$ | $1.858\times10^{-9}$ | $1.858\times10^{-9}$ |
| $q_{31a}$ | $2.804\times10^{-8}$ | $2.804\times10^{-8}$ |
| $q_{31b}$ | $5.438\times10^{-9}$ | $5.438\times10^{-9}$ |
| $q_{31s}$ ($n^1L$, $n \geq 3$) | $6.539\times10^{-10}$ | $6.539\times10^{-10}$ |
| $q_{33}$ ($n^3L$) | $2.022\times10^{-7}$ | — |
| $C_3$ | $3.311\times10^{-10}$ | — |
| $Q_{31}$ | $1.272\times10^{-9}$ | — |

EXHALE evaluates the same Bray tables independently (its own transcription, identical in every column used here); its tabulated quartic fits for the feed and the $n \geq 3$ singlets sit 0.9% and 1.1% off the sums above, which its own reference

sums reproduce. The feed is 10.6% of $q_{13}$ at $10^4$ K and 33% at $2\times10^4$ K; $q_{31s}$ is 2.0% of $q_{31a}+q_{31b}$ at $10^4$ K and 7.9% at $2\times10^4$ K. $C_3$, Black (1981, MNRAS 197, 553) eq. (12), $8.38\times10^{-10}\,T^{1/2}e^{-55338/T}$ from the Taylor, Kingston & Bell (1979) cross sections over $10^3$–$10^5$ K, and $Q_{31}$ below are evaluated at $T$ clamped to $[10^2, 3\times10^4]$ K.
*Penning and associative ionization*, $\mathrm{He}(2^3\mathrm{S})+\mathrm{H}^0$, from García Muñoz (2025, A&A 698, A199, Fig. 5), the Maxwellian average of the Movre & Meyer (1997) total ionization cross sections,

$$Q_{31} = 10^{-9}\exp\left(c/T + d_1\ln T + d_2(\ln T)^2 + d_3(\ln T)^3\right)\ \mathrm{cm^3\,s^{-1}}, \tag{22}$$

$c=-86.4804$, $d_1=-0.286766$, $d_2=0.0868445$, $d_3=-0.00573001$, which reproduces the paper's Table A.5 (1.02, 1.32, 1.35, $1.27\times10^{-9}$ at 500, 2000, 5000, $10^4$ K). It replaced (2026-09-24) the Taylor et al. (2025, ApJ 989, 68) two-branch power law, which jumps by 1.57× at 4000 K. The same paper partitions the total 0.9 : 0.1 between Penning ($\mathrm{He}+\mathrm{H}^+ + e$) and associative ($\mathrm{HeH}^+ + e$) ionization, "an average". MoCHII carries no $\mathrm{HeH}^+$, so both branches are counted as $\mathrm{H}^0 \to \mathrm{H}^+$ (the molecular ion carries the hydrogen's charge) and both deposit $E_3 - \mathrm{IP(H)} = 6.22$ eV; for the associative 10% that omits the $\mathrm{HeH}^+$ binding.

**The $2^3$S photoionization rate.** $\Phi_3 = \int 4\pi J_\nu\,\sigma_3(\nu)\,d\nu/(h\nu)$ over the transported band, accumulated during transport as a path estimator, $\sum \ell_{\mathrm{eff}}\,w\,L_{\mathrm{pkt}}\,\sigma_3(E)/(h\nu)$ over the packet segments crossing the leaf, at each packet's own energy (`hei_metastable_path_score`, called from `ion_score_mod` whenever `hei_metastable` or `hei_metastable_coupling` is on), and normalized by the leaf volume exactly as the $\Gamma_i$ are. Its heating companion $\Gamma_3 = \sum \ell_{\mathrm{eff}}\,w\,L_{\mathrm{pkt}}\,\sigma_3(E)\,(1-\mathrm{IP}_3/E)$, $\mathrm{IP}_3 = \mathrm{IP(He)} - E_3 = 4.768$ eV, is accumulated beside it. *Band limitation:* the threshold is 4.78 eV (the edge of the $\sigma_3$ fit; $\mathrm{IP}_3 = 4.768$ eV), but without `add_fuv` the transported band starts at 13.6 eV, so [4.78, 13.6) eV is missing from both; $\sigma_3$ falls from $4.7\times10^{-18}$ cm$^2$ at the edge to $1.2\times10^{-18}$ at 13.6 eV, $5.9\times10^{-19}$ at 20 eV and $2.2\times10^{-19}$ at the Cooper minimum near 34.7 eV, then rises to $2.6\times10^{-18}$ at the bump near 45.6 eV, so the ionizing band alone gives $\Phi_3 > 0$ wherever a field exists, and `add_fuv` with `efuv_min`= 4.78 adds the soft-UV part. That part dominates, so an input that leaves `add_fuv` and `efuv_min` out gets both whenever this population is used, launched as its own stratum (Section 4.2). Measured on the `tests/hei_meta` spheres (40 kK, coupling and thermal balance on, two seeds each; the FUV extension against none): the $n_e$-weighted $\Phi_3$ rises 13.0-fold at $n_{\mathrm{H}} = 100$ and 13.4-fold at $10^4$ cm$^{-3}$. At $n_{\mathrm{H}} = 100$ the $2^3$S level still empties by $A_{31}$ ($\Phi_3 = 1.8\times10^{-7}$ against $A_{31} = 1.27\times10^{-4}$ s$^{-1}$), so the population moves by −0.05% (central column −0.8%, at the seed scatter) and $T_e$ by −2 K. At $10^4$ cm$^{-3}$ photoionization is a main exit: $N(2^3\mathrm{S})$ falls 13.9%, the central column 27.8%, and the photoionization heating raises $\langle T_e\rangle_{n_e}$ by 2.8% (525 K), with $x_{\mathrm{HI}}$ −0.46% and $x_{\mathrm{HeI}}$ −0.74% (seed scatter $\le 3\times10^{-4}$ on every one of these). The runs cost 22% and 46% more wall time. The $2^3$S opacity is not carried in the transport, so the photons $\Phi_3$ absorbs are not removed from the field. Below 13.6 eV $\mathrm{H}^0$ does not compete, so the comparison that bounds this is the $2^3$S column itself: through the densest standing case (`examples/hei_metastable/hei_high_density.in`, largest column $N(2^3\mathrm{S}) = 2.87\times10^{15}$ cm$^{-2}$) it is $\tau_3 = 0.013$ at 4.8 eV, where $\sigma_3$ is largest. $\sigma_3(E)$ is two VFKY96 wings joined by a log-linear bridge, a ∼4% fit to Norcross (1971) + TOPbase.

**Coupling into the ionization balance.** Ionization of $2^3$S by electrons and photons returns the atom to $\mathrm{He}^+$: it undoes the recombination that fed it. `solve_ion_cell` adds $(n_e C_3 + \Phi_3)\,n_3$ to the $\mathrm{He}^0 \to \mathrm{He}^+$ rate. Because $n_3$ is linear in $n_{\mathrm{HeI}}$

and $n_{\mathrm{HeII}}$ this is solved *jointly* in the closed form rather than lagged: with $u = (n_e C_3 + \Phi_3)/D$, the share of $2^3$S atoms that end ionized,

$$\frac{n_{\mathrm{HeII}}}{n_{\mathrm{HeI}}} = \frac{\Gamma_{\mathrm{HeI}} + C_{\mathrm{HeI}} n_e + u\,(q_{13} + q_{\mathrm{feed}})\, n_e\ [+\mathrm{CX}]}{(\alpha_{\mathrm{He}} - u\,\alpha_3)\, n_e\ [+\mathrm{CX}]}, \tag{23}$$

i.e. a share $u$ of the triplet recombinations is undone and a share $u$ of the collisional triplet excitations becomes a two-step ionization; $\alpha_{\mathrm{He}} - u\alpha_3 > 0$ always. $D$ is taken at the iterate's $n_e$ and $n_{\mathrm{HI}}$. Penning ionization adds $n_3 Q_{31}$ to the $\mathrm{H}^0$ ionization rate; there $n_3$ uses the previous iterate's helium fractions, the lag the He–H charge-exchange terms already carry, and the loop's exit test then also requires $x_{\mathrm{HI}}$ and $x_{\mathrm{HeII}}$ to have stopped moving. The gate solves partially neutral cells ($x_{\mathrm{HII}} = 0.006$–$0.87$, with and without He–H charge exchange, Penning up to 0.7% of $\Gamma_{\mathrm{H}}$) and finds both balances closed, *with* the $2^3$S terms at the solved state, to $\leq 1.1\times10^{-10}$ inside one call. `hydrogen_balance_residual` carries the Penning term too.

**Coupling into the thermal balance: the energy ledger.** Energies above $1^1$S, $E_3 = E(2^3\mathrm{S}) = 19.8196$ eV. Per event, the gas

| process | rate [cm$^{-3}$ s$^{-1}$] | gas energy |
|---|---|---|
| excitation $1^1\mathrm{S} \to 2^3\mathrm{S}, n^3L$ | $n_e n_{\mathrm{HeI}} q$ | $-E(n^3L)$ $\quad[E(n^3L) - E_3$ leaves as triplet lines] |
| super-elastic $2^3\mathrm{S} \to 1^1\mathrm{S}$ | $n_e n_3 q_{31g}$ | $+E_3$ |
| $2^3\mathrm{S} \to 2^1\mathrm{S}, 2^1\mathrm{P}, n^1L, n^3L$ | $n_e n_3 q$ | $-\Delta E$ $\quad$[the atom then radiates] |
| electron-impact ionization | $n_e n_3 C_3$ | $-\mathrm{IP}_3$ |
| photoionization | $n_3 \Phi_3$ | $+(h\nu - \mathrm{IP}_3)$, i.e. $n_3\Gamma_3$ |
| Penning / associative | $n_{\mathrm{HI}} n_3 Q_{31}$ | $+(E_3 - \mathrm{IP(H)})$ |
| radiative decay | $n_3 A_{31}$ | 0 |

so `hei_triplet_net_heating`, added to the net rate of `thermal_mod::net_rate`, is

$$H_3 = n_3 \left[ n_e \left( q_{31g} E_3 - \sum_{31a,31b,31s,33} q\,\Delta E - C_3\,\mathrm{IP}_3 \right) + \Gamma_3 + n_{\mathrm{HI}} Q_{31}\,(E_3 - \mathrm{IP(H)}) \right] - n_e n_{\mathrm{HeI}} \sum_{13,\,\mathrm{feed}} q\,E(n^3L). \tag{24}$$

It closes: with $n_3$ steady, recombination $E_3$ plus the photon energy $\Phi_3$ absorbs equals $H_3$ plus the photon energy the levels emit plus the ionization energy stored (IP(He) for $C_3$ and $\Phi_3$, IP(H) for $Q_{31}$); the gate assembles the photon side from the rows' own energies and finds closure to $4\times10^{-16}$. Not in it: excitation of the singlets from $1^1$S (the open He I collisional-excitation item) and the recombination kinetic energy, already in `cooling_mod`'s $\beta$ on the balance's own $\alpha$. The $2^1$S/$2^1$P populations are not solved: a transfer into them is taken to decay radiatively ($A_{2q} = 51$ s$^{-1}$ against $n_e q(2^1\mathrm{S} \to 2^3\mathrm{S}) \sim 2\times10^{-7} n_e$ s$^{-1}$).

**Coupling into the He I diffuse channel.** See Section 4.10: the 19.82 eV photons are $n_3 A_{31}$, and the collisional transfers feed the two-photon and 584 Å branches.

**Measured sizes (2026-09-25).** On the converged Lexington states (`results/continuous_energy/m5_converged/`), restarted for one iteration with the coupling on and off at 36 ranks, volume-integrated and relative to the H+He photoheating:

| | HII40 | HII20 |
|---|---|---|
| excitation from $1^1$S (direct + feed) | $-4.7\times10^{-8}$ | $-6.6\times10^{-10}$ |
| super-elastic $q_{31g}$ | $+4.4\times10^{-4}$ | $+5.4\times10^{-5}$ |
| transfers to the singlets | $-2.7\times10^{-4}$ | $-2.7\times10^{-5}$ |
| triplet excitations ($q_{33}$, 10830 etc.) | $-1.58\times10^{-3}$ | $-1.36\times10^{-4}$ |
| $C_3$ | $-3.6\times10^{-6}$ | $-1.3\times10^{-7}$ |
| $\Phi_3$ heating | $+4.0\times10^{-5}$ | $+8.5\times10^{-6}$ |
| Penning | $+6.4\times10^{-8}$ | $+1.7\times10^{-9}$ |
| net $H_3$ | $-1.37\times10^{-3}$ | $-1.01\times10^{-4}$ |
| Penning / photoionization of H | $3.2\times10^{-8}$ | $4.2\times10^{-10}$ |
| undone share of triplet recombinations | $3.1\times10^{-4}$ | $5.6\times10^{-4}$ |
| of which $\Phi_3$ / $C_3$ | 0.81 / 0.19 | 0.97 / 0.03 |
| $2^3$S decaying by $A_{31}$ | 0.974 | 0.977 |
| channel 4: 19.82 eV / 584 Å / $2^1$S energy, coupled/uncoupled | 0.974 / 1.014 / 1.080 | 0.977 / 1.011 / 1.064 |
| channel 4 total energy | −1.27% | −1.20% |
| $\langle T[\mathrm{N_pN_e}]\rangle$ after the one update, on − off | –9.7 K | –0.7 K |

The net term is a cooling dominated by the collisional excitation of $2^3$S within the triplets, which no other term of the thermal balance carries; the population itself moves by a few percent because 2.0–2.6% of the $2^3$S atoms leave by $q_{31a}$ and $q_{31b}$ instead of decaying. The leaf temperatures of the two runs also differ by $\pm80$ K (1st/99th percentile): the coupled diffuse channel changes the diffuse packets, so the two transport realizations differ; the –9.7 K is the ionized-gas mean ($x_{\mathrm{HI}} < 0.01$ leaves: median −10.0 K) and the full effect needs a converged run. The cell maximum of $|H_3|$ against photoheating (11 at HII40) is set by leaves whose tallied photoheating is Monte Carlo zero.

On the three `tests/hei_meta` inputs (fixed $T_e = 10^4$ K, case B, no diffuse field, so only the ionization coupling acts; 24 ranks, same seed), coupling on against off moves the total $2^3$S column by +0.018% (`hei_lowd`), +1.09% (`hei_highd`) and +0.091% (`hei_fuv`), the mass-weighted $x_{\mathrm{HeII}}$ by +0.019%, +1.12% and +0.095%, and in the leaves where helium is mostly ionized the median $x_{\mathrm{HeI}}$ by –0.061%, –5.4% and –0.33% (median $u$ there $3.2\times10^{-4}$, $1.9\times10^{-2}$, $1.3\times10^{-3}$, with $\Phi_3$ supplying 15%, 64% and 78% of it). The coefficient change alone (Bray feed and $n \geq 3$ singlets, $\alpha_3 = f_{3S}\alpha_B$), evaluated on the same coupling-off state, moves the column by +1.42%, +0.13% and +1.42%.

**10833 Å absorption (Stage 2).** From $n_3$ the line-center opacity of each fine-structure component is

$$\kappa_{0,i} = \frac{\sqrt{\pi}\,e^2}{m_e c} f_i \frac{\lambda_i}{b} n_3, \qquad b = \sqrt{2kT_e/m_{\mathrm{He}} + v_{\mathrm{turb}}^2}, \tag{25}$$

the classical integrated cross section $\pi e^2/(m_e c) = 0.02654$ cm$^2$ Hz divided by $\sqrt{\pi}\,b$ for the Gaussian peak. The three $2^3\mathrm{S} \to 2^3\mathrm{P}_J$ components (vacuum 10832.06 / 10833.22 / 10833.31 Å; $f = 0.0599/0.1797/0.2996$, ratio 1:3:5; $A = 1.0216\times10^7$ s$^{-1}$ each) take the NIST ASD values (Drake 2007), cross-checked against p-winds; the CHIANTI `he_1.wgfa` $A \simeq 1.15\times10^7$ (∼13% high) is *not* used. The $J' = 1,2$ pair sits 0.089 Å apart, far inside the ∼0.4 Å thermal Doppler width at $10^4$ K, so they blend into the observed "doublet" (their line-center opacities are summed); $J' = 0$ at 10832.06 Å

is resolved. The Python `HeiMetaData` reader deposits $n_3$ and $\kappa_0$ along the line of sight (the flux-conserving deposit of the emissivity maps) into column [cm$^{-2}$] and $\tau_0 = \int \kappa_0 \, \mathrm{d}l$ maps, and warns when line-center $\tau_0 > 1$.
Caveats recorded at the code site: (i) *local equilibrium* — the advection term $v \, \mathrm{d}f_3/\mathrm{d}r$ is absent, valid where the $2^3$S destruction time is short against any flow time (dense H II), not a faithful outflow benchmark; (ii) 10833 can be optically thick in dense metastable columns, where the optically-thin $\tau_0$ diagnostic breaks down (line transfer is out of scope).

# 4. Algorithms

## 4.1 The iteration loop

read grid → persist gas leaf state → fill `kap_ion`
**do** iter = 1, `gas_niter`
  zero the band tally $J_b$
  sample and transport stellar packets (analytic edge walk, Eq. 4)
  [diffuse] rebuild the diffuse emission table from the current state;
    sample and transport $L_{\mathrm{dif}}/L_{\mathrm{pkt}}$ diffuse LyC packets
  ALLREDUCE $J_b$; rate integrals (Eq. 5); metal Γ's
  solve each leaf: ionization only or coupled $(x, T_e)$
  refill `kap_ion` from the new state (the stellar tally is
    recomputed from zero each iteration — no reuse)
  converged by `par%conv_crit`: cell-max or volume-integrated tests (below)
**end do**
outputs: rates + state maps; line luminosities; nebular continuum

Every rank runs the identical cell solves (inputs are the reduced tallies), so all ranks agree on convergence; only the node master writes the shared state arrays. Under-relaxation on $x$ (`par%ion_relax`) is available for sharp I-fronts. Practical lessons recorded: a fully neutral start advances the front only ∼one cell per iteration (the photons die in the first neutral cell), while an ionized start converges globally; and $\max|\Delta x_{\mathrm{HII}}|$ stalls at the front-cell Monte-Carlo noise floor ($\sim$2–4$\times10^{-2}$ at $4\times10^7$ packets), so tolerances below that are unreachable by that criterion — the diffuse field, interestingly, smooths the front and converges to $3\times10^{-3}$.

## 4.2 Stellar packet launch

For each direct packet, MoCHII first samples its energy with Eq. (1); it then samples an isotropic point-source direction with $\mu = 2u_\mu - 1$ and $\phi = 2\pi u_\phi$. The source position, the exact energy, the direction, and the common packet luminosity are placed in the packet before the octree edge walk begins. Thus energy sampling is not conditional on the diagnostic binning. At every traversed cell the path estimator uses the packet's exact-energy opacity and cross sections; the energy-bin index is accumulated only alongside this estimator to form the reported spectrum. Rebuilding the direct tally from zero on every nonlinear iteration means that the changing opacity is always applied to a new sample of the source spectrum rather than to a reweighted old tally.

For ordinary pseudo-random launching the inverse-CDF variate and the two direction variates are independent Mersenne-Twister draws. The same probability law is used by the Sobol option described next. The packet weight is not divided by the probability density: equal packet luminosity combined with inverse-CDF sampling is the importance sampling rule that recovers the source luminosity spectrum in expectation.

**Band strata with the FUV extension.** With `par%add_fuv` the band $[E_{\rm FUV}, E_{\rm max}]$ has two parts that do different physics: above $E_{\rm min}$ the photons ionize H and He, below it they heat grains, ionize the low-threshold metals and the He I $2^3$S level, and nothing else. Drawn from one CDF, a packet lands in the ionizing part with probability equal to that part's luminosity share, $f_{\rm ion} = 1 - F(E_{\rm min})$: 0.448 for a 40 kK blackbody from 4.78 eV, 0.064 for 20 kK. Every hydrogen and helium rate then rests on $f_{\rm ion}N$ packets, and its Monte Carlo noise grows by $f_{\rm ion}^{-1/2}$ — 1.5 and 3.9 for those two stars — over a run without the extension. `par%fuv_sampling = 'separate'` (the default) launches the two parts as strata. For source component $c$ with band luminosity $L_c$ and CDF $F_c$, the parts carry

$$L_{{\rm ion},c} = L_c\,[1 - F_c(E_{\rm min})], \qquad L_{{\rm FUV},c} = L_c\,F_c(E_{\rm min}), \tag{26}$$

and the ionizing stratum launches $N$ = `no_photons` packets, the FUV stratum $N_{\rm FUV}$ = `no_photons_fuv` (default $N/4$). Within a stratum the component is chosen with probability $L_{s,c}/\sum_{c'} L_{s,c'}$, so a component with no luminosity there launches nothing in it, and the energy comes from its conditional CDF,

$$E = F_c^{-1}\big(F_c(E_{\rm min}) + u\,[1 - F_c(E_{\rm min})]\big) \quad \text{(ionizing)}, \qquad E = F_c^{-1}\big(u\,F_c(E_{\rm min})\big) \quad \text{(FUV)}, \tag{27}$$

the same piecewise-linear CDF the whole-band draw inverts ($E_{\rm min}$ is a knot of every Planck sampler, and the tabulated CDF is exact at any energy). A packet of stratum $s$ carries $L_s/N_s$ with $L_s = \sum_c L_{s,c}$; its weight starts at 1 as before, so the Russian roulette of the dust scattering acts relative to the launch weight. Each stratum then reproduces its own luminosity exactly, and the two sum to the band luminosity. Every estimator downstream — the rate and heating path estimators, $\Phi_3$ and its heating, the dust absorption, the $J_\nu$ tally and hence $G_0$, the peel-off images, the slab boundary tally, the energy ledger — sums $L_{\rm pkt}\,w\,\ell$ over segments, so none of them needs to know which stratum a packet came from. The diffuse recombination packets carry the luminosity of one *ionizing* source packet, $L_{\rm ion}/N$ ($L/N$ without the strata), so the field that drives the ionization and the field it produces are sampled equally finely whatever the FUV share. With `launch_sequence = 'sobol'` the ionizing stratum uses the stellar stream at indices $0 \ldots N-1$ — the points a run without the extension uses — and the FUV stratum a third scramble stream of its own, indexed from 0.

Measured on one fixed state (`tests/fuv_stratified/`, a dusty 4 pc sphere, $N = 10^6$, four seeds per mode): the volume integrals of $n_{\rm HI}\Gamma_{\rm HI}$, $\Phi_3$, $G_0$ and the grain heating agree between `'separate'` and `'proportional'` within $1.6\sigma$ ($\leq 1.1 \times 10^{-3}$); the leaf noise of $\Gamma_{\rm HI}$ in the ionized gas is that of a run without the extension, and `'proportional'`'s is 1.51 and 4.07 times larger for the 40 and 20 kK stars (predicted 1.49, 3.95). The FUV side pays for it: with $N/4$ FUV packets the leaf noise of $G_0$ is 1.5 (40 kK) and 1.9 (20 kK) times `'proportional'`'s. An FUV packet is absorbed by nothing in ionized gas without dust or metals and crosses more of the grid than an ionizing one — 1.6 times the path segments, 2.0 with dust scattering, in that geometry — so the default $N/4$ adds 40–50% to a pass, about what `'proportional'` costs at the same $N$ for a 40 kK star. `'proportional'` is kept so runs made with it reproduce bit for bit (verified against the previous binary, all `rates.h5` datasets and the line files, with and without the extension, pseudo-random and Sobol, single, multiple and preset sources).

**Convergence measures.** Two pairs are computed every iteration and `par%conv_crit` selects which gates the loop: the cell-max tests ($\max|\Delta x_{\rm HII}| <$ `gas_tol`, $\max|\Delta T_e|/T_e <$ `gas_tol_te`) and the volume-integrated tests

$$\frac{\sum_{\rm leaf} V\,|\Delta x_{\rm HII}|}{\sum_{\rm leaf} V\,x_{\rm HII}} < \texttt{gas_tol}, \qquad \frac{\sum_{\rm leaf} V\,n_e\,|\Delta T_e|}{\sum_{\rm leaf} V\,n_e\,T_e} < \texttt{gas_tol_te}. \tag{28}$$

The volume measures fix both stall modes of the cell-max tests: front jitter occupies little volume (the front shell is a few thousand cells against $\sim 10^5$ ionized ones, so its noise enters the $L_1$ norm suppressed by that ratio), and the $n_e$ weight removes the vacuum and no-heating cells whose `te_min`-pinned oscillation dominates $\max|\Delta T_e|$ in FUV/PDR configurations. `'vol'` is the recommended production setting; `'cell'` (the historical behavior) remains the default so the recorded gates reproduce. *The drift test (2026-09-25).* Under `'vol'` the ionization measures are no longer tested on their last step: with $d_n$ the step and $\rho$ its decay ratio, a measure passes when the drift still to come, $d_n\rho/(1-\rho)$, is below `gas_tol`, or — once the steps no longer fall (the Monte Carlo floor) — when $d_n$ is. *Revised 2026-09-26:* $\rho$ is the least-squares slope of $\ln d$ over the last 8 iterations (it was the median of the last three ratios), and the floor is accepted only when the signed step

$$s_n = \frac{\sum_{\rm leaf} V\,(x_{{\rm HII},n} - x_{{\rm HII},n-1})}{\sum_{\rm leaf} V\,x_{{\rm HII},n}} \tag{29}$$

(and its helium counterparts) has a mean over those 8 iterations that does not stand out of its scatter, $|\bar{s}| < 2.365\,\sigma_s/\sqrt{8}$ (Student's $t$, two-sided 95%, 7 degrees of freedom). On the floor the leaves fluctuate about their fixed point and their changes cancel in $s$; a front still moving changes the ionized volume with one sign every iteration. The floor had been read off the step ratios alone (any of the last three $\geq 1$), which a slow decay under Monte Carlo noise also satisfies: a restart of HII40 onto the hydrogenic $\alpha_B$ (−1.1%) passed after 4 iterations with its H front still moving outward, $\rho \approx 0.97$. Helium is tested alongside hydrogen, through $\sum V|\Delta x_{\rm HeI}|/\sum V(1-x_{\rm HeI})$ and, where $He^{2+}$ is at least 1% of the ionized helium, the $He^{2+}$ counterpart (`fixed_point_distance_mod`, gate `tests/convergence/`). The step test stopped the HII40 benchmark at iteration 37: $x_{\rm HII}$ was decaying with $\rho \approx 0.86$, so six times the last step was still to come, and the helium measure was five times larger again. Run for a fixed 80 iterations the same input settles by iteration 60 — He I front 4.271 pc against 4.103 at the early stop, $\langle T[N_pN_e]\rangle$ 8210.5 against 8175.3 K — and the drift test stops at 58, at that answer. Measured on the FUV dusty Strömgren case at $8\times10^6$ packets (where the cell-max tests sat at 0.1 and 6.4 indefinitely): the volume measures decay smoothly to well-defined Monte Carlo floors of $2.7\times10^{-3}$ ($x$) and $6.6\times10^{-3}$ ($T_e$) by iteration $\sim$27, and running 30 further iterations leaves the solution unchanged ($V_{\rm ion}$ to 0.014%) — the floor IS convergence. Set the tolerances just above the floor for the packet count in use (here $5\times10^{-3}$ and $10^{-2}$ would stop the loop at iteration $\sim$23); the floors scale as $1/\sqrt{N_{\rm packets}}$. With `par%require_convergence` (default off) a non-converged gas iteration prints a warning and, when the switch is set, aborts before writing output; either way the rates-file header records the convergence status in the keywords `CONVERGD`, `NITERDON`, `FINALDX`, and `FINALDTE`.

## 4.3 Quasi-random photon launching (optional)

With `par%launch_sequence = 'sobol'` the packet launch draws its variables — source component where relevant, energy, direction, and any entry-surface coordinates — from an Owen-scrambled Sobol net (`src/qmc_mod.f90`) instead of the Mersenne Twister, while every post-launch transport event stays pseudo-random. The Sobol point is addressed by the *global* photon number, so the launch set is independent of the MPI task count, and `par%qmc_seed` is held fixed across

the nonlinear iterations (common random numbers). Because the band has no scattering, the direct field is an analytic function of the launch variables (Eq. 4), so a low-discrepancy launch set reduces its noise directly: on the fixed-state analytic-attenuation gate the cell-wise $\Gamma_{\rm HI}$ error scales as $\sim 1/N$ (Monte Carlo $1/\sqrt{N}$), with effective photon gains of $29\times$ at $N = 2^{17}$ and $222\times$ at $2^{20}$, at unchanged wall time and no significant bias. The method, the fixed dimension layouts, and all gates are documented in the companion memo `MoCHII_QMC`. In the validated single point-source continuous mode the three Sobol coordinates are $(u_E, u_\mu, u_\phi)$: $u_E$ is inverted through the source CDF and the remaining coordinates set the isotropic direction. QMC therefore reduces the discrepancy of the energy samples as well as of the angular samples; it does not replace continuous energy sampling by bin selection. Diffuse packets use a separate scrambled stream and the nine coordinates listed in §4.10.

## 4.4 Ionization equilibrium

Per leaf at temperature $T$ (`solve_ion_cell`): with $R_i = \Gamma_i + n_e C_i(T)$ the ionization rates and $\alpha_i$ the case-A/B recombination rates, $x_{\rm HI} = \alpha n_e/(R_{\rm HI} + \alpha n_e)$ and the He stage chain $n_{i+1}/n_i = R_i/(\alpha_{i+1} n_e)$, closed by $n_e = n_{\rm H}[x_{\rm HII} + y_{\rm He}(x_{\rm HeII} + 2x_{\rm HeIII})]$ and solved by a damped fixed point on $n_e$ (mixing $\frac{1}{2}$, converged at $10^{-10}$, $\leq 200$ iterations).

*Two-way charge exchange* (`par%charge_exchange = 'two_way'`, the default). Each reaction $X^{i+} + H^+ \leftrightarrow X^{(i+1)+} + H^0$ (§3.8) moves one unit of charge between a metal and hydrogen, so hydrogen's balance carries it too:

$$n_{\rm HI}\,(\Gamma_{\rm H} + n_e C_{\rm H} + R^{\rm out}_{\rm cx}) \;=\; (\alpha_{\rm H} n_e + R^{\rm in}_{\rm cx})\, n_{\rm HII}, \tag{30}$$

with the sums running over every registry element and transition,

$$R^{\rm in}_{\rm cx} = \sum_{\rm el} \sum_i k^{\rm CX}_{{\rm ion},i}\, n(X^{i+}), \qquad R^{\rm out}_{\rm cx} = \sum_{\rm el} \sum_i k^{\rm CX}_{{\rm rec},i}\, n(X^{(i+1)+}), \tag{31}$$

in $\mathrm{s}^{-1}$: $R^{\rm in}_{\rm cx}$ neutralizes an $H^+$, $R^{\rm out}_{\rm cx}$ ionizes an $H^0$. Eliminating $n_{\rm HII} = n_{\rm H}(1 - x_{\rm HI})$ keeps the closed form,

$$x_{\rm HI} = \frac{\alpha_{\rm H} n_e + R^{\rm in}_{\rm cx}}{\Gamma_{\rm H} + (C_{\rm H} + \alpha_{\rm H}) n_e + R^{\rm in}_{\rm cx} + R^{\rm out}_{\rm cx}}, \tag{32}$$

so no matrix and no Newton step are needed; *both* charge-exchange terms appear in the denominator, $R^{\rm in}_{\rm cx}$ for the same reason $\alpha_{\rm H} n_e$ does. The two sums come from the same walk of the stage chain that produces the metal electrons (`species_metal_sums`) and read the same cached coefficients the cascade reads, so the charge each reaction moves is conserved in the reacting pair by construction rather than to a tolerance; the gate `tests/charge_exchange/` forms all 29 active reactions from both sides and finds them equal to $6 \times 10^{-16}$ relative. The stage densities are taken from the previous iterate of the $n_e$ fixed point, since they depend on $n_{\rm HI}$ and $n_{\rm HII}$ themselves. The value `'metal_only'` keeps the charge exchange in the metal cascade alone — the earlier bookkeeping, which does not conserve charge — and is retained only to reproduce the recorded gates.

The correction is small on these benchmarks, and the reason is worth stating because it is not the usual abundance argument. Where the near-resonant $O^0 + H^+$ pair locks oxygen to hydrogen — that is, where $\Gamma_{\rm O}$ and $n_e \alpha_{\rm O}$ are negligible beside the charge-exchange terms — detailed balance of the pair gives $R_{\rm out}/R_{\rm in} = n_{\rm HII}/n_{\rm HI}$, and substituting that into Eq. (32) returns $x_{\rm HI} = \alpha_H n_e/[\Gamma_H + (C_H + \alpha_H) n_e]$ exactly: the two terms cancel identically. What survives is only the departure from that lock. Measured with one switch changed and the same Sobol seed, $\langle T[N_{\rm p} N_{\rm e}] \rangle$ moves by −1.41 K at HII40 and −0.13 K at HII20, $L(\mathrm{H}\beta)$ by +0.045% and +0.007%, and $r(\mathrm{He}^0 = 0.5)$ by −0.0002 and −0.0000 pc. All

of these are inside the tolerance the runs were converged to (`par%gas_tol_te` = $10^{-2}$, i.e. 82 K at HII40) by factors of 58 and 533, so the recorded benchmark tables are not re-baselined on them. Correcting the interchanged oxygen rows of §3.8 is a separate change and moves the temperatures the *other* way: measured from the same converged state, $\langle T[N_\mathrm{p}N_\mathrm{e}]\rangle$ rises by +2.2 K at HII40 and +2.1 K at HII20 (to 8170 and 6926 K, before the 2026-09-24 update that gives the current 8177 and 6936 K), $L(\mathrm{H}\beta)$ by +0.005%, and the ionization fronts by $\leq 4\times10^{-4}$ pc.

The same cancellation has a numerical consequence, and it dictates how the balance is solved. Substituting into Eq. (32) with the charge-exchange terms lagged gives a map whose slope at the fixed point is $R_\mathrm{in}/(R_\mathrm{in}+\alpha_H n_e)$, which tends to unity wherever $R_\mathrm{in} \gg \alpha_H n_e$ — that is, throughout partially neutral gas, where the iteration becomes marginally stable. At $T = 10^4$ K, $n_\mathrm{H} = 100$ and the HII20 abundances it then needs 124 passes at $x_\mathrm{HII} = 0.18$, 239 at 0.10 and 1285 at 0.021, against a 200-pass cap, and a capped cell leaves with $x_\mathrm{HII}$ high by about 8%.

MoCHII therefore does not substitute. `hydrogen_neutral_fraction` solves the balance at the current $n_e$ for its root, with the whole metal cascade re-evaluated at each trial $x_\mathrm{HI}$. The root is bracketed in $[0,1]$ *by construction* — $R(0) = -(\alpha_H n_e + R_\mathrm{in}) < 0$ and $R(1) = \Gamma_H + (C_H + \alpha_H)n_e + R_\mathrm{out} > 0$ — so Illinois iteration keeps the bracket at every step and the fraction cannot leave the physical range, which is the one property the plain substitution had and a bare secant would forfeit. One call now reaches the converged state to $3.6\times10^{-11}$ at every $x_\mathrm{HII}$ down to 0.02, against $1.8\times10^{-3}$ for substitution, and on HII20 the printed balance residual falls from a maximum of $4.6\times10^{-3}$ to $1.3\times10^{-6}$ with the volume-weighted mean from $4.7\times10^{-4}$ to $4.5\times10^{-9}$. The benchmark numbers above move by 0.01 K, so the method changed the path to the root and not the root. Extrapolating the $n_e$ sequence instead — Aitken — was tried and made the pass count worse, because a jump in one component leaves the rest of the coupled state off the slow manifold; that is recorded with its measurements in `docs/ION_BALANCE_CONVERGENCE_PLAN.md`.

*Helium–hydrogen charge exchange* (`par%helium_charge_exchange`, default on; rates in §3.8). With $k_{21}$, $k_{12}$, $k_{32}$, $k_{23}$ the coefficients of $\mathrm{He}^+ + \mathrm{H} \rightarrow \mathrm{He} + \mathrm{H}^+$, its reverse, $\mathrm{He}^{2+} + \mathrm{H} \rightarrow \mathrm{He}^+ + \mathrm{H}^+$ and its reverse, hydrogen's balance Eq. (30) gains, for each H,

$$R^\mathrm{out}_\mathrm{He} = n_\mathrm{He}\,(x_\mathrm{HeII}k_{21} + x_\mathrm{HeIII}k_{32}), \qquad R^\mathrm{in}_\mathrm{He} = n_\mathrm{He}\,(x_\mathrm{HeI}k_{12} + x_\mathrm{HeII}k_{23}), \tag{33}$$

added to $R^\mathrm{out}_\mathrm{cx}$ and $R^\mathrm{in}_\mathrm{cx}$, and the helium chain gains the same reactions from the other side,

$$\frac{n_\mathrm{HeII}}{n_\mathrm{HeI}} = \frac{\Gamma_\mathrm{HeI} + n_e C_\mathrm{HeI} + n_\mathrm{HII}k_{12}}{\alpha_\mathrm{HeII}n_e + n_\mathrm{HI}k_{21}}, \qquad \frac{n_\mathrm{HeIII}}{n_\mathrm{HeII}} = \frac{\Gamma_\mathrm{HeII} + n_e C_\mathrm{HeII} + n_\mathrm{HII}k_{23}}{\alpha_\mathrm{HeIII}n_e + n_\mathrm{HI}k_{32}}, \tag{34}$$

so each exchange is counted once in each balance and charge is conserved in the pair (`solve_ion_cell`, and `hydrogen_balance_residual`, which evaluates the same terms on the written He fractions). The He fractions in $R_\mathrm{He}$ are those of the previous iterate, and when the metal terms are also present they are held constant through `hydrogen_neutral_fraction`'s root find. Because an exchange moves charge without changing $n_e$, the fixed point on $n_e$ alone could stop while $x_\mathrm{HI}$ and $x_\mathrm{HeII}$ are still moving, so with the switch on the loop also requires both to have changed by less than $10^{-10}$ relative before it exits. Measured on the production benchmark inputs (`results/continuous_energy/m2_atomic`, the same binary with the exchange on and off): $\langle T[N_\mathrm{p}N_\mathrm{e}]\rangle$ moves by +0.011 K at HII40 and +0.003 K at HII20 and $L(\mathrm{H}\beta)$ by $-5\times10^{-7}$ and $-1.3\times10^{-6}$ (the Kimura $\mathrm{He}+\mathrm{H}^+$ rate, briefly adopted, gave −0.005 K and +0.010 K; `m3_evening`) — the level at which two runs differ anyway; in these soft-spectrum nebulae $\mathrm{He}^+$ and $\mathrm{H}^0$ barely overlap. On the 40 kK PDR test, whose $\mathrm{He}^+$ zone reaches the partly neutral edge, the whole 2026-09-24 update lowers the $\mathrm{He}^+$ fraction by 1.0% and the He II recombination lines by 6%; that run was not repeated

with the exchange off, so the share of the exchange in those two numbers is not separated.

Because the written state is under-relaxed (`par%ion_relax`) while the metal stage fractions written alongside are evaluated at the written $n_e$ and $T_e$, the residual of Eq. (30) on that state is not identically zero; `hydrogen_balance_residual` reports it each gas iteration as the maximum over leaves and the volume-weighted mean. It is a printed diagnostic and does not gate convergence.

## 4.5 Thermal balance

When `par%solve_te` is set, `gas_thermal_update` replaces the fixed-$T$ `gas_equilibrium_update` (§4.4) and updates each leaf's temperature once per iteration, from that iteration's radiation field, by finding the equilibrium $\mathcal{H}(T) = \mathcal{C}(T)$ — the root of the net rate

$$\mathcal{N}(T) = \mathcal{H}(T) - \mathcal{C}(T), \qquad \mathcal{H} = n_{\rm HI}\mathcal{H}_{\rm HI} + y_{\rm He} n_{\rm H}(x_{\rm HeI}\mathcal{H}_{\rm HeI} + x_{\rm HeII}\mathcal{H}_{\rm HeII}), \tag{35}$$

with the photoheating rates $\mathcal{H}_i$ (per atom) from Eq. (5). The ionization state is *re-solved* (`solve_ion_cell`, §4.4) at every trial $T$: both $\mathcal{H}$ and $\mathcal{C}$ depend on the fractions $(x_{\rm HI}, x_{\rm HeI}, x_{\rm HeII}, n_e)$, which themselves depend on $T$. The ionization response is mild, so $\mathcal{N}(T)$ is monotone through its root and bisection on $\log T_e$ in [`te_min`, `te_max`] is safe: each step takes the geometric midpoint $\sqrt{T_{\rm lo} T_{\rm hi}}$ (up to 60 steps, stopping at $T_{\rm hi}/T_{\rm lo} - 1 < 10^{-5}$), and if cooling already wins at `te_min` ($\mathcal{N} \le 0$) the leaf is pinned to `te_min` while if heating still wins at `te_max` it is pinned to `te_max`. With `charge_exchange_heating` (on by default) $\mathcal{H}$ also carries the kinetic energy the charge-exchange reactions release or take up (Eq. 15, §3.8), evaluated with the fractions of the same trial $T$.

The cooling $\mathcal{C}(T)$ (`cooling_mod`) assembles recombination (Eq. 11), free-free ($1.42554\times10^{-27}\sqrt{T}\,Z^2\,\bar{g}\,n_e n_{\rm ion}$, summed over H II, He II ($Z = 1$) and He III ($Z = 2$) with $Z$ the *net* ion charge; under `par%metal_freefree` (on by default) the trace metals are added with $Z_{\rm ion}$ = (stage −1), not the nuclear $Z$, using $\bar{g}(T, Z_{\rm ion})$ tabulated for $Z = 1$–4 — singly-ionized metals dominate this small $\sim10^{-4}$-of-the-H/He term), collisional ionization ($k_{\rm ci}\times$ ionization potential), H I line cooling, and the trace-metal sum $\sum n_e\, n_{\rm ion}\Lambda_{\rm ion}(T, n_e)$ with the cascade fractions (§4.6) re-evaluated at each trial $T$. The line-cooling coefficient $\Lambda_{\rm ion}(T, n_e)$ is the density-suppressed value of Eq. (19) by default (`par%cooling_model = local_ne`), the plain fit $\Lambda_{\rm ion}(T)$ under `low_density`. With `par%use_metals` the cooling also carries, whatever heats the gas, the atomic-hydrogen impact on the ground-term fine structure of C I, C II and O I (`metal_cooling_H`, §4.18): the increment of the radiated power when $n_{\rm HI} q^{\rm H}$ joins $n_e q^e$ in one statistical equilibrium, added to the electron share taken from the table. Two further terms enter $\mathcal{N}$: metal photoheating (`par%metal_heat`, on by default, with the shell-resolved heating coefficient of §3.1); and, for PDR runs, grain photoelectric heating with grain recombination cooling (Bakes & Tielens 1994, `par%grain_pe`, off by default, driven by the local FUV field $G_0$, both weighted by $x_{\rm HI}$ and evaluated at $\min(T, 10^4\,{\rm K})$ so the fits stay in range — two guards the PDR gate needed against a 12–35 kK runaway).

Under `par%use_sec_ion` a photoelectron of kinetic energy $E_0 = h\nu - E_{{\rm th},s}$ from absorber $s$ (H I, He I, He II) with $E_0 > 30$ eV is partitioned between heat, the ionization of H I and He I, and excitation that escapes as line radiation. Two published determinations enter, each for what it provides. Shull & van Steenberg (1985, ApJ 298, 268; SvS85, Table 2, read from the published scan) give the energy budget of the two ionization channels, $\phi_{\rm HI}(x) = 0.3908(1 - x^{0.4092})^{1.7592}$ and $\phi_{\rm HeI}(x) = 0.0554(1 - x^{0.4614})^{1.6660}$, which vanish at $x = 1$ but, by their own footnote, are "only accurate for $E_0 > 100$ eV". Dalgarno, Yan & Liu (1999, ApJS 125, 237; DYL99, read from the published PDF) resolve $E_0$ over 30–1000 eV through

the mean energy per ion pair of their H–He mixture, $W(x,E_0) = W_0(1+Cx^a)$ (eq. 13, Table 4), and the heating efficiency $\eta(x,E_0) = 1+(\eta_0-1)/(1+Cx^a)$ (eq. 14, Table 7), both stated for $x \le 0.1$. The ionization budget is

$$f_{\rm HI}(x,E_0) = \phi_{\rm HI}(x)\,\frac{W_{\rm H^+}(x,1\,{\rm keV})}{W_{\rm H^+}(x,E_0)}, \qquad f_{\rm HeI}(x,E_0) = \phi_{\rm HeI}(x)\,\frac{W_{\rm He^+}(x,1\,{\rm keV})}{W_{\rm He^+}(x,E_0)}, \tag{36}$$

exactly SvS85 at 1 keV, and smaller for slower electrons ($W_{\rm He^+}$ rises from 487 eV at 1 keV to 6740 eV at 30 eV). The budget $f_{\rm ion} = f_{\rm HI} + f_{\rm HeI}$ is divided among the neutrals actually present on SvS85's reference $n_{\rm HeI}/n_{\rm HI} = 0.1$, $c_{\rm HI} = f_{\rm ion}\,0.1 f_{\rm HI}/w$, $c_{\rm HeI} = f_{\rm ion} f_{\rm HeI}/w$ with $w = 0.1 f_{\rm HI} n_{\rm HI} + f_{\rm HeI} n_{\rm HeI}$, so that $n_{\rm HI}c_{\rm HI} + n_{\rm HeI}c_{\rm HeI} = f_{\rm ion}$ at every composition, and the heat share is $f_{\rm heat} = 1-(1-\eta)(1-x)$, clamped to $[0, 1-f_{\rm ion}]$; the $(1-x)$ closure (EXHALE's, in neither paper) removes the $(1-\eta_0)/(1+C) = 10\%$ that eq. (14) alone leaves unthermalized in a fully ionized gas at 1 keV. Here $x = n_e/(n_{\rm H}+n_{\rm He})$, the free electrons that compete with the neutrals for the photoelectron's energy (SvS85's $n({\rm H^+})/n({\rm H_{tot}})$ under their equal-ionization assumption), taken from the current iterate of the cell solve. In the transport the deposited energy $\sigma_s(1-E_{{\rm th},s}/E)$ above 30 eV is scored at each packet's energy onto the six DYL99 nodes with weights linear in $\log E_0$ (clamped at 1 keV); the weights do not depend on the gas, so the partition is applied to those node integrals exactly inside `solve_ion_cell` (the secondary rates) and in the thermal sum (the heat withheld). Approximations: below 30 eV the energy is all heat (no tabulation covers 13.6–30 eV), so the partition steps there; above $x = 0.1$ DYL99 is extrapolated (EXHALE measured eq. 14 with the closure against SvS85's own 3 keV Monte Carlo over $x > 0.1$ to within 5.6%); both calculations carry He/H = 0.1; metal photoelectrons are all heat. The previous implementation (SvS85 alone, from 40 eV, with $x$ the ionized fraction of the H+He nuclei, the rates evaluated on the previous iterate's state and no redistribution by composition) used a fit outside its stated range below 100 eV: at $E_0 = 100$ eV and $x = 0.1$ it counted 1.21 secondary H ionizations against 0.55 now. Gate `tests/secondary_ionization/` (the 1 keV limit reproduces SvS85 Table 2 to $2\times10^{-16}$, energy is conserved at any He/H, the heat share is Table 7's $\eta_0$ at $x \to 0$ and exactly 1 at $x = 1$). Default is off. The module is `photoelectron_degradation_mod`, rewritten on 2026-09-24 after the EXHALE module `electron_energy_degradation`. Measured on the HII20 shell grid under a $10^5$ K blackbody (`results/secondary_ionization/`, 80 iterations, same seed): inside the H II region, where $x \to 1$ and the SvS85 budget vanishes, the run with secondaries reproduces the run without them; the whole effect sits in the partly neutral front, where $T_e$ drops by up to 205 K at $x({\rm H^0}) \approx 0.9$ (the previous form: 120 K) and $x({\rm H^0})$ by 0.18%; $\langle T[N_pN_e]\rangle$ moves by −0.5 K and $L({\rm H}\beta)$ by +0.014%.

The solved $T_e$ is written directly (only the ionization fractions carry the under-relaxation `par%ion_relax`), $n_e$ is rebuilt from the new state — including the trace-metal electrons under `par%metal_ne` — and, as in the ionization-only path, only the node-local rank 0 runs the bisection (the reduced band tally is identical on every node master) and writes the shared state, with the convergence metrics broadcast within the node. The loop is gated by the $T_e$ convergence measures defined above (`gas_tol_te`, cell-max or volume-integrated per `par%conv_crit`).

## 4.6 The trace-metal cascade

The registry (`species_mod`) activates an element when its abundance is positive. Stage fractions follow the product chain

$$\frac{n_{i+1}}{n_i} = \frac{\Gamma_i + n_e C_i + n_{\rm HII}\,k^{\rm CX}_{{\rm ion},i}}{n_e\,[\alpha_{\rm RR}+\alpha_{\rm DR}]_{i+1} + n_{\rm HI}\,k^{\rm CX}_{{\rm rec},i+1}}, \tag{37}$$

with metal $\Gamma_i$ computed once per iteration from the same band tally. The chain is evaluated on a known H/He and thermal state rather than solved jointly with it, but the metals are not excluded from the ionizing opacity (§4.7), from $n_e$ (`par%metal_ne`), or from hydrogen's own balance: the two $k^{\mathrm{CX}}$ terms above enter Eq. (30) as well (§4.4). Adding an ion is one entry in the data generators plus its cooling and $n$-level files.

## 4.7 Metal opacity in the He-ionizing window

*Record (2026-09-25).* The front shifts and bounds in this part were measured on runs stopped by the earlier convergence rule (§5.4), which ended the HII40 benchmark 0.17 pc short of its converged He I front; they document the search for a residual that was mostly that early stop, and are not current bounds. The converged residual is 0.034 pc.

"Trace" names the *data* treatment, not a switched-off feedback. The founding plan lists metals feeding back on $n_e$ and on the opacity as a non-goal, but the production inputs run with `par%metal_ne`, `par%ion_metal_abs` and `par%metal_heat` all true, so both feedbacks are active — and since 2026-07-31 the first and third are also the shipped defaults, for the reason measured below.

Table 4: What the two metal feedbacks are worth on the benchmarks. Each run changes only the named switch; same inputs, same Sobol seed. Fronts from the volume-weighted 0.01-pc shell diagnostic.

| `metal_ne` | `metal_heat` | $\langle T[N_{\mathrm{p}}N_{\mathrm{e}}]\rangle$ [K] | $L(\mathrm{H}\beta)$ | $r(\mathrm{He}^0{=}0.5)$ | $r(\mathrm{H}^0{=}0.1)$ |
|---|---|---|---|---|---|
| *HII40* | | | | | |
| on | on | 8168.9 | 1.9293e37 | 4.1278 | 4.6111 |
| on | off | 8147.9 | 1.9306e37 | 4.1248 | 4.6090 |
| off | on | 8168.7 | 1.9291e37 | 4.1291 | 4.6123 |
| off | off | 8147.6 | 1.9304e37 | 4.1259 | 4.6100 |
| *HII20* | | | | | |
| on | on | 6924.5 | 4.6837e36 | 1.2408 | 2.8352 |
| on | off | 6918.8 | 4.6839e36 | 1.2406 | 2.8345 |
| off | on | 6924.4 | 4.6836e36 | 1.2408 | 2.8358 |
| off | off | 6918.7 | 4.6838e36 | 1.2407 | 2.8352 |

The two switches separate cleanly and add. Switching both off costs 21.3 K (0.26%) at HII40 and 5.8 K (0.084%) at HII20; `metal_heat` carries nearly all of it (−21.1 and −5.7 K) and `metal_ne` almost none (−0.2 and −0.1 K), with the fronts moving by at most 0.003 pc. The smallness of `metal_ne` is not an argument that metal electrons do not matter: the metal stages hold about 0.1% of the electrons in the ionized gas, and $\langle T[N_{\mathrm{p}}N_{\mathrm{e}}]\rangle$ is weighted by $N_{\mathrm{p}}N_{\mathrm{e}}$, which suppresses exactly the neutral gas where the metal stages are the *only* electron source and $n_e$ would collapse without them. That is why the bound above does not transfer past the ionization front, and why both now default on: an approximation is worth keeping only where its cost is known, and beyond the front it is not.

Concretely, `species_cached_opacity` sums over *all* active stages, with the cross sections evaluated at each packet's own energy by `species_packet_cross_sections` and the self-consistent stage fractions, and the sum is iterated together with the opacity it changes. The cross sections themselves are not in question. For the eight ions that carry the He-ionizing window — C I–III, N II, O II, Ne I, S II, S III — MoCHII reproduces Verner's own `phfit2.f` to $10^{-6}$

relative, the single-precision round-off of `phfit2` itself; since 2026-09-24 the gate `tests/photo_shells/` extends that check to every registry ion and to the inner-shell branch (worst $2.3\times10^{-6}$, §3.1). None of these eight has an inner edge below 100 eV, so the inner-shell correction of §3.1 does not reach the measurements of this subsection.

What is genuinely absent is metal recombination continua as a diffuse *source*: the diffuse field has exactly four channels (`dif_ch(4, nleaf)`) and all four are H/He. Summing the metal ground-state recombination continua whose photons lie above 24.587 eV (C IV, N III, N IV, O III, O IV, Ne III, Ne IV, S III, S IV) gives 0.29% of the He I ground-continuum channel, about 0.001 pc of front position. Metal photoelectrons are not partitioned under `par%use_sec_ion` either (their energy is all heat), but that switch is off in the benchmarks.

**The measured lever.** Scaling the summed metal cross section on the converged HII40 model (68 ranks, same Sobol seed, 0.01-pc shell diagnostic) gives

| variant | $r(\mathrm{He}^0{=}0.5)$ [pc] | $\Delta r$ | pc per unit fraction |
|---|---|---|---|
| production | 4.11970 | — | — |
| metal $\sigma \times 0.70$ | 4.04764 | −0.07206 | 0.240 |
| metal $\sigma \times 1.30$ | 4.15063 | +0.03093 | 0.103 |
| metal absorption off | 3.50470 | −0.61500 | 0.615 |
| metals off entirely | 1.93335 | −2.18635 | (also removes all metal cooling; $T_e$ 8162 → 18681 K) |

The response is strongly asymmetric and saturating: removing metal opacity costs 0.615 pc per unit while adding it gains only 0.103 pc per unit. Extrapolating the upward branch, closing the 0.178 pc Cloudy residual of §4.10 would take +180% metal opacity (×2.8), and the saturation means still more. That is not available, and it retires the metal opacity as a single-parameter explanation.

The prediction recorded before these runs was "metal opacity ×1.30 moves $r(\mathrm{He}^0{=}0.5)$ outward, ×0.70 inward, of order 0.1–0.2 pc". The *sign* was right both ways; the *magnitude* was wrong by factors of 1.5–6, because the prediction extrapolated the on/off bound of 0.615 pc as if the response were symmetric. An on/off bound measures what removal costs and says little about what addition could gain. The companion statement that "a ∼30% error in the metal opacity would be worth ∼0.18 pc" is accordingly withdrawn: 30% is worth 0.031–0.072 pc.

**Mechanism: spectral hardening, not cooling.** The metal stages with thresholds inside 21–40 eV (Ne I 21.56, S II 23.33, C II 24.38, N II 29.60, S III 34.83, O II 35.12 eV) preferentially remove the *softest* He-ionizing photons, and $\sigma(\mathrm{He\,I})/\sigma(\mathrm{H\,I})$ rises steeply with energy — 6.1 at 25 eV, 9.0 at 35, 13.9 at 55, 19.9 at 95 eV. The hardened surviving field is therefore taken up by $\mathrm{He}^0$ rather than $\mathrm{H}^0$ and the $\mathrm{He}^+$ zone extends; remove the metal opacity and the field stays soft, $\mathrm{H}^0$ takes a larger share, and the zone collapses. The alternative readings are excluded by the same runs: between production and `ion_metal_abs = .false.` the hydrogen structure and the mean temperature barely move ($r(\mathrm{H}^0{=}0.1) = 4.61633$ vs. 4.61551 pc; $T_e$ 8162 vs. 8183 K), so the 0.615 pc is neither a thermal effect nor a change in the hydrogen solution.

**What Cloudy has here that MoCHII does not.** Cloudy's base is the same Verner `phfit`. Its `opacity_createall.cpp` then adds channels MoCHII has no equivalent for: $\mathrm{N}^+$ excited-level photoionization (Henry

1970), O I valence replaced wholesale by `ofit` when PHFIT96 is in use, O I $^1$S, $O^{2+}$ $^1$D and $^1$S, photoionization into excited states of $O^+$, and a $Mg^+$ excited level, several of them fitted to Opacity Project data. So Cloudy photoionizes metals out of excited levels and MoCHII only out of the ground level. As opacity these are small: the excited-level populations are $10^{-5}$–$10^{-4}$ of the ground, so the $O^{2+}$ $^1$D opacity is of order $10^{-4}$ of the $O^+$ ground opacity.
An independent Opacity Project comparison is possible for C, N and O through the TOPbase tables carried by `sirocco`. Photon-weighted with a 40-kK blackbody over 24.587–54.4 eV, OP/VFKY96 is 0.946 (C II), 0.978 (N II) and 0.863 (O II) — OP is 2–14% *lower*. With the measured upward derivative that is −0.005 to −0.03 pc, the wrong direction for the residual. Ne and S are absent from that table and were not checked.

**A justification that had to be retracted in the code.** Two comments — at the `ion_metal_abs` declaration in `define.f90` and above the metal opacity routine in `species_mod` (then `species_opacity_add`, now `species_cached_opacity`) — used to justify the metal opacity as "negligible next to H/He above 13.6 eV". The measurement above contradicts that, and both now carry the measured derivatives instead: the opacity is small in *magnitude* but not in *effect*, because it acts on the spectral shape in the band where $He^0$ and $H^0$ compete. An on/off statement is the form such a justification has to take here, since the response is asymmetric and a single number in either direction misstates it.

## 4.8 Case B on the spot: the He I recombination photons

With `par%recombination_continuum = 'on_the_spot'` (the default until 2026-09-26, since then `'transported'`) the recombination photons are not transported but absorbed where they are made. For hydrogen that is case B itself. For helium it was, until 2026-09-25, only half of the approximation: every $He^+ \to He^0$ recombination photon was given back to helium and none ionized hydrogen, although the whole excited cascade lies below the $He^0$ edge and only hydrogen can take it. With `par%hei_on_the_spot` (default `.true.`; `hei_recombination_photons_mod`) the two balances are those of Osterbrock & Ferland (2006) eqs. 2.24–2.25,

$$n(\mathrm{H}^0)\,(\Gamma_\mathrm{H} + C_\mathrm{H} n_e) + \left[y\,\alpha_1(\mathrm{He}^0) + p\,\alpha_B(\mathrm{He}^0)\right] n_e n(\mathrm{He}^+) = \alpha_B(\mathrm{H}^0)\, n_e n_\mathrm{p}, \tag{38}$$

$$n(\mathrm{He}^0)\,(\Gamma_\mathrm{He} + C_\mathrm{He} n_e) = \left[\alpha_B(\mathrm{He}^0) + y\,\alpha_1(\mathrm{He}^0)\right] n_e n(\mathrm{He}^+). \tag{39}$$

$y$ is the share of the ground-capture photons that $H^0$ absorbs, $n(\mathrm{H}^0)\sigma_\mathrm{H}/[n(\mathrm{H}^0)\sigma_\mathrm{H} + n(\mathrm{He}^0)\sigma_\mathrm{He}]$ averaged over the Milne spectrum $E^2\sigma_\mathrm{He}(E)e^{-(E-E_\mathrm{th})/kT}$ by a four-point Gauss–Laguerre rule on the transport cross sections (within $4\times10^{-8}$ of a direct quadrature). $p$ counts the H-ionizing photons of one case-B recombination with the same exits the diffuse field's channel 4 uses — one routine serves both: $p = f_{3S} + f_{1P}\,p_{584} + [f_{1S} + f_{1P}(1-p_{584})]N_{2q}$, with the branch fractions of Section 4.10, $p_{584}$ of Wood, Mathis & Ercolano (2004) and $N_{2q} = 0.556$ H-ionizing photons per $2^1$S decay; with the $2^3$S population coupled the triplet term is $n_3 A_{31}$ and the collisional transfers feed the singlet exits. Osterbrock & Ferland assume every 584 Å photon ionizes H ($p_{584} = 1$) and find $p \simeq 0.96$; the same limit here gives 0.966. Each photon $H^0$ absorbs heats the gas by $E - 13.598$ eV; for the ground captures the capture's own kinetic-energy loss, which the case-B recombination cooling omits, cancels the photon's excess over threshold, leaving 10.989 eV. Dust and the metals are given no share of these photons. Hydrogen's source enters its photoionization rate per $H^0$ at the previous iterate of the cell

solve; it is at most ∼15% of hydrogen's recombinations, so that map converges fast. Gate `tests/hei_on_the_spot/`. `.false.` reproduces the earlier case B bit for bit.

The omission was not small. On the on-the-spot Lexington runs (`results/continuous_energy/m7_hei_on_the_spot/`, restarted from converged states), turning the He I photons over to hydrogen raises $\langle T[\mathrm{N}^{+}n_e]\rangle$ by 178 K at HII40 (7959 → 8137 K), $L(\mathrm{H}\beta)$ by 7.8% and moves the H front out by 0.16 pc. Against the run with the continuum transported, the on-the-spot result then misses by 62 K and 0.023 pc instead of 240 K and 0.18 pc. At HII20 it adds 21 K overall and ∼550 K inside the $\mathrm{He}^{+}$ zone. What remains between the two treatments is the locality of the on-the-spot approximation itself.

## 4.9 Case B on the spot: the He II recombination photons

Where $\mathrm{He}^{2+}$ exists the same omission is larger. Osterbrock & Ferland (2006, sec. 2.5) name the coupling: the $\mathrm{He}^{2+}$ zone is also an $\mathrm{H}^{+}$ zone, and the He II Ly$\alpha$ (40.8 eV), the $2^2$S two-photon continuum and the Balmer continuum (whose threshold, 13.604 eV, is the hydrogen edge) all ionize $\mathrm{H}^0$ — their Table 2.6 gives 0.64 + 0.36 + 0.20 H-ionizing photons per H recombination at $n(\mathrm{He}^{2+})/n_\mathrm{p} = 0.15$ and $10^4$ K. Until 2026-09-25 the on-the-spot mode recombined $\mathrm{He}^{2+}$ at $\alpha_B$ and let none of those photons ionize anything. With `par%heii_on_the_spot` (default `.true.`; `heii_recombination_photons_mod`) the three balances read

$$n(\mathrm{H}^0)(\Gamma_\mathrm{H} + C_\mathrm{H} n_e) + S^{\mathrm{I}}_\mathrm{H} + S^{\mathrm{II}}_\mathrm{H} = \alpha_B(\mathrm{H}^0)\, n_e n_\mathrm{p}, \tag{40}$$

$$n(\mathrm{He}^0)(\Gamma_\mathrm{He} + C_\mathrm{He} n_e) + S^{\mathrm{II}}_\mathrm{He} = \left[\alpha_B(\mathrm{He}^0) + y\,\alpha_1(\mathrm{He}^0)\right] n_e n(\mathrm{He}^{+}), \tag{41}$$

$$n(\mathrm{He}^{+})(\Gamma_{\mathrm{He}^{+}} + C_{\mathrm{He}^{+}} n_e) = \left[\alpha_B(\mathrm{He}^{+}) + (y_\mathrm{H} + y_\mathrm{He})\,\alpha_1(\mathrm{He}^{+})\right] n_e n(\mathrm{He}^{2+}), \tag{42}$$

with $S^{\mathrm{I}}_\mathrm{H}$ the He I source of Section 4.8 and

$$S^{\mathrm{II}}_\mathrm{H} = n_e n(\mathrm{He}^{2+})\left[y_\mathrm{H}\,\alpha_1(\mathrm{He}^{+}) + \sum_k q_k\, f_{\mathrm{H},k}\right], \qquad S^{\mathrm{II}}_\mathrm{He} = n_e n(\mathrm{He}^{2+})\left[y_\mathrm{He}\,\alpha_1(\mathrm{He}^{+}) + \sum_k q_k\, f_{\mathrm{He},k}\right]. \tag{43}$$

*Ground captures.* The $\mathrm{He}^{2+} \to \mathrm{He}^{+}(1s)$ photons, $E \geq 54.416$ eV with the Milne spectrum $E^2\sigma_{\mathrm{He}^{+}}(E)e^{-(E-E_\mathrm{th})/kT}$, are shared by opacity among $\mathrm{He}^{+}$, $\mathrm{He}^0$ and $\mathrm{H}^0$; $y_\mathrm{H}$ and $y_\mathrm{He}$ are $n_X\sigma_X / \sum_{X'} n_{X'}\sigma_{X'}$ averaged by the same four-point Gauss–Laguerre rule as the He I captures (within $8.5 \times 10^{-8}$ of a direct quadrature). $\mathrm{He}^{+}$'s share re-ionizes it — case B — and changes nothing. *Excited cascade.* $q_k$ counts the photons, per $n_e n(\mathrm{He}^{2+})$, of the three paths channel 5 transports (Section 4.10.1): free–bound into $n \geq 2$, the 40.813 eV Ly$\alpha$ and the in-band $2^2$S pair, taken from the *same* branch luminosities and split into a part below the $\mathrm{He}^0$ edge and eight groups above it (boundaries quadratic in the group index, the first 0.47 eV wide for the free–bound). None reaches 54.4 eV, so $\mathrm{He}^{+}$ takes none; below 24.587 eV $f_{\mathrm{H},k} = 1$, above it $\mathrm{H}^0$ and $\mathrm{He}^0$ share each group by opacity at its photon-mean energy (within $5.2 \times 10^{-4}$ of a photon-by-photon quadrature). The energy of each path closes on the transported channel's luminosity to $2 \times 10^{-16}$, and the photon count on what its samplers imply, $L\langle 1/E\rangle$, to $2.1 \times 10^{-5}$. At $n(\mathrm{He}^{2+})/n_\mathrm{p} = 0.15$ the paths give 0.648 (Ly$\alpha$), 0.339 (two-photon) and 0.199 (free–bound) H-ionizing photons per H recombination at 10.5 kK, against Osterbrock & Ferland's 0.64, 0.36, 0.20 at 10 kK. The tables are built over the whole spectrum the gas emits whatever `par%eion_max` is: the photons are made by the gas, not by the band. *Heat.* An absorber $X$ gains $E - \mathrm{IP}(X)$ per photon. For the excited photons that is the whole of it, since the kinetic energy the captures into $n \geq 2$ remove is in the case-B recombination cooling ($\beta$ from $\alpha_B(\mathrm{He}^{+})$). The

ground captures are not in it: each removes its electron's kinetic energy $K$ and emits $E = \mathrm{IP(He^+)} + K$, so an absorption by $X$ nets $\mathrm{IP(He^+)} - \mathrm{IP}(X)$ per photon whatever its energy — 40.819 eV for $H^0$, 29.830 eV for $He^0$, and nothing for $He^+$. *Iteration.* Both sources join the photoionization rates *per absorber* at the previous iterate of the cell solve. For $He^0$ this form is required rather than convenient: its share of an excited group falls with $n(He^0)$, so the rate per $He^0$ stays finite as $He^0$ is depleted, whereas a source count held at the previous iterate's share can exceed the $He^+ \rightarrow He^0$ recombinations that replenish $He^0$, and the closed form then has no positive solution. *Validity.* Absorption by H, $He^0$ and $He^+$ only (no dust, no metals), in the leaf that made the photon. Osterbrock & Ferland note that Ly$\alpha$ (resonance-scattered by $He^+$) and the Balmer continuum (just above the H edge) travel little, but that most of the two-photon continuum escapes the $He^{2+}$ zone into the $He^+$ zone; on the spot those photons are misplaced, not lost. Gate `tests/heii_on_the_spot/`. `.false.` reproduces the earlier case B bit for bit.

Measured on a $10^5$ K blackbody in the HII40 geometry (`results/heii_on_the_spot/`: gas to 5.2 pc at $n_\mathrm{H} = 100$, $L = 9.3598 \times 10^{38}$ erg s$^{-1}$ so that $Q$(H) equals HII40's, $Q(\mathrm{He^+})/Q(\mathrm{H}) = 0.058$; $2^{22}$ Sobol packets, 20 ranks, thermal balance on, C/N/O/Ne/S at the HII20 abundances). The three runs share every input but the treatment of the recombination continuum: on the spot with `heii_on_the_spot` on and off (both converged by the drift test, 35 and 34 iterations), and transported with `hei_diffuse` and `heii_diffuse` on, which stopped at 80 iterations with the H and $He^0$ drift at the Monte Carlo floor and the $He^{2+}$ drift at $5.9 \times 10^{-3}$ against the $5 \times 10^{-3}$ tolerance.

| | on the spot, on | on the spot, off | transported | on/off vs transported |
|---|---|---|---|---|
| $\langle T[n_\mathrm{p} n_e] \rangle$ [K] | 11799 | 11577 | 11704 | +95 / −128 K |
| $L(\mathrm{H}\beta)$ [$10^{36}$ erg s$^{-1}$] | 9.2438 | 8.5419 | 9.2540 | −0.11% / −7.7% |
| He II 4686 [$10^{35}$ erg s$^{-1}$] | 10.069 | 10.736 | 10.529 | −4.4% / +2.0% |
| H front, $x(\mathrm{H^0}) = 0.5$ [pc] | 4.049 | 3.932 | 4.056 | −0.006 / −0.124 |
| $He^{2+}$ front, $x(\mathrm{He^{2+}}) = 0.5$ [pc] | 1.997 | 1.904 | 1.926 | +0.071 / −0.022 |

(Fronts from volume-weighted spherical shells.) Everything outside the $He^{2+}$ zone moves onto the transported run: $L(\mathrm{H}\beta)$ from 7.7% low to 0.11%, the H front from 0.124 pc inside to 0.006 pc. Inside the $He^{2+}$ zone the on-the-spot run is now the one that is off (Figure 1): it deposits the whole cascade there, $x(\mathrm{H^0})$ falls eightfold, $T_e$ rises to 15.2 kK against the transported 12.8 kK and the unheated 11.3 kK, and the hotter gas recombines more slowly, moving the $He^{2+}$ front 0.071 pc out and lowering 4686. In the gas the source replaces 0.71 of hydrogen's recombinations there (volume mean, at most 0.82), and the local balance closes. That is not guaranteed: $H^0$'s share of the cascade does not fall with $n(H^0)$ (below the $He^0$ edge nothing else absorbs, above it $He^0$ is as scarce as $H^0$ in the $He^{2+}$ zone), so it enters hydrogen's balance as an absolute source, its share evaluated at every trial $x_\mathrm{HI}$ of the root find (`hydrogen_fraction_with_cascade`), and a cell can hold only as many absorptions as the recombinations that make $H^0$. A weakly lit fully $He^{2+}$ cell at He/H = 0.1 reaches 0.92 and closes to $10^{-11}$; at He/H = 0.15 ($1.2$–$1.8 \times 10^4$ K) the cascades outnumber the supply and no local balance exists. There the cell settles at $x_\mathrm{HI} \approx 0$, the heat is limited to the photons the supply can absorb (`f_h_only`, e.g. 0.66 at He/H = 0.2, $1.8 \times 10^4$ K), the H balance residual printed each iteration shows the leaf, and setup prints a note for `abund_He` $> 0.12$: such nebulae (Type I planetary nebulae) need the transported continuum. Where both $H^0$ and $He^0$ are trace absorbers of the cascade, the two shares are solved together (since 2026-09-26, `joint_neutral_fractions` in `solve_ion_cell`, used where $x(\mathrm{He^{2+}}) > 10^{-3}$): $x_\mathrm{HeI}$ is the root of helium's balance with hydrogen's own root evaluated

at each trial $n(\mathrm{He}^0)$. $\mathrm{He}^0$'s share vanishes with $n(\mathrm{He}^0)$ and grows as $\mathrm{H}^0$'s falls, so that root is unique. Before it, $\mathrm{He}^0$'s share entered its balance as a rate per $\mathrm{He}^0$ at the previous iterate, the two shares traded places from pass to pass, and a cell could leave the loop with hydrogen's balance open by $O(1)$: 0.89 in the $\mathrm{He}^{2+}$ core of the G2a run at He/H = 0.1, and 0.46 at He/H = 0.2 in the gate, which a change of $\alpha_B$ by 1% had moved off the state it happened to settle in before. Now the gate's He/H = 0.2 cell closes to $2\times10^{-16}$ with $f = 0.661$. Gate `tests/heii_on_the_spot/` (g). How much of that is the approximation and how much the reference: in the transported run the photons cross the $\mathrm{He}^{2+}$ zone almost freely — at 40.8 eV hydrogen there gives $\tau \approx 0.01$–$0.04$ per pc — and its $x(\mathrm{H}^0)$ there stays within 25% of the run without them. For the two-photon and free–bound photons that is what Osterbrock & Ferland describe; Ly$\alpha$, two thirds of the cascade, is transported as a continuum photon, whereas they note it is resonance-scattered and absorbed close to where it is made. The $\mathrm{He}^{2+}$ zone of a real nebula therefore lies between the two runs, and neither is a reference for it.

Where the $\mathrm{He}^{2+}$ zone is optically thin at 40.8 eV the on-the-spot treatment of the cascade fails outright, not by degree. In the G2a sphere ($4\times10^4$ K star, $n_\mathrm{H} = 100$) hydrogen gives $\tau = 2\times10^{-6}$ at 40.8 eV from the center to 0.24 pc, yet the cascade absorbed in place there deposits 6 times the stellar photoheating at 0.1 pc and 4 times at 0.2 pc, and $T_e$ reaches `te_max` = $5\times10^4$ K in the run and in its 1D reference alike. The mode keeps its definition, and two things are offered with it (2026-09-26). Each iteration `thermal_mod` counts, and prints when there are any, the leaves held at `te_max` and the leaves where the on-the-spot He II photons (the $\mathrm{He}^{2+}$ ground captures, and the cascade unless it is transported) out-heat the transported field while the leaf is thin at 40.8 eV ($\tau_\mathrm{leaf} < 0.01$). And `par%heii_diffuse`, set in the input, transports the cascade alone as channel 5 while every other recombination photon stays on the spot: `heii_on_the_spot` then keeps only the $\mathrm{He}^{2+}$ ground captures (`heii_cascade_mod::heii_cascade_transported`, which `heii_ots_nodes` consults, so the sources, the heat and the absorption limit drop the cascade together; gate `tests/heii_on_the_spot/` (h)). Left out of the input it stays off in this mode, so existing on-the-spot runs do not change.

## 4.10 Explicit diffuse LyC field and packet sampling

Case B on the spot is replaced by case A rates plus explicit emission of ground-recombination packets. For each leaf and each of the three ground continua H II→H I($1s$), He II→He I($1^1S$), and He III→He II($1s$), the emissivity uses $\alpha_1(T) = \alpha_A(T) - \alpha_B(T)$. At the start of every gas iteration the code rebuilds the energy-luminosity table

$$L_{\mathrm{ch},l} = n_{e,l}\, n_{\mathrm{rec},l}\, \alpha_{1,\mathrm{ch}}(T_l)\, \langle E \rangle_{n,\mathrm{ch}}(T_l)\, V_l, \qquad L_\mathrm{dif} = \sum_{l,\mathrm{ch}} L_{\mathrm{ch},l}, \tag{44}$$

with $\langle E \rangle_n$ the photon-number-weighted mean energy of the emitted continuum.

**Ground-continuum spectrum (`par%diffuse_energy_model`).** Detailed balance between photoionization and radiative recombination fixes the free–bound spectrum. Per unit photon energy the photon-number emissivity of each ground continuum is

$$n(E) \;\propto\; E^2\, \sigma_\mathrm{pi}(E) \exp\!\left[-\frac{E - E_\mathrm{th}}{kT_e}\right], \qquad E \geq E_\mathrm{th}, \tag{45}$$

evaluated with the *same* ground-state cross section the transport absorbs with, so a packet is absorbed with the cross section it was emitted with. This is the default, `'milne'`. Restoring the Milne prefactor turns the same integral into the

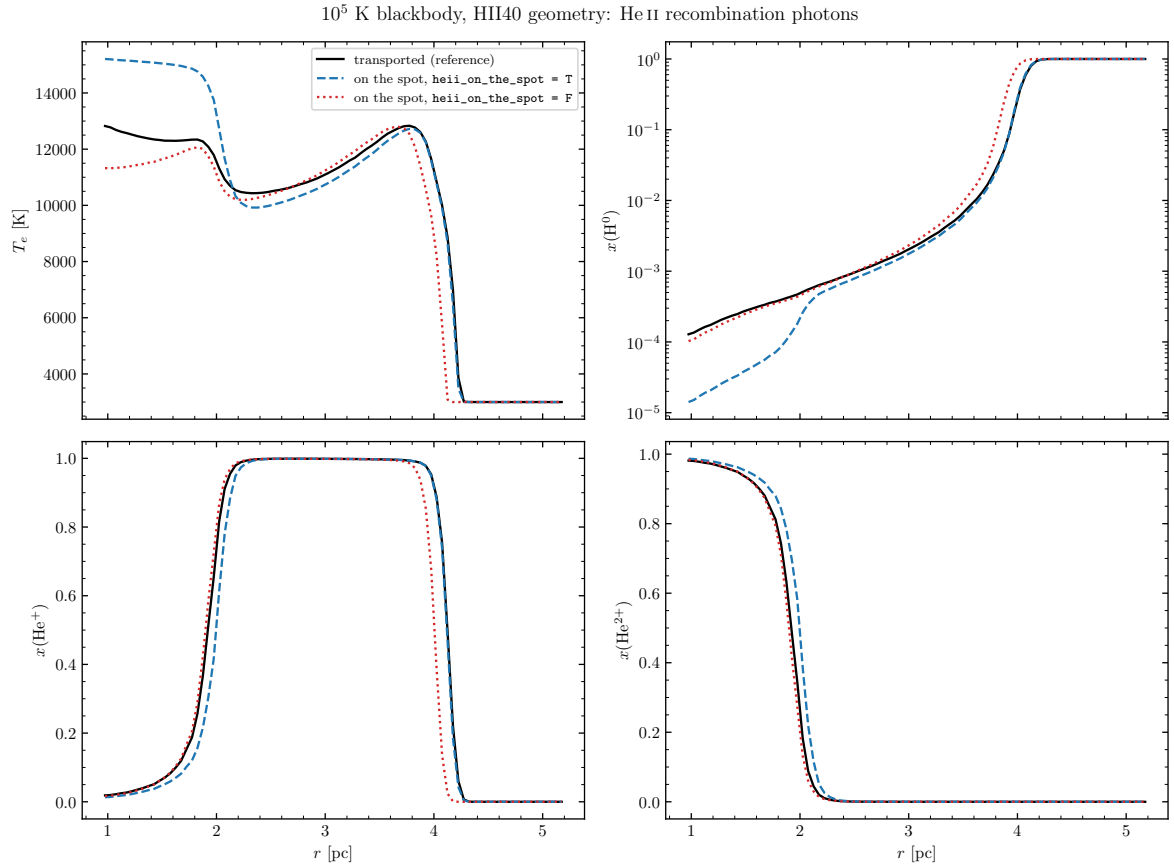



Figure 1: Spherically averaged $T_e$, $x(H^0)$, $x(He^+)$ and $x(He^{2+})$ of the three runs of Section 4.9 (`results/heii_on_the_spot/plot_profiles.py`).

ground-state recombination coefficient,

$$\alpha_1(T) = \frac{g_n}{2g_+}\frac{2}{\sqrt{\pi}}(kT)^{-3/2}\sqrt{\frac{2}{m_e}}\frac{1}{m_e c^2}\int_{E_{\rm th}}^{\infty} E^2\sigma_{\rm pi}(E)e^{-(E-E_{\rm th})/kT}\,dE, \tag{46}$$

so the spectrum and the emission rate are not independent assumptions. Equation (46) reproduces $\alpha_{1s}(\mathrm{H\,I}, 10^4\ \mathrm{K}) = 1.579\times10^{-13}\ \mathrm{cm^3\ s^{-1}}$ against the tabulated $1.58\times10^{-13}$. The code reports that integral at start-up against the $\alpha_A-\alpha_B$ that sets the packet count and rescales neither side. Since the correction described next, the largest disagreement over the three channels is 0.01%. It was 3.44% before, all of it in He I, and that number is what exposed the defect.

**The $\alpha_1$ source-mixing defect.** `recomb_mod` took $\alpha_A$ from the Badnell total, $\alpha_1$ from the *level-resolved* Mao & Kaastra fit, and then defined $\alpha_B = \alpha_A - \alpha_1$. That subtraction crosses two unrelated determinations. Nothing in the construction forces the Mao $n$=1 value to be the ground term of the Badnell total it is subtracted from, and whatever separates them lands entirely in $\alpha_B$ — and, through $\alpha_1$, in the diffuse packet count as well.

The cross-section side is not the error. MoCHII's `sigma_HeI` reproduces Verner's own `phfit2.f` to a ratio of 1.00000 at every energy over 24.6–105 eV, and its threshold value, 7.436 Mb, sits 0.5% above the measured $7.40\pm0.07$ Mb of Samson et al. (1994), inside the experimental uncertainty. Nor is the start-up comparison between unlike quantities: because $\alpha_B = \alpha_A - \alpha_1$, the "rates" side of it *is* $\alpha_1$(He I) itself, so the 3.44% separated two determinations of one number.

**Cloudy settles which side is right.** Cloudy c25.00 derives its H-like and He-like recombination level by level from its *own* photoionization cross sections through the exact Milne relation (`source/iso_radiative_recomb.cpp`), caching the result in `data/h_iso_recomb.dat` and `data/he_iso_recomb.dat`. In those files row 0 of each element is the ground level and the last row is the total — verified on hydrogen, where row 0 is $1.5840\times10^{-13}$ and the last row $4.1642\times10^{-13}$ against the known $\alpha_{1s} = 1.58\times10^{-13}$ and $\alpha_A = 4.18\times10^{-13}\,\mathrm{cm^3\,s^{-1}}$. At $10^4$ K, in $10^{-13}\,\mathrm{cm^3\,s^{-1}}$:

| | $\alpha_A$ | | $\alpha_1$ | | |
|---|---|---|---|---|---|
| channel | MoCHII | Cloudy | Mao & Kaastra | Milne integral | Cloudy |
| H I | 4.1933 | 4.1642 | 1.5873 | 1.5790 | 1.5840 |
| He I | 4.3719 | 4.3583 | **1.5650** | 1.6086 | 1.6032 |
| He II | 21.895 | 21.817 | 6.5366 | 6.5116 | 6.5145 |

Badnell's totals are accurate to 0.3–0.7% on all three channels, and the Mao $n$=1 values to 0.2–0.3% for H I and He II — but He I is 2.38% low, and the subtraction propagates that into $\alpha_B$(He I) as +1.88%. MoCHII's own Milne integral matches Cloudy to 0.3%, 0.34% and 0.04%. The Milne relation is exact, Cloudy applies it to a $He^0$ ground cross section that agrees with Verner's to the 0.4% implied by the $\alpha_1$ agreement, and so *the Milne route is the correct one and the level-resolved fit is the low side*.

**An earlier reading of the same numbers, withdrawn.** This document previously concluded the opposite — that the Milne integral was the outlier — from the third determination available at the time, Benjamin, Skillman & Smits (1999) with J. S. Mathis as carried by K. Wood's `ionize`, $\alpha_1(\mathrm{He\,I}) = 1.54\times10^{-13}(T_4)^{-0.486}\ \mathrm{cm^3\,s^{-1}}$:

| $T$ | BSS99 + JSM | Mao & Kaastra | Milne integral | Cloudy c25.00 |
|---|---|---|---|---|
| 5000 K | 2.157 | 2.184 (+1.3%) | 2.259 (+4.8%) | — |
| $10^4$ K | 1.540 | 1.565 (+1.6%) | 1.609 (+4.5%) | 1.603 (+4.1%) |
| $2\times10^4$ K | 1.100 | 1.128 (+2.6%) | 1.151 (+4.7%) | — |

In $10^{-13}\,\mathrm{cm^3\,s^{-1}}$, percentages against the BSS99 column. The arithmetic was right — the Milne integral does sit 4.5–4.8% above BSS99 and 2.8% above Mao while the two fits cluster within 2.6% of each other — but the inference was a majority vote taken on the mutual closeness of two fits, which is not evidence that either reproduces the cross sections in use. Cloudy, computing the same quantity level by level from its own cross sections, lands within 0.34% of the Milne integral. The earlier conclusion is withdrawn.

**The correction.** `tools/fitting/fit_alpha1_milne.py` evaluates Eq. (46) on the cross sections the transport actually uses — exact hydrogenic for H I and He II (`sigma_hydrogenic`, $Z$ = 1 and 2; note that `sigma_HI` in `src/photo_xsec.f90` is the exact hydrogenic form, not VFKY96) and VFKY96 for He I — at 241 temperatures over $10^3$–$10^5$ K, and refits the same seven coefficients of Eq. (9), so that no Fortran structure changes:

| function | $(a_0, b_0, c_0, a_1, b_1, a_2, b_2)$ |
|---|---|
| `alpha1_HII` | $1.87007\times10^{-2}$, 1.29077, $-2.17465\times10^{-3}$, $1.29189\times10^{1}$, $8.05060\times10^{-1}$, $8.44399\times10^{-2}$, $5.93940\times10^{-4}$ |
| `alpha1_HeII` | $3.26862\times10^{-3}$, $6.97361\times10^{-1}$, $3.14907\times10^{-4}$, $5.76341\times10^{1}$, 1.18534, $2.58417\times10^{1}$, $9.90134\times10^{-1}$ |
| `alpha1_HeIII` | $2.61525\times10^{-1}$, 1.39527, $-3.22079\times10^{-4}$, $6.40708\times10^{1}$, $8.97582\times10^{-1}$, $5.01551\times10^{-1}$, $1.21533\times10^{-3}$ |

The largest $|\mathrm{fit}/\mathrm{Milne}-1|$ is 0.019% (H I), 0.001% (He I) and 0.006% (He II) over $10^3$–$10^5$ K, and 0.0075%, 0.0004%, 0.0021% over the production range $3\times10^3$–$3\times10^4$ K, so the functional form is not the limiting approximation. At $10^4$ K the fitted $\alpha_1$ is 1.5791, 1.6086 and $6.5117\times10^{-13}\,\mathrm{cm^3\,s^{-1}}$: within 0.35% of Cloudy on all three channels, with He I corrected from −2.38% to +0.34%. The code's own cross-check confirms it, `milne_setup` reporting a largest $\alpha_1$ disagreement of 3.44% before and 0.01% after.

The gate is `tests/milne/check_alpha1.f90`, wired into `tests/milne/run_check.sh`. It re-implements the Milne quadrature independently from `photo_xsec`, checks the three channels at 81 temperatures over $10^3$–$10^5$ K to $10^{-3}$, checks $\alpha_1 < \alpha_A$ everywhere (largest $\alpha_1/\alpha_A$ 0.560, 0.496, 0.445), and checks all three against the hardcoded Cloudy c25.00 values to 1%. Reinstating the old Mao He I coefficients makes it fail on both counts ($3.886\times10^{-2}$ from the quadrature, 2.383% from Cloudy) and it passes again after reverting.

**Effect on the benchmarks, graded against the prediction.** What was recorded before the runs: "$\alpha_1$(He I) rises 2.4%, so channel 2 emits more; $\alpha_B$(He I) falls 1.9%, so there is less case-B recombination. Both push the front outward, toward Cloudy, by of order 0.015–0.02 pc." Measured on the paired HII40/HII20 production runs (68 ranks, same Sobol seed, converged in 38 and 27 iterations as before):

| | HII40 | HII20 |
|---|---|---|
| $\Delta r(\mathrm{He}^0{=}0.5)$ | +0.00811 pc | +0.00210 pc |
| Cloudy residual | −0.18561 → −0.17750 pc | — |
| $\Delta L(\mathrm{H}\beta)$ | −0.3713% | −0.4159% |
| mean $T_e(x_{\mathrm{HI}} < 0.95)$ | 9053.4 → 9051.7 K | 8522.3 → 8528.7 K |

The *sign* was right, outward and toward Cloudy on both benchmarks. The *magnitude* was about half the low end of the predicted range, +0.0081 against 0.015–0.02 pc, so the prediction was roughly a factor two high. `tests/continuous_energy/check_hii_production.py` passes on both, with energy closure $1.5\times10^{-16}$ and 0. The published benchmark quantities move by less than the width of the published spread: $\langle T[N_pN_e]\rangle$ from 8162 to 8169 K at HII40 and 6917 to 6925 K at HII20, $T_{\mathrm{inner}}$ from 7535 to 7536 and 6936 to 6938 K, and $R_{\mathrm{out}}$ from 8.89 to $8.88\times10^{18}$ cm at HII20 with $1.44\times10^{19}$ cm unchanged at HII40. A later, independent correction — exchanging the oxygen charge-exchange rows of §3.8 — then carries these a further +2.2 K at HII40 and +2.1 K at HII20, to 8170 and 6926 K (8177 and 6936 K after the 2026-09-24 update).

$L(\mathrm{H}\beta)$ moves the *wrong* way. It falls 0.37–0.42% ($1.94 \to 1.93\times10^{37}$ and $4.70 \to 4.68\times10^{36}\,\mathrm{erg\,s^{-1}}$), taking MoCHII from about 3% below the published $2.01$–$2.10\times10^{37}$ at HII40 to slightly further below. The likely route is channel 1:

$\alpha_1$(H I) drops 0.52% from the Mao value to the Milne one, so the H I ground continuum recycles marginally fewer H-ionizing photons. This is not an argument against the change — the acceptance criterion is physical correctness, and an exact relation was being violated — but it is recorded rather than buried.

Two legacy options remain. `'exponential'` is the historical near-threshold approximation

$$E_{\rm pkt} = E_{\rm th} + kT_e[-\ln(\max(u, \epsilon))], \tag{47}$$

i.e. Eq. (45) with the $E^2\sigma_{\rm pi}$ weight dropped; it is kept to reproduce earlier runs. `'threshold'` emits at $E_{\rm th}$ and is a sensitivity bracket, not a model. The three thresholds are 13.598, 24.587, and 54.416 eV, and every model is a sampled photon energy, not a bin-center approximation.

**Sampling weight.** MoCHII emits equal-energy packets and fixes their number from the channel's *energy* luminosity. The packet energies must therefore follow the energy-luminosity density $E\,n(E)$, not $n(E)$ itself. With that pairing the represented photon rate is exactly the physical one,

$$N_{\rm pkt}L_{\rm pkt}\left\langle E^{-1}\right\rangle_{En} = R\left\langle E\right\rangle_n \frac{1}{\left\langle E\right\rangle_n} = R, \tag{48}$$

whereas drawing from $n(E)$ while counting packets from the energy luminosity inflates it by $\left\langle E\right\rangle_n\left\langle E^{-1}\right\rangle_n > 1$ (1.002 at 8000 K, 1.010 at $2\times10^4$ K). The CDF is tabulated in $x = (E - E_{\rm th})/kT$, interpolated in $\log T$ and inverted once, rather than inverting two neighboring temperature rows and interpolating the resulting energies.

**Effect.** Against the exponential approximation the converged HII40 calculation moves $r(\mathrm{He}^0 = 0.5)$ outward by 0.00063 pc and $r(\mathrm{C}^{2+} = 0.5)$ by 0.00052 pc, closing 0.3% of the residual against Cloudy c25. The two continua move in opposite directions because each follows its own mean energy: the H I continuum softens (14.460 → 14.427 eV at $10^4$ K) since the hydrogenic cross section falls as $E^{-3}$, while the He I continuum hardens (25.449 → 25.460 eV) because the Verner He I fit falls more slowly near threshold. The exact spectrum is therefore a correctness fix, not the explanation of the Cloudy front residual. The same holds for the three He I cascade corrections described below: the two-photon shape moves $r(\mathrm{He}^0 = 0.5)$ by −0.00030 pc, the branch fractions by +0.00024 pc, and the fate of the 584 Å photon by −0.00363 pc. All four corrections to the emitted diffuse spectrum together move it −0.00306 pc, i.e. 1.7% of the 0.18 pc residual and *away* from Cloudy rather than toward it. Switching the whole fourth channel off moves it −0.103 pc, so the channel's presence matters and its internal detail does not.

*Record (2026-09-25).* The front shifts and bounds in this part were measured on runs stopped by the earlier convergence rule (§5.4), which ended the HII40 benchmark 0.17 pc short of its converged He I front; they document the search for a residual that was mostly that early stop, and are not current bounds. The converged residual is 0.034 pc.

**Where the Cloudy front residual is not.** Each of those four corrections is independently correct — three of them reproduce an external reference implementation and the first an exact detailed-balance relation — and the acceptance criterion is physical correctness, not agreement with a particular reference calculation, so a net displacement in the unhelpful direction is not an argument against any of them. What the sum does settle is the diagnosis: *the* 0.18 *pc residual is not in what the diffuse field emits.* Four independent corrections, spanning the emitted spectrum's shape, its branch weights and the fate of its strongest line, together account for 1.7% of the gap and with the wrong sign.

**Bounding the remaining candidates.** The quantity to be explained is $r(\mathrm{He}^0 = 0.5) = 4.12781$ pc against Cloudy c25.00's 4.30531 pc, i.e. −0.17750 pc after the $\alpha_1$ correction above. The sensitivity runs below were measured against the pre-correction production point, 4.11970 pc; the 0.008 pc between the two baselines changes no lever, and the raw front positions are quoted as measured. The front's sensitivity to the diffuse field is calibrated two ways from the 584 Å measurement above: differentially (channel 4 lost 8.87% of its energy, is 17.6% of the diffuse field, and moved the front 0.00363 pc) and by switching that channel off entirely (17.6% of the field, 0.103 pc), giving 0.233 and 0.585 pc per unit fractional change in the total diffuse band luminosity. Two further sensitivity runs (HII40, 68 ranks, same Sobol stream, 38 iterations) scale the He I atomic data by 3%: reducing $\sigma_{\mathrm{pi}}$(He I) moves the front +0.00799 pc and reducing the He I recombination rate moves it +0.02410 pc. Every candidate then has a number attached to it, and every number is too small:

| axis | bound measured | to reach 0.178 pc |
|---|---|---|
| what the diffuse field emits | four corrections, net −0.00306 pc | wrong sign |
| explicit transport vs. on-the-spot[†] | 4.11970 to 4.18159 pc | the whole range is 0.062 pc |
| total diffuse band luminosity | 0.233–0.585 pc per unit fraction | 30–76% |
| $\sigma_{\mathrm{pi}}$(He I) | +0.00799 pc per −3% (0.27 pc/unit) | ∼70% |
| $\alpha$(He I) | +0.02410 pc per −3% (0.80 pc/unit) | ∼23% |
| $\alpha_A$, $\alpha_1$, $\alpha_B$(He I) | 0.3%, 0.34%, 1.9% from Cloudy (§3.4) | already there; code spread is 11% |
| metal opacity | 0.103 pc/unit upward, 0.615 downward (§4.7) | +180%, and saturating |
| metal recombination continua | 0.29% of channel 2 | ∼0.001 pc |
| He I excited-level ionization | 0.2–0.6% extra, and below the band floor | structurally absent |
| dust | `dust_density_law='none'` in the benchmark | excluded by construction |
| source photon budget | $Q$ ratios 1.00005 and 1.00003 | 14% in $Q(\mathrm{He}^0)$ |
| Cloudy-side resolution | median zone width 0.0106 pc at the front | 17× finer already |

[†] Measured before the on-the-spot He I photons of Section 4.8 existed: the on-the-spot end then gave every He recombination photon back to helium.

The transport entry is the decisive one, because it was the leading candidate: switching the whole diffuse treatment between its extremes — explicit transport at 4.11970 pc and the on-the-spot approximation at 4.18159 pc — spans only 0.062 pc, and Cloudy sits 0.124 pc beyond the more favorable end of that range. No choice about how diffuse photons are transported can reach it. The photon budget is verified against the same 40-kK blackbody Cloudy was given: $Q(>13.598\ \mathrm{eV}) = 4.2656\times10^{49}$ s$^{-1}$ against $4.2653\times10^{49}$ and $Q(>24.587\ \mathrm{eV}) = 4.6082\times10^{48}$ against $4.6081\times10^{48}$, ratios 1.00005 and 1.00003. Cloudy resolves the front with 40 zones between 4.0 and 4.6 pc, median width 0.0106 pc and maximum 0.039 pc, so the reference position is not a resolution artifact either.

**The He-ionizing photon budget.** Where the sensitivity lives is visible in the budget itself. Integrated over the $\mathrm{He}^+$ zone of the converged HII40 model:

| HII40 | $s^{-1}$ |
|---|---|
| stellar $Q(>24.587$ eV) | $4.6082\times10^{48}$ |
| diffuse He I ground continuum, recycled | $1.6135\times10^{48}$ |
| total He-ionizing supply | $6.2217\times10^{48}$ |
| $He^0$ photoionizations = case-A recombinations | $4.7008\times10^{48}$ |
| absorbed by $H^0$ or metals, or escaped | $1.5210\times10^{48}$ = 24.4% of the supply |

Nearly a quarter of the He-ionizing supply is lost to absorbers other than $He^0$, and halving that loss fraction alone would move the front 0.211 pc. The $H^0/He^0$ competition is therefore the one place where a few-percent difference has the leverage to matter, which is consistent with the radial signature below and with the metal-opacity mechanism of §4.7. The budget also closes geometrically: with the leaf volume computed correctly the summed gas volume matches the geometric shell 0.972–4.729 pc to 0.25%. (`LeafSize` in the output is the leaf *full* width, and the residual analysis script had used $(2\,\mathrm{size})^3$, eight times the cell volume. Because it enters only as a relative weight within a shell, every front position reported from it is unaffected and re-running after the fix reproduces all of them identically; the expression was nevertheless wrong and is now $\mathrm{size}^3$. The table generator, the Python readers and the g1/g4 gates all used the convention correctly and were never affected.)

**The radial signature.** What the residual does have is a shape. The hydrogen structure agrees with Cloudy throughout the ionized zone; only the helium structure drifts, and it drifts monotonically outward:

| $r$ [pc] | 1.6 | 2.0 | 2.5 | 3.0 | 3.5 | 3.8 |
|---|---|---|---|---|---|---|
| $x(He^0)$, MoCHII/Cloudy | 1.007 | 1.015 | 1.031 | 1.059 | 1.109 | 1.196 |
| $x(H^0)$, MoCHII/Cloudy | 0.981 | 0.984 | 0.989 | 1.000 | 1.005 | 1.009 |

A normalization difference in a rate or a cross section would be radius-independent; this one compounds with depth. That is what a 0.7% seed at 1.6 pc must do when $He^0$ carries almost all of the opacity above 24.6 eV: at 30 eV, $\kappa(He^0)/[\kappa(He^0)+\kappa(H^0)]$ rises from 0.843 at 1.6 pc to 0.956 at 4.0 pc, so any excess $He^0$ shortens the mean free path of the very photons that would have removed it. The signature therefore identifies a small helium-specific seed, amplified geometrically, rather than a large error anywhere.

Closing 0.178 pc is 4.1% in radius and 13% in volume, which asks for $Q(He^0)$ higher by $\sim$14% or $\alpha_B$(He I) lower by $\sim$20%. $Q(He^0)$ is verified to 0.003%, and MoCHII's $\alpha_A$, $\alpha_1$ and $\alpha_B$(He I) now sit within 0.3%, 0.34% and 1.9% of Cloudy's own level-resolved values, with no published determination anywhere near 20% away. *The residual is therefore unexplained*, and what has been achieved is the systematic closing of the space of simple explanations: it is not in what the diffuse field emits, not in how those photons are transported, not in the He I photoionization cross section, not in the He I recombination rate, not in the metal opacity, not in the metal recombination continua, not in He I excited-level ionization, not in dust, not in the source photon budget, and not in the reference calculation's resolution.

The two candidates named in earlier revisions have both been examined and neither survives as a single parameter. The helium ionization *network* has been compared code to code (§3.4): every code sums the full cascade, MoCHII's $\alpha_A$(He I) agrees with Cloudy's explicit level sum to 0.3%, and the He I excited levels that a resolved model atom carries are worth 0.2–0.6% of extra ionization. The metal opacity has been measured (§4.7): its upward derivative is 0.103 pc per unit,

so closing the gap would need +180% against cross sections that reproduce Verner's to $10^{-6}$ and an Opacity Project comparison that runs 2–14% the other way.

What remains is the one hypothesis that no single-parameter experiment can reproduce: several differences of a few percent each — in the He I recombination rate, in the shape of $\sigma_{\rm pi}$(He I), in the metal opacity of the 21–40 eV window, in Cloudy-side modeling choices not yet identified — acting together through the $He^0$ optical depth, which compounds them with depth exactly as the radial signature shows. Each is individually below its own measured lever; their product over 24.4% of the He-ionizing supply lost to competing absorbers is not bounded by any of the experiments above. Testing it requires varying several ingredients jointly, not one at a time, and that is the open item.

The number of diffuse packets is $N_{\rm dif} = \max[1, \mathrm{nint}(L_{\rm dif}/L_{\rm pkt})]$, with $L_{\rm pkt}$ the luminosity of one ionizing source packet ($L_{\rm ion}/N$; with the FUV extension the ionizing stratum's), and each carries $L_{\rm dif}/N_{\rm dif}$, so the emitted luminosity is exactly $L_{\rm dif}$ (until 2026-09-25 each carried $L_{\rm pkt}$, off by up to half a packet, and a field under half a packet was dropped). A packet first selects a leaf from the CDF of $\sum_{\rm ch} L_{{\rm ch},l}$, then a channel from that leaf's luminosity weights over the five channels. Its emission point is uniform in the selected cubic leaf and its direction is isotropic. It is then transported with the same exact-energy edge walk and path estimator as a stellar packet. Hence diffuse luminosity is represented by packet count, with a packet weight within half a packet of the stellar one. At HII40 the diffuse band luminosity converges to $6.8\times10^{38}$ erg/s = $0.50\,L_\star$.

For output bookkeeping, the energy from Eq. (47) is mapped by binary search of `ion_eedge` (`ion_bin_of`). This works for both threshold-aligned and logarithmic diagnostic grids; an energy on an edge falls in the bin above it. The assignment is a tally address only: it does not modify $E_{\rm pkt}$ or its cross sections. The binary search also corrects the former closed-form log-grid mapping, which could place a 24.6-eV He I recombination photon below the He I threshold on an aligned grid.

**He I excited-level recombination (`par%hei_diffuse`).** A fourth channel adds the He I *excited-level* recombination radiation (previously those photons simply vanished). Every case-B He II recombination reaches $1^1$S through exactly one $n = 2$ term, so what the channel needs is not three absolute rates but three *branch fractions*; the absolute rate is already fixed by the $\alpha_B$(He II) that sets the packet count. MoCHII takes all three from one internally consistent set of effective recombination coefficients,

$$\begin{aligned}
\alpha_{\rm eff}(2^3{\rm S}) &= 2.10\times10^{-13}\,(T_4)^{-0.381},\\
\alpha_{\rm eff}(2^1{\rm S}) &= 2.06\times10^{-14}\,(T_4)^{-0.451},\\
\alpha_{\rm eff}(2^1{\rm P}) &= 4.17\times10^{-14}\,(T_4)^{-0.695},
\end{aligned} \tag{49}$$

in $\mathrm{cm^3\,s^{-1}}$ — Benjamin, Skillman & Smits (1999) Table 1 with J. S. Mathis, in the form carried by K. Wood's `ionize` — and normalizes by their sum. That all three come from one source matters more than any individual value: a fraction assembled from fits with unrelated temperature dependences is not a fraction of anything. Normalizing by the sum is the source's own choice, made for the reason it states, that the fitting formulae carry different temperature dependences. Outside $3\times10^3$–$2\times10^4$ K the fits are clamped, not extrapolated.

The closure

$$\alpha_{\rm eff}(2^3{\rm S}) + \alpha_{\rm eff}(2^1{\rm S}) + \alpha_{\rm eff}(2^1{\rm P}) = \alpha_B({\rm He\,II}) \tag{50}$$

is physically true, but MoCHII does *not* use it to derive a branch. Its two sides are independent atomic data: the three fits

of Eq. (49) account for $0.90\,\alpha_B$ at 8000 K and $0.97\,\alpha_B$ at $10^4$ K, and the ratio is reported at start-up rather than absorbed. Forcing it onto $\alpha_B$ would dump the entire mismatch between two data sets into whichever branch was solved for last. The resulting fractions, as the code prints them, are

| $T$ [K] | $2^3$S | $2^1$S | $2^1$P | $\sum\alpha_{\rm eff}/\alpha_B$ |
|---|---|---|---|---|
| $5\times10^3$ | 0.74084 | 0.07629 | 0.18288 | 0.767 |
| $8\times10^3$ | 0.76183 | 0.07591 | 0.16226 | 0.896 |
| $10^4$ | 0.77121 | 0.07565 | 0.15314 | 0.970 |
| $1.5\times10^4$ | 0.78729 | 0.07507 | 0.13764 | 1.134 |
| statistical | 0.75000 | 0.08333 | 0.16667 | — |

against the AGN3 low-density statistical weights of the last row, which the code used before. The correction is a moderate temperature-dependent shift, not a reordering.

Two provenance points belong with these numbers. First, $2.06\times10^{-14}+4.17\times10^{-14}=6.23\times10^{-14}$ exactly, so the $6.23\times10^{-14}(T_4)^{-0.827}$ that `nebcont_mod` took from MOCASSIN for the He I two-photon continuum (Section 4.20) is the *singlet total*, not $\alpha_{\rm eff}(2^1{\rm S})$. An earlier version of this section read it as the latter and closed $2^1$P against $\alpha_B$, obtaining 0.746 : 0.224 : 0.031 together with an argument that case-B trapping inverts the singlet split; both the numbers and the argument are withdrawn. Second, the $2^3$S population of `hei_metastable_mod` (Section 3.13) takes its recombination source from this set: $\alpha_3 = f_{3S}\,\alpha_B({\rm He\,II})$. It used to carry its own Oklopcic & Hirata (2018) fit, $2.10\times10^{-13}(T_4)^{-0.778}$, which an earlier version of this paragraph defended as a separate "absolute production rate". That defense is withdrawn (2026-09-25): the absolute rate of triplet recombinations is fixed once the balance's $\alpha_B$ and the branch fraction are, and a second normalization supplied beside them is the defect of lesson (4).

The $2^1$S photons follow the tabulated Drake, Victor & Dalgarno (1969) distribution (`data/HeI2phot.dat`, the same table the nebular continuum uses) rather than a flat draw. That shape fixes the H-ionizing yield at 0.55644 photons and 8.9646 eV per decay, with a mean in-band energy of 16.110 eV, against the 0.56 and 17.109 eV of the flat surrogate. The two corrections are individually small and act in opposite directions. Per case-B He II recombination the channel emits

| | H-ionizing energy [eV] | photons |
|---|---|---|
| statistical weights, flat two-photon | 19.200 | 0.9634 |
| statistical weights, Drake shape | 19.149 | 0.9633 |
| Eq. (49), Drake shape, 8000 K | 19.223 | 0.9663 |

i.e. –0.27% in energy from the two-photon shape, then +0.39% in energy and +0.34% in photons from the branch fractions, for a net +0.12% and +0.30% against the statistical weights. In transport (68 ranks, same Sobol stream, branch fractions against the Drake-only state) the He I front moves by +0.00024 pc at HII40 and +0.00006 pc at HII20, $r(C^{2+}=0.5)$ by +0.00015 and 0.00000 pc, and $L(\mathrm{H}\beta)$ by +0.0121% and +0.0027%; both benchmarks still pass `tests/continuous_energy/check_hii_production.py`. Like the Milne ground continua, this is a correctness fix and not the explanation of the 0.18 pc Cloudy residual.

**The fate of the 584 Å photon.** The $2^1$P branch decays to the ground state on He I Ly$\alpha$, 584 Å = 21.22 eV, which is both an H-ionizing photon and a resonance photon in a medium optically thick to it in $He^0$. It therefore scatters many times before anything irreversible happens, and two irreversible outcomes compete: photoionization of a hydrogen atom (depositing $21.22-13.60 = 7.62$ eV of heat), or the 2.06 $\mu$m decay $2^1\mathrm{P} \rightarrow 2^1\mathrm{S}$ of the last He atom that absorbed it ($A \approx 2\times10^6$ s$^{-1}$ against $1.8\times10^9$ s$^{-1}$ at 584 Å, about one scattering in a thousand; electron impact is slower still at nebular densities), after which the decay is the two-photon continuum instead. Which one wins is set by the $H^0$ and $He^0$ opacities the trapped photon sees, so it is a local, density- and ionization-dependent branching and not a constant. MoCHII resolves it with the escape-probability closure of Wood, Mathis & Ercolano (2004), Section 3.3,

$$p_{584} = \left[1 + \frac{77\,(y_{\mathrm{He}}/0.1)\,x(\mathrm{He}^0)}{\sqrt{T}\,x(\mathrm{H}^0)}\right]^{-1}, \tag{51}$$

with $y_{\mathrm{He}} = n_{\mathrm{He}}/n_{\mathrm{H}}$. Their eq. 17 competes $n(\mathrm{H}^0)a_\nu(\mathrm{H}^0)$ against $n(\mathrm{He}^0)a_\nu(\mathrm{He}^0)/914$ and writes the result in neutral fractions "assuming He/H = 0.1", so the 77 carries that abundance; the factor $y_{\mathrm{He}}/0.1$ (since 2026-09-26) is exactly 1 for every run at He/H = 0.1. the probability that the 584 Å photon ends up ionizing hydrogen (`hei_584_ionizes_H` in `hei_cascade_mod`). Its two limits are the correct ones: $x(\mathrm{H}^0) \rightarrow 0$ gives $p_{584} \rightarrow 0$, nothing left to ionize and everything converted, while $x(\mathrm{He}^0) \rightarrow 0$ gives $p_{584} \rightarrow 1$, nothing left to convert. K. Wood's `ionize` and CMacIonize carry the same closure in two algebraically different arrangements, and the gate `tests/hei_cascade/check_hei_584.f90` reproduces both to $2.2\times10^{-16}$ over a grid in $(T, x_{\mathrm{H}^0}, x_{\mathrm{He}^0})$. The four codes in the reference trees sit at three different points on the question:

| code | cascade coefficients | fate of the 584 Å photon | transported? |
|---|---|---|---|
| Wood `ionize` | Eq. (49), sum-normalized | $p_{584}$ ionizes H; $1-p_{584}$ converts | no, on the spot |
| CMacIonize | the same set and normalization | the same split | no, on the spot |
| MOCASSIN | singlet total $6.23\times10^{-14}(T_4)^{-0.827}$ | all converted ($p_{584} = 0$) | emissivity only |
| MoCHII | Eq. (49), sum-normalized | Eq. (51) split | yes, full transport |

CMacIonize is also where MoCHII's flat-surrogate constant 0.56 came from, as $2 \times 28\%$ of the two-photon distribution above the Lyman limit; Eq. (49) and the Drake table replaced both of those constants above. MOCASSIN does not resolve the competition at all: its comment states the assumption that all He I singlets end in $2^1$S, and the $6.23\times10^{-14}$ singlet total is that $p_{584} \rightarrow 0$ limit written as one coefficient. MoCHII is now the only one of the four that applies the branching and then *transports* what it produces: Wood and CMacIonize destroy the 584 Å photon where it was created, so the deposition of whatever the branch yields is assumed local, while MoCHII carries the branch product through the grid with the same edge walk as any other packet and lets the grid decide where it is absorbed.

Because $p_{584}$ depends only on the emitting leaf's own $T$, $x(\mathrm{H}^0)$ and $x(\mathrm{He}^0)$, the split folds into the branch weights of `hei_branch_energies` at build time and needs no random variate of its own, so the nine-coordinate Sobol assignment below is unchanged:

$$\begin{aligned} e(2^3\mathrm{S}) &= f_{2^3\mathrm{S}} \times 19.82 \text{ eV}, \\ e(2^1\mathrm{P}) &= f_{2^1\mathrm{P}}\, p_{584} \times 21.22 \text{ eV}, \\ e(2^1\mathrm{S}) &= \left[f_{2^1\mathrm{S}} + f_{2^1\mathrm{P}}(1-p_{584})\right] \times 8.9646 \text{ eV}. \end{aligned} \tag{52}$$

A converted decay radiates the two-photon pair, 20.62 eV, and not 21.22 eV; the difference of 0.60 eV leaves as $2^1\mathrm{P} \rightarrow 2^1\mathrm{S}$ at 2.06 $\mu$m, which is non-ionizing and therefore correctly absent from the band. The He I two-photon nebular continuum takes its $2^1\mathrm{S}$ decays from the same exits (`hei_twophoton_decays`, Section 4.20), which closes an internal inconsistency: `nebcont_mod` had inherited MOCASSIN's singlet total, the $p_{584} \rightarrow 0$ limit, while `diffuse_mod` transported every $2^1\mathrm{P}$ decay as a 21.22-eV packet, the $p_{584} \rightarrow 1$ limit, so the two modules disagreed about the fate of the same decay. The domain of validity is recorded at the code site: this is an escape-probability closure for a 584 Å line assumed optically thick, static and dust free. Dust competes for the trapped photon and is not in the closure — a small correction at the Lexington dust content, not a small one at higher dust-to-gas ratios.

**Effect of the 584 Å split.** Weighted by the $2^1\mathrm{P}$ emission rate on the converged grids, $\langle p_{584} \rangle = 0.117$ at HII40 and 0.0044 at HII20: 88% and 99.6% of that branch converts to the two-photon continuum, and the channel's output per case-B He II recombination falls from 19.22 to 17.47 eV at HII40 (−9.1%) and from 19.24 to 17.12 eV at HII20 (−11.0%), with the photon count down 6.6% and 7.9%. In transport (68 ranks, same Sobol stream, against the branch-fraction state):

| | HII40 | HII20 |
|---|---:|---:|
| $\Delta r(\mathrm{He}^0 = 0.5)$ [pc] | −0.00363 | −0.00022 |
| $\Delta r(\mathrm{C}^{2+} = 0.5)$ [pc] | −0.00332 | −0.00001 |
| $\Delta r(\mathrm{H}^0 = 0.1)$ [pc] | −0.01002 | (density bounded) |
| $\Delta L(\mathrm{H}\beta)$ | −0.1997% | −0.0704% |
| $\langle T_e \rangle$ $(x_{\mathrm{H}^0} < 0.95)$ [K] | 9080.2 → 9053.4 | 8523.0 → 8522.3 |
| diffuse band luminosity | −1.56% | −0.22% |

Both benchmarks still pass `tests/continuous_energy/check_hii_production.py`. This is the largest of the four corrections to the emitted diffuse spectrum — six times the Milne ground continua and twelve times the branch fractions — and it still moves the He I front only 2% of the way to the Cloudy residual, in the wrong direction.

Three predictions were recorded before these runs, and grading them is worth keeping. The sign was predicted correctly: losing channel-4 photons pulls the fronts inward. The magnitude, of order 0.01 pc, was correct. The third, that HII20 would move more than HII40, was *wrong* — HII40 moved sixteen times further. The prediction had been read off $p_{584}$ alone, which measures only what fraction of the $2^1\mathrm{P}$ branch converts; front displacement instead scales with that branch's *absolute* contribution to the ionizing budget. Inverting the measured luminosity drops against the change per recombination puts channel 4 at 17.6% of the HII40 diffuse field and 2.0% of the HII20 one: a 20 kK spectrum makes few photons above 24.6 eV, so $\mathrm{He}^+$ is scarce and the channel fed by its recombinations is small. A larger converted fraction of a much smaller channel moves the front less. The lesson generalizes — a local branching probability predicts what happens per event, never how far a front moves; that second factor is the channel's share of the budget, set by the hardness of the source.

**The $2^3\mathrm{S}$ population in channel 4 (`par%hei_metastable_coupling`).** The branch weights above send every triplet recombination out as a 19.82 eV photon. That is the low-density limit only: the $2^3\mathrm{S}$ atom is destroyed by electrons,

photons and hydrogen atoms long before it decays at $n_e \gtrsim n_{\rm crit}$ (Section 3.13), and it is also fed by collisional excitation from $1^1$S. With the coupling on, `hei_branch_energies_coupled` builds the three weights per unit volume from the population $n_3$ of Eq. (21):

$$\begin{aligned} e(2^3\mathrm{S}) &= n_3 A_{31} \times 19.82\ \mathrm{eV}, \\ N_{1S} &= n_e n_{\mathrm{HeII}} \alpha_B f_{1S} + n_e n_3 q_{31a} + n_e n_3 q_{31s} f_{1S}/(f_{1S}+f_{1P}), \\ N_{1P} &= n_e n_{\mathrm{HeII}} \alpha_B f_{1P} + n_e n_3 q_{31b} + n_e n_3 q_{31s} f_{1P}/(f_{1S}+f_{1P}), \\ e(2^1\mathrm{P}) &= N_{1P}\, p_{584} \times 21.22\ \mathrm{eV}, \qquad e(2^1\mathrm{S}) = [N_{1S} + N_{1P}(1-p_{584})] \times 8.9646\ \mathrm{eV}. \end{aligned} \tag{53}$$

The destruction channels without a photon — $q_{31g}$, $C_3$, $\Phi_3$, Penning — emit nothing; the singlet recombination branches are unchanged. The transfers into $n^1L$, $n \geq 3$, are divided between $2^1$S and $2^1$P in the ratio of the singlet recombination branches, an approximation that gives a collisionally populated $n^1L$ term the cascade of a recombining one. In the limit $n_e \to 0$ with $\Phi_3 = 0$, $n_{\rm HI} = 0$ and no excitation from $1^1$S, $n_3 A_{31} = f_{3S}\alpha_B n_e n_{\rm HeII}$ and the weights reduce to $n_e n_{\rm HeII}\alpha_B$ times the uncoupled ones; the gate finds $2.5\times10^{-13}$ at $n_e = 10^{-10}$ cm$^{-3}$. At $10^4$ K with $n_{\rm HeII} = 0.1\,n_e$, $n_{\rm HI} = n_{\rm HeI} = 10^{-3} n_e$ and $\Phi_3 = 0$ (gate case), the 19.82 eV / 584 Å/ two-photon weights are $0.972/1.023/1.093$ times the uncoupled ones at $n_e = 100$ cm$^{-3}$ and $0.259/1.603/3.491$ at $10^4$ cm$^{-3}$. Because the weights depend on $\Phi_3$ they are stored per leaf at build time and read by both launchers; the Sobol coordinates are unchanged.

With `launch_sequence='sobol'`, diffuse packets use a separate Owen-scrambled Sobol stream so that their changing count does not disturb the direct-source sequence. Its fixed nine coordinates are: one for the leaf CDF; three for the position within that leaf; two for $(\mu, \phi)$; one for the channel CDF; one for the ground-continuum energy draw (or the excited-He branch, or the He II cascade path); and one for the two-photon energy, drawn from the tabulated Drake shape, or for the channel-5 free–bound or two-photon energy. Coordinates that are irrelevant to a selected channel are still reserved, preserving the global packet-indexed layout across iterations and MPI task counts. The fifth channel adds no coordinate: its path pick and its energy draw are the same two decisions channel 4 makes.

### 4.10.1 The He III cascade (channel 5)

Channel 3 emits the He II *ground* continuum, $\alpha_1 = \alpha_A - \alpha_B$, above 54.416 eV. Everything else a He III recombination radiates on its way down through He II was dropped, and unlike the He I cascade that omission is not a small correction: the He II $n = 2$ threshold sits at 13.604 eV, a hair *above* the hydrogen edge, so the entire Balmer continuum photoionizes hydrogen, and the $2^2$S pair spans 40.813 eV rather than He I's 20.62 eV. `par%heii_diffuse` adds the three paths (`heii_cascade_mod`):

| path | coefficient | share of the channel |
|---|---|---|
| free–bound, $n \geq 2$ | $\int(\gamma_{\rm tab} - \gamma_{\rm ff})\mathrm{d}\nu$, 13.598–54.416 eV | 7.5–10.2% |
| Ly$\alpha$, $2^2\mathrm{P} \to 1^2\mathrm{S}$, 40.813 eV | $\alpha_B\,(1 - f_{2s})$ (case B) | 64.2–70.0% |
| two-photon, $2^2\mathrm{S} \to 1^2\mathrm{S}$ | $\alpha_B\, f_{2s}$, 36.090 eV in band | 22.5–25.6% |

over $10^4$–$3\times10^4$ K in case B. The two-photon term — the only one of the three the written continuum ever carried — is under a quarter of the channel, which is why adding it alone would have been the wrong repair.

The free–bound term is emitted at the *capture*, not at the $n = 2$ exit, so it carries its own coefficient instead of a branch of $\alpha_B$. It is read off the same fb+ff tables the written continuum uses (`recomb_continuum_mod`, shared by both consumers for exactly this reason) with the free–free term removed, integrated from the hydrogen edge to $E_{\rm th}$(He II) where channel 3 takes over. Nothing is double counted: the tabulated range stops at the He II ground edge, so the $n = 1$ continuum is structurally absent from it. The integral is Simpson on a 512-point grid, and each panel is broken at the table's frequency nodes: the He II $n = 2$ edge (0.9999 Ryd) sits inside the first grid cell above the hydrogen edge, and a panel across that jump undercounted the free–bound by 0.87% at 9.5 kK falling to 0.26% at 37.5 kK (corrected 2026-09-25; the channel luminosity moves by <0.07%).

The exit branching, $f_{2s} = \alpha_{\rm eff}(2^2{\rm S})/\alpha_B$ and the Lyman-line shares, comes with *both* parts from one zero-density *nl*-resolved cascade (`hydrogenic_recombination_mod`, Section 4.20), in the case of the recombination lines, `par%lyman_lines`. Taking them from one calculation makes each share a closure rather than a difference of determinations, the error that put a 2.4% He I bias into the diffuse packet count when $\alpha_B$ was formed as $\alpha_A - \alpha_1$ across sources. In case B the only Lyman exit is Ly$\alpha$, $1 - f_{2s}$, with $f_{2s} = 0.241$ at 5 kK rising to 0.311 at 30 kK. In case A, when the Lyman lines escape, every $n$p $\to$ 1s is an exit: at $10^4$ K Ly$\alpha$ 0.706, Ly$\beta$ 0.110, Ly$\gamma$ 0.038, $n$ = 5–30 0.054 and $n \geq 31$ 0.0014 (put at the $n = 31$ energy, within 0.06 eV), with $f_{2s} = 0.091$; the shares sum to 1 to $2\times10^{-8}$. Each line is transported at its own energy, 40.8–54.4 eV, the line picked by its energy share so that the equal-energy packets carry the photon rate. Until 2026-09-26 the channel was case B whatever `par%lyman_lines` said, and $f_{2s}$ came from SH95's $\alpha_{\rm eff}(2^2{\rm S})$ at $n_e = 100$ on four anchors, 5000–30000 K, held at the end values outside them (24% low at 3000 K, 62% high at $6\times10^4$ K), over the SH95 alpha-tot; inside the anchors the two agree to 0.3% at $10^4$ K and 0.6% at 5000 K. The *rate* that branching multiplies is MoCHII's own $\alpha_B$(He III), the one the ionization balance uses, exactly as channel 4 applies its fractions to $\alpha_B$(He II). The one independent check available on the $2^2$P branch is MOCASSIN's He II Ly$\alpha$ emissivity fit, $10^{-0.792\log_{10}T_e+6.01}\times10^{-25}\,{\rm erg\,cm^3\,s^{-1}}$, which the case-B closure reproduces to 2.3–6.2% over 5–30 kK (`tests/heii_cascade/`).

Two approximations, with their range. He II Ly$\alpha$ is emitted at 40.813 eV and transported as a continuum photon; in reality it resonantly scatters off He II many times first, but since 40.8 eV cannot photoionize He II (54.4 eV) the scattering only diffuses it spatially before hydrogen or He I absorbs it, and the $He^{2+}$ zone is compact wherever it exists. This is what MOCASSIN does with its `HeIILyman` lines. That holds only while the continuum photon is itself absorbed near its origin; where the $He^{2+}$ zone is large and highly ionized it is not — on a $10^5$ K blackbody in the HII40 geometry hydrogen there gives $\tau \approx 0.01$–$0.04$ per pc at 40.8 eV across a 2 pc zone — and the transported Ly$\alpha$ leaves a zone that resonance scattering would hold it in (Section 4.9). The same holds for each higher Lyman line in case A. Second, in case B only Ly$\alpha$ is emitted: every cascade is routed through $n = 2$, so Ly$\beta$ and above are converted rather than emitted. Lines ending on $n \geq 2$ all fall below the 13.604 eV series limit and are correctly absent from the band.

Measured end to end on a $10^5$ K blackbody in the HII40 geometry, where $He^{2+}$ is abundant, switching the channel on raises the diffuse-field luminosity by 44% ($2.32 \to 3.33\times10^{38}$ erg s$^{-1}$) and $L$(H$\beta$) by 0.98%, leaves the He II lines within 0.11% in absolute terms, raises the He I lines by 0.2–0.3% (the 40.8 eV photons ionize $He^0$ as well), and drops [C I] 9827, 9853 by 9.0% as the extra ionizing photons reach the partly neutral edge. Since 2026-09-25 it is on whenever the continuum is transported and the band reaches 40.813 eV (the tests and results that predate that pin it off); the hard planetary-nebula and AGN spectra of Section 6. are where it matters. Both Lexington benchmarks have essentially no

$He^{2+}$ and are unaffected either way.

## 4.11 Solution-driven I-front re-refinement

At iteration `refine_iter` the octree is rebuilt from the current solution: an old leaf is flagged as front when $\epsilon < x_{\rm HI} < 1-\epsilon$ or a face neighbor differs by more than `refine_dx`; the new tree carries level `refine_lmax` wherever a cell overlaps a front-flagged leaf and `refine_lbase` elsewhere. The gas state and `rhokap` map over by position (each new leaf samples the old leaf containing its center), and every dependent array is recreated (the band opacity, tallies, metal $\Gamma$ blocks, diffuse tables, rate arrays — the last was the one initially missed, an allocate-once array that faulted when $n_{\rm leaf}$ grew). All ranks rebuild identically from the shared state, and the old grid's shared-memory windows are *recycled*: `destroy_shared_mem` (the collective counterpart of `create_shared_mem`) frees the 18 old windows right after the position sampling — the old tree's last reader — so repeated re-refinement neither leaks nor holds old and new grids in memory simultaneously.

## 4.12 Dust in the ionizing band

The three parts of the dust coupling, now complete:

- **Absorption competition.** With `par%ion_add_dust`, each band bin adds $\kappa_{\rm d}(b,\mathrm{leaf}) = \rho\kappa_{\rm leaf}\, s_{\rm abs}(b)$ where $s_{\rm abs}(b) = (1-a(E_b))C_{\rm ext}(E_b)/C_{\rm ext}(\lambda_{\rm ref})$. By default these come from the grain model named by `par%dust_model`, so that the power a cell absorbs is set by the same grains that reradiate it in Section 4.13. No model is extended: the astrodust and DL07 grids come from the random-orientation T-matrix table, which spans $1.0\times10^{-4}$–$3.981\times10^{4}\,\mu$m in 1762 wavelengths, i.e. 12.4 keV down to the far infrared, so below the Lyman limit the optics are the same $b/a = 1.4$ oblate spheroid on the DH21 dielectric function as above it (with the Rayleigh, $x < 0.1$, and geometric-optics, $x > 50$, limits wherever the size parameter leaves the T-matrix range, exactly as the rest of the table does) rather than a volume-equivalent-sphere approximation. `grain_model_mod` still asks for the grid down to $1.24/E_{\rm max}$, but SEDust prepends points only when that is *shorter* than the table's first wavelength: at $E_{\rm max}$ = 100 eV it is not ($0.0124\,\mu$m against $1.0\times10^{-4}\,\mu$m), and an extension can arise only above $E_{\rm max}$ = 12398.4 eV. There astrodust refuses to build one — that band's optics are the same spheroid, and this tree carries the Q table without the T-matrix solver — and the refusal is turned into a message naming `par%eion_max`; a clamped refractive index or a silent equal-volume sphere is never passed off as physics. The Zubko model needs no extension either: its own DustEM optics tables span $0.001$–$10^{4}\,\mu$m (1.24 keV down to $1.24\times10^{-4}$ eV), so the whole ionizing band is already tabulated, which is why `build_zubko` takes no `lam_min`. Whichever model is named, `gas_opacity_mod` verifies at setup that its grid spans the whole transported band — $1.24/E_{\rm max}$ to $1.24/E_{\rm min}$, the latter taken from `efuv_min` when `par%add_fuv` — together with $\lambda_{\rm ref}$, and stops the run if it does not (the interpolators clamp at the grid edges, so an unchecked short grid would silently freeze the EUV optics). Setting `par%ion_dust_kext` overrides the model with an explicit table covering the EUV (the WD01/D03 $R_V$=3.1 table, 1 Å–$10^{4}\,\mu$m: $C_{\rm abs}/{\rm H} = 1.83\times10^{-21}$ at 13.6 eV falling to $3.96\times10^{-22}\,{\rm cm}^2$ at 100 eV). A trap worth recording: the D03 table is *not monotonic* in wavelength (an out-of-order seam) — without sorting, the linear interpolation search silently returned $s_{\rm abs}(100\,{\rm eV}) = 0.903$ instead of 0.832 and shifted the dusty Strömgren radius by +1.6%.

- **Scattering.** With `par%ion_dust_scatter` the scattering coefficient joins the extinction ($s_{\rm sca}(b) = a\,C_{\rm ext}/C_{\rm ext,ref}$; in-band Henyey–Greenstein asymmetry $g(E_b) = 0.67$–$0.99$ from the table $\langle\cos\rangle$ column) and interactions are sampled: after the analytic zero-variance direct walk, a first interaction is forced along the ray with weight $w_1 = 1 - e^{-\tau_{\rm edge}}$, the packet keeps the albedo fraction $a_{\rm eff} = \kappa_{\rm sca,d}/\kappa_{\rm ext}$ at each interaction, scatters into an HG direction, and continues with unforced flights (whose segments tally into $J_b$) under Russian roulette ($w_{\rm min} = 10^{-4}$). Stellar and diffuse packets share the walk; gas ionization, gas heating, and grain heating all derive from the same tally, so the energy split is bookkept by construction.

- **Live ionization-dependent dust with a PAH split.** `dust_density_law='laursen09_live'` recomputes the dust density from the *computed* $x_{\rm HI}$ every iteration, $n_{\rm d} \propto (Z/Z_{\rm ref})\,n_{\rm H}\left[(1-f_{\rm PAH})(x_{\rm HI} + f_{\rm ion}x_{\rm HII}) + f_{\rm PAH}(x_{\rm HI} + f_{\rm ion,PAH}\,x_{\rm HII})\right]$, the PAH share carrying its own ionized-gas survival ($f_{\rm ion,PAH} = 0$ default: PAHs destroyed in ionized gas) — the generalization including its PAH-specific option.

- **Grain heating and the equilibrium temperature.** `Heat_dust(leaf)` $= \sum_b [4\pi J \Delta\nu]_b\,\kappa_{\rm abs,d}$ [erg s$^{-1}$ cm$^{-3}$] is consumed by `dust_temp_mod`: $f_{\rm emit}(T) = 4\pi \int s_{\rm abs}(\nu) B_\nu(T)\,d\nu$ is precomputed from the same kext table (log-$T$ grid, 3.2–3200 K) and $T_{\rm d}$(leaf) solves Heat_dust $= (\rho\kappa/d_{\rm cm}) f_{\rm emit}(T_{\rm d})$; outputs are the $T_{\rm d}$ leaf map and the grid-integrated modified-blackbody IR spectrum (1–1000 $\mu$m). FUV grain heating is transported by the `par%add_fuv` band extension (Section 2.1): gas cross sections vanish below their thresholds automatically, so no new transport code is involved, and the two-segment grid keeps the ionizing bins at full resolution. This is the single equilibrium *mixture* temperature; the size- and material-resolved treatment is the SEDust coupling below.

**Where the size-integrated extinction curve comes from.** The curve above is not integrated over grain size when the transport asks for it. SEDust's `dust_extinction` returns the *precomputed* size-integrated curve of a `SEDust/data/kext_*.dat` table, which the builder loaded, interpolated onto the model's own wavelength grid (log $\lambda$–log $C$ for the cross sections, linear in log $\lambda$ for $\langle\cos\rangle$, and the table value itself wherever a model wavelength coincides with a table wavelength). The size integral

$$C_{\rm ext}(\lambda) = \sum_{\rm pop}\sum_{a} {\rm d}n(a)\left[C_{\rm abs}(\lambda,a) + C_{\rm sca}(\lambda,a)\right], \qquad a_\lambda = \frac{C_{\rm sca}}{C_{\rm ext}}, \qquad \langle\cos\rangle = \frac{\sum {\rm d}n\,C_{\rm sca}\,g}{\sum {\rm d}n\,C_{\rm sca}} \tag{54}$$

is the calculation that *wrote* those tables; it remains reachable as SEDust's `size_integrated_extinction`. Which table a run serves is `par%sed_kext`: blank (the default) takes SEDust's own standard table for the named `par%dust_model`, so those file names are written down in one place; naming a file instead makes it mandatory, and the build stops with that file named if it cannot be read. The standard tables are `kext_astrodust_MW_euv.dat` (1762 wavelengths, $1.0\times10^{-4}$–$3.981\times10^{4}$ $\mu$m), `kext_dl07_MW_euv.dat` (1823, $6.2054\times10^{-5}$–$3.981\times10^{4}$ $\mu$m), and `kext_zubko_BARE_GR_S.dat` (1201, $10^{-3}$–$10^{4}$ $\mu$m).

Serving a table instead of redoing the integral is exact where the model grid and the table grid coincide and interpolated where they do not, and the gate `tests/grain_kext` measures both against `size_integrated_extinction` over the model as MoCHII builds it. Cross sections are compared relative to themselves (they set rates); the albedo $a$ and the product $a\langle\cos\rangle$ absolutely, because the transport uses them as the probability that an interaction scatters and the forward displacement it carries. The worst departure on each side of the Lyman limit 0.0912 $\mu$m, which the gate keeps as its reporting boundary:

| | $\lambda \geq 0.0912\,\mu$m | | $\lambda < 0.0912\,\mu$m | | band | |
|---|---|---|---|---|---|---|
| model as built | $C_{\rm ext}, C_{\rm abs}$ | $a$, $a\langle\cos\rangle$ | $C_{\rm ext}, C_{\rm abs}$ | $a$, $a\langle\cos\rangle$ | $s_{\rm abs}$ | $C_{\rm ext}(\lambda_{\rm ref})$ |
| astrodust, $E_{\rm max}$ = 100 eV | 4.91e–13 | 4.24e–13 | 4.84e–13 | 8.13e–13 | 1.95e–14 | 2.83e–14 |
| astrodust, $E_{\rm max}$ = 150 eV | 4.91e–13 | 4.24e–13 | 4.84e–13 | 8.13e–13 | 1.95e–14 | 2.83e–14 |
| astrodust, no `lam_min` | 4.91e–13 | 4.24e–13 | 4.84e–13 | 8.13e–13 | 1.95e–14 | 2.83e–14 |
| dl07, $E_{\rm max}$ = 100 eV | 4.65e–13 | 3.07e–13 | 3.21e–12 | 1.50e–12 | 2.95e–14 | 2.12e–14 |
| zubko | 4.91e–13 | 2.72e–13 | 4.83e–13 | 2.77e–13 | 2.89e–14 | 1.42e–14 |
| tolerance | 1e–10 | 1e–10 | 1e–9 | 1e–9 | 1e–4 | 1e–9 |

Nothing here is interpolation any more: the model grid *is* the table grid on both sides of the Lyman limit, so the served value is the table value and the only difference is the 13 significant digits the tables are written with. The three astrodust rows are identical because they are the same grid — asking for $1.24/E_{\rm max}$ at 100 eV, at 150 eV, or not asking at all all return the native 1762-point table. DL07 is the one case where the two grids are not the same object: its kext table is the finer of the two (1823 points from $6.2054\times10^{-5}\,\mu$m against the model's 1762 from $1.0\times10^{-4}\,\mu$m), but the model's nodes fall on table nodes to rounding, which is why its numbers sit at $10^{-12}$ rather than $10^{-13}$. The two integrals the transport is actually built on follow: the band-averaged $s_{\rm abs}$ over 13.598–100 eV agrees to $2\text{–}3\times10^{-14}$ and the normalization $C_{\rm ext}(\lambda_{\rm ref} = 0.55\,\mu\text{m})$ to $1\text{–}3\times10^{-14}$. Because no interpolation is left to bound, the tolerance below the Lyman limit was tightened from $3\times10^{-2}$ to $10^{-9}$.

What the table presumes is that it was computed from that model's *default* inputs. Changing `par%sed_sizedist`, `par%sed_qtable`, `par%sed_dl07_sdindex` or `par%sed_zubko_config` moves the grains that emit without moving the curve that extinguishes, and the two then no longer describe the same dust. That is exactly what the gate catches: a table belonging to other grains departs by $O(1)$ where the two are otherwise equal to 13 digits.

**What the longer Q table changed.** The astrodust table used to stop at the Lyman limit (169 radii × 1129 wavelengths), and everything below it was built at run time by extending the model with Mie theory for the volume-equivalent sphere. It now runs to $1.0\times10^{-4}\,\mu$m in 1762 wavelengths, so that region is the spheroid itself. Above $0.0912\,\mu$m the size-integrated curve is unchanged to rounding (worst $1.4\times10^{-10}$ on $C_{\rm ext}$); below it, it was recomputed. Over the band MoCHII transports, 13.598–100 eV (137 grid points), the new curve departs from the old by at most $1.010\times10^{-2}$ ($C_{\rm ext}$), $3.798\times10^{-2}$ ($C_{\rm abs}$) and $4.645\times10^{-2}$ ($C_{\rm sca}$), with medians $5.44\times10^{-3}$, $1.11\times10^{-2}$ and $9.48\times10^{-3}$; the band integrals move +0.459% ($C_{\rm ext}$) and +0.910% ($C_{\rm abs}$). Far larger differences, up to a factor 1300, sit at $1.0\text{–}1.1\times10^{-4}\,\mu$m — the 12.4 keV end of the table, outside anything MoCHII transports — and the two must not be quoted together. Building the longer grid costs 156.4 s (astrodust), about 73 s (DL07) and 0.55 s (Zubko) at `par%sed_NT` = 200, paid once per rank at startup.

## 4.13 Stochastic dust emission (SEDust coupling)

With `par%dust_sed`, `sedust_mod` (adapted from the MoCafe `dustemis_mod`) computes the grid-integrated dust SED with size-resolved stochastic heating and PAH bands, using the in-tree SEDust library (built from source under `SEDust/sed` by `build_lib.sh`; astrodust, DL07, and Zubko grain models). The coupling follows the MoCafe convention that makes the normalization exact by construction:

- **Shape from the local field.** For each leaf, the band tally $J_b$ is converted to $J_\lambda$ on the SEDust wavelength grid

— for astrodust and DL07 the native 1762-point Q-table grid, $1.0\times10^{-4}$–$3.981\times10^{4}$ $\mu$m, and for Zubko its own 1201-point DustEM grid, $10^{-3}$–$10^{4}$ $\mu$m. The whole transported band falls on the grid either way, with or without `par%ion_add_dust`, so the grains absorb it with their own cross sections at their own wavelengths and no part of $J$ has to be deposited into the shortest SEDust bin. (While the astrodust grid stopped at the Lyman limit, an emission-only run did exactly that: energy-conserving, but it underestimated the hardness of the stochastic temperature spikes.) SEDust then returns the emission spectrum of the grain population in this field: equilibrium large grains plus the full temperature-distribution solve for small grains and PAHs.

- **The single-photon bound is a property of the field.** The enthalpy window of the stochastic solve is capped by the hardest photon the *radiation field* carries, `hardest_photon_energy`$(\lambda, J_\lambda) = hc/\lambda_{\rm short}$ with $\lambda_{\rm short}$ the shortest wavelength at which $J_\lambda$ exceeds $10^{-12}$ of its peak, not by the short end $hc/\lambda(1)$ of the model's own grid. The two are not the same thing: a model grid is the grid of its optics tables and can reach far past the illuminated band — the Zubko/ZDA tables start at $10^{-3}$ $\mu$m (1.24 keV), 91 times harder than a field cut off at the Lyman limit, and the astrodust and DL07 Q-table grid now starts a further decade up, at $1.0\times10^{-4}$ $\mu$m (12.4 keV) — and measuring the grid there would raise the bound by that factor with no photon to justify it, coarsening the enthalpy bins (a fixed bin count spread over a correspondingly wider window). The threshold is deliberately one-sided: both refinement loops expand a window whose top bin is still populated, so an underestimate costs iterations and is recovered, while their contraction is guarded and capped.

- **One CMB temperature.** The net cooling of a grain is $\kappa B(T_{\rm d}) - \kappa_{\rm CMB}$, the second term being $C_{\rm abs}$ integrated against a CMB blackbody, and it is what keeps a grain from cooling below its surroundings. That temperature is now taken from `radfield::cmb_temperature()` — 2.725 K, the same value `J_Mathis` puts into the field the grains absorb — instead of a hard-coded 2.9 K in the cooling term alone. While the two differed, the solver held grains up against photons the field never supplied. Setting `use_mathis_corrected = .false.` recovers the literal Mathis (1983) 2.9 K together with that paper's 4000 K dilution factor, so the pair moves together.

- **Normalization from the transport.** The SEDust spectrum is used only as a shape: each leaf's emission is normalized to its absorbed power Heat_dust $\cdot V$ from the Monte Carlo tally (radiative equilibrium). H II regions are optically thin to the re-emitted IR — but that is a result, not an assumption: the emitted infrared is transported (Section 4.14) and what it deposits is returned to Heat_dust before this spectrum is written. The grid-integrated $L_\lambda$ is accumulated over leaves distributed round-robin (an `MPI_ALLREDUCE`), and $\int L_\lambda\, d\lambda = \sum$ Heat_dust $V$ holds by construction and is written into the `_dustsed.txt` header.

- **Cost.** The exact stochastic solve costs $\sim$0.1–0.2 s for each leaf with nonzero grain heating; it runs once, at output time, after convergence (the dusty Strömgren smoke test: $\sim$1.9 hours on 8 ranks for a $64^3$ grid — the solve, not the transport, dominates).

- **Smoke result.** Dusty Strömgren at $4\times10^{4}$ K: $\int L_\lambda d\lambda = 4.296\times10^{38}$ erg/s $= \sum$ Heat_dust $V$ to 4 digits; the SED shows the PAH bands at 3.3/6.2/7.7/8.6/11.3/12.7/17 $\mu$m over a warm 20–60 $\mu$m plateau with the FIR falloff — the canonical stochastically heated spectrum for a hot compact field.

- **Live PAH share** (`par%sed_pah_live`): `dust_emission`'s channel output (astrodust: 'AD' + 'PAH') lets each leaf's spectrum be assembled with the PAH channel weighted by the leaf's PAH survival ($x_{\rm HI} + f_{\rm ion,pah} x_{\rm HII}$), divided under `laursen09_live` by the dust survival already carried by `rhokap`; the Heat_dust $\cdot V$ normalization keeps energy closure (the PAH energy shifts to the large grains). Smoke: the PAH bands drop to 0.26 (7.7 $\mu$m) and 0.15 (11.3 $\mu$m) of the PAH-everywhere reference, the FIR rises by 1.06–1.36×, and the 8-$\mu$m map turns from centrally peaked into a *ring* at the ionization front — the classic Spitzer/JWST bubble morphology.

## 4.14 Transport of the reradiated infrared

With `par%dust_emis_transport` (on by default) the infrared the grains reradiate is not merely tallied: it is launched as packets and carried through the grid, and whatever the grains reabsorb is returned to Heat_dust, so a grain is heated by the nebula's own infrared as well as by the star. Because the optics come from the same grain model that produced the emission (Section 4.12), absorption and reemission refer to one population.

- **Launch.** A leaf is chosen in proportion to its emitted power Heat_dust$V$, the position uniformly inside it, the direction isotropically, and the wavelength from that leaf's own emission spectrum: the bin from the trapezoid cumulative and the wavelength *inside* the bin in closed form from the same piecewise-linear density, so the sampled wavelength is continuous and no bin center enters. Packets are allocated by stratified sampling of the cumulative leaf luminosity with one shared offset, so the counts telescope to exactly $N$ and the launched luminosity equals $\sum$Heat_dust$V$ identically. Each rank walks only its own leaves and reuses one spectrum buffer, so nothing of size leaf×wavelength is ever stored.

- **Transport.** The packets ride the same routine as the ionizing ones. No separate infrared opacity is needed: the photoionization cross sections vanish below their thresholds, so an infrared packet automatically sees grain absorption and scattering alone, evaluated at its exact wavelength. The emission is cut at the first wavelength that clears every gas threshold in the registry, and the estimators are switched by band, so no infrared segment can enter the ionizing-band tallies — without that switch the infrared would have been counted a second time into Heat_dust and the whole infrared field would have entered the Habing $G_0$ integral.

- **Self-consistency.** Passes repeat until the total emitted luminosity changes by less than `par%dust_tol`, at most `par%dust_niter` times. The reabsorbed power enters the next pass exactly, through Heat_dust; it does not enter the local mean intensity that sets the SEDust spectral *shape*, which remains the stellar one — an approximation valid while $\tau_{\rm IR} \ll 1$, whereas the emitted *power* carries no such error at any optical depth.

- **What it measures.** In an H II region the answer is that the nebula is indeed thin to its own infrared: a dusty Strömgren model at DGR = 1 reabsorbs 0.62–0.85% of the emitted infrared and converges in two passes, the second pass changing the emitted luminosity by 3–6$\times 10^{-5}$. The infrared energy budget closes to $\sim 10^{-16}$, and the extra luminosity of the second pass equals the power extinguished in the first. The escaping spectrum is written to `_dustesc.txt`; its ratio to the emitted spectrum is the infrared escape fraction.

## 4.15 Peel-off imaging

With `par%ion_peel`, one extra transport pass runs on the converged state with observer peeling (the MoCafe peeling-off convention, adapted to the band): every emitted packet — stellar and diffuse — peels a direct contribution $L_{\rm pkt} w\, e^{-\tau}/(4\pi r^2)$ toward each observer, and, when dust scattering is sampled, every interaction peels a scattered contribution with the Henyey–Greenstein phase function of the packet's bin (after the albedo weight, before the new direction is sampled). The optical depth to the box edge along the observer direction is integrated through `kap_ion` at the packet's bin by a tally-free walk (tallying the virtual peel ray would double count), so the images carry exactly the transport's extinction: gas, dust absorption, and dust scattering. Images are TAN projections on the MoCafe observer geometry (`obsx/y/z` or angle lists, rotation matrix, auto-sized pixel scale), written as surface brightness [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] split into direct/scattered and ionizing/FUV channels. The band tally is rebuilt by the imaging pass, so the written rates reflect the same field the images carry. With `par%peel_bins` the channel axis becomes the band bin — `direct_cube`/`scatt_cube` ($n_x, n_y, n_\nu$) — for hardness maps and synthetic EUV/FUV photometry (the band-integrated channel images are still written, derived from the cube; the gate shows the expected color: the FUV scattered halo shrinks from a 15.5-px to a 12.0-px half-light radius between 6.2 and 13.2 eV).

**External-field peel.** The direct peel of an external packet (Section 2.2) replaces the isotropic $1/(4\pi r^2)$ with its Lambert emission density about the entry-surface inward normal $\hat{n}_{\rm in}$, $p(\Omega) = (\hat{k}\cdot\hat{n}_{\rm in})/\pi$, so the contribution is $L_{\rm pkt}\, w\, \max(\hat{k}_{\rm obs}\cdot \hat{n}_{\rm in}, 0)\, e^{-\tau}/(\pi r^2)$: only the far-side surface (inward normal toward the observer) contributes, and the image is the background field seen through the medium — the cloud in silhouette. In the optically thin limit the image-integrated flux is $F = JA_{\rm proj}/d^2$, and with `save_direc0` the ratio `direc0/direct` $= e^{+\tau}$ is again the line-of-sight optical-depth map. The scattered peel is origin-agnostic and unchanged.
For the *dust* bands no transport is needed — the nebula is optically thin to its own IR — so `par%dust_emis_bands` instead stores SEDust band emissivities of each leaf (`_dustemis`, the same block layout as the `_emis` file) and the Python map maker projects them into flux-conserving dust-band images exactly like line maps.

## 4.16 Cartesian (car) grids

`par%grid_type = 'car'` builds a single-level Cartesian grid and stores its leaves in raster order: leaf = cell = $1 + i_x + n_x(i_y + n_y i_z)$. No tree topology, no neighbor table, and no geometry arrays are built. (`'uniform'` is accepted as a backward-compatible alias.) The leaf list comes either from a file (`par%amr_file`, single level) or — with `amr_file` empty — directly from the namelist: `par%nx/ny/nz` cells over the box $[-x_{\rm max}, x_{\rm max}]\times\ldots$ (cubic cells required).

**Uniform density and sphere/shell geometry.** A gas-density model is applied on the leaf centers before either grid (`'car'` or `'amr'`) is built: `par%nH_const` overrides every leaf's $n_{\rm H}$ with one uniform value, and `par%rmax/rmin` zero $n_{\rm H}$ outside a sphere of radius $r_{\rm max}$ / inside $r_{\rm min}$ (a shell). Because the cut is on the leaf centers, a sharp sphere or shell edge can be resolved by the octree (`'amr'` with a refined leaf list) while the density stays exactly uniform — the case that motivates uniform density on `'amr'` as well as `'car'`.

**Analytic geometry.** All geometry queries go through accessor functions in `octree_mod` (`cell_cx/cy/cz/ch`, `leaf_cell`, `leaf_cx/cy/cz/half`). On the octree these are plain array reads; on the car grid they decode the raster index and evaluate $c_x = x_{\min} + (i_x + \frac{1}{2})\,\Delta$ — the same floating-point expression that previously filled the arrays, so nothing changes numerically. The car build therefore allocates *none* of the tree arrays: not only `children` (8×), `neighbor` (6×), and `parent`, but also the cell centers, half-widths, and the leaf↔cell maps. At $512^3$ this removes the entire ∼10 GB of shared tree memory; the grid costs only its physical fields. All physics modules (thermal, ionization, diffuse, lines, dust) read geometry exclusively through the accessors, so they run unchanged on either grid.

**Traversal (`par%car_walk`).** The car grid is not a physics choice — both walks discretize the same transport integral and agree to rounding — so `par%car_walk` selects only *how* the ray is stepped:

- `'dda'` (default): dedicated incremental Amanatides–Woo walks, the natural car traversal. Each ray precomputes, once, the parametric distance to the next face crossing per axis ($t_{\max,x/y/z}$) and the crossing period ($t_{\Delta,x/y/z} = \Delta/|k_{x/y/z}|$); the inner loop advances by comparisons and additions alone — no exit-face solve at each step, no leaf re-lookup. The tally expressions (analytic first-flight $w(e^{-\tau_{\rm in}} - e^{-\tau_{\rm out}})/\kappa$, pathlength sums, $\tau$-target stop) are identical to the `'amr'` walks. It is the fastest mode: ×1.64 over the `'amr'` walk, ×2.0 over the octree (total Strömgren run time 13.0 → 6.5 minutes).
- `'amr'` (verification): the octree band walks (`raytrace_ion_to_edge_amr`, `_to_tau_amr`, `_tau_only_amr`) run unchanged; only their two geometry primitives branch on the grid mode — the leaf finder `amr_find_leaf` becomes three integer divisions (`car_index`) and the face-neighbor step `amr_next_leaf` becomes pure integer arithmetic on the raster index (`car_next`). The exit face and distance are still recomputed each step from the cell center and half-width (`amr_cell_exit`), exactly as on the octree. Because the floating-point order along each ray is then unchanged, the same-seed Strömgren run is *bit-identical* to the single-level-octree run ($\max|\Delta x_{\rm HI}| = 0$) — the cross-check the DDA walk cannot provide (it differs at rounding by design, $\max|\Delta x_{\rm HI}| = 2.7 \times 10^{-15}$, $V_{\rm ion}$ identical). This certifies the raster backend rather than serving production (it is ×1.25 over the octree, slower than DDA).

Everything downstream is untouched (the gas modules only consume leaf-indexed arrays) and `refine_front` is octree-only.

## 4.17 Plane-parallel slab illumination

An *xy-periodic slab* models a medium that is horizontally infinite: a packet that leaves an $x$ or $y$ face re-enters at the opposite face, while the $z$ faces escape. The medium may vary only in $z$ (a 1D column, `nx = ny = 1`) or be an arbitrary 3D tile (`nx, ny` $> 1$) repeated periodically. The wrap is source-agnostic — it is a boundary condition on the transport, so an internal source under `xy_periodic` is a horizontally periodic array of sources, and the external slab illumination below rides on the same wrap. In the octree walks the wrap re-enters the packet at the opposite face's exact boundary coordinate (not by $x \pm x_{\rm range}$, which can land a rounding-epsilon outside the box and have the leaf lookup drop the packet); the incremental DDA wraps the integer index (`modulo`), the parametric $t$-coordinates being translation-invariant on a uniform grid.

External illumination (`source_geometry = 'slab'`) enters the top ($+z$) and/or bottom ($-z$) face, each as a collimated beam at incidence angle $\theta$ or an isotropic (Lambert) field. The physical normalization is the flux crossing the horizontal face,

$$F_z = \int_{\text{hemisphere}} I_{\text{in}} \cos\theta \, \mathrm{d}\Omega = \begin{cases} \mu F_n, & \text{beam } (\mu = \cos\theta,\ F_n \text{ normal to the beam}) \\ \pi I, & \text{isotropic } (I \text{ the specific intensity}), \end{cases} \tag{55}$$

so the user inputs the source strength ($F_n$ or $I$) and the geometric factor ($\mu$ or $\pi$) gives $F_z$. The entering luminosity of one periodic tile is $L = F_z A_{\text{zface}}$ (the packet weight is $L/N_{\text{phot}}$); $A_{\text{zface}}$ cancels in every result expressed per unit volume, which therefore scales linearly with $F_z$ and is independent of the tile area and lateral resolution. The two faces compose additively (both lit $\Rightarrow L = (F_{z,\text{top}} + F_{z,\text{bot}})A$).

The emergent observable is the boundary intensity: a packet escaping a $z$-face carries its surviving luminosity into a ($\mu = |k_z|$, face) histogram, written as $I(\mu)$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] with $F = 2\pi \int I \mu \, \mathrm{d}\mu$, together with the reflected/transmitted/absorbed energy budget. Both the direct beam (the analytic edge walk) and, with dust scattering, the scattered/diffuse flights contribute, so the budget closes to unity. Point-observer peel-off is disabled under `xy_periodic` (the periodic images admit no finite-distance observer). The quasi-random (`sobol`) launch covers the slab in both modes (a six-dimensional layout: frequency, face, the two in-face coordinates, and the Lambert $\mu, \phi$).

## 4.18 PDR-zone physics

Three switches extend the FUV zone beyond ion structure; the first two default on since 2026-07-31 (§4.7), `grain_pe` off:

- `par%metal_ne`: electrons from the metal cascade join the $n_e$ closure of the cell solve, $n_e = n_{\text{H}}[x_{\text{HII}} + y_{\text{He}}(\ldots)] + \sum_{\text{el}} A_{\text{el}} n_{\text{H}} \sum_i (i-1) f_i$ — a $\lesssim 10^{-3}$ correction inside the H II region but the *only* electrons beyond the front ($n_e \simeq n_{\text{CII}} \approx A_{\text{C}} n_{\text{H}}$ in the PDR zone), where they control the [C II]/Mg II excitation and the photoelectric efficiency.

- `par%metal_heat`: metal photoheating enters the thermal balance; the heating integrals $\mathcal{H}_i = \int 4\pi J_\nu \sum_s \sigma_s (1 - E_{\text{th},s}/h\nu) \, d\nu$ (the subshell sum of §3.1, one term below the first inner edge) are accumulated as path estimators at each packet's own energy, together with the metal $\Gamma$'s.

- `par%grain_pe`: grain photoelectric heating and grain recombination cooling with the Bakes & Tielens (1994) fits, $\Gamma_{\text{pe}} = 1.3\times10^{-24}\, \epsilon\, G_0\, n_{\text{H}}$ with $\epsilon = 4.87\times10^{-2}/[1 + 4\times10^{-3} x^{0.73}] + 3.65\times10^{-2}(T/10^4)^{0.7}/[1 + 2\times10^{-4} x]$, $x = G_0\sqrt{T}/n_e$, and $\Lambda_{\text{rec}} = 4.65\times10^{-30} T^{0.94} x^{\beta} n_e n_{\text{H}}$, $\beta = 0.74/T^{0.068}$, both scaled by `pe_scale` times the leaf's dust amount relative to Milky-Way dust, measured from the leaf opacity itself: $(\rho\kappa/d_{\text{cm}})/(n_{\text{H}} C_{\text{ext,MW}}(V))$ with $C_{\text{ext,MW}}(V) = 4.868\times10^{-22}\,\text{cm}^2/\text{H}$ — robust against the DGR-versus-`cext_dust` input conventions (the first gate run had DGR = 1 with the MW extinction folded into `cext_dust`; a naive DGR$/0.01$ scale made the heating $100\times$ too strong and drove the zone to a false Ly$\alpha$-capped 12 kK equilibrium). $G_0$ is the local FUV field in Habing units summed from the FUV bins of the tally (`add_fuv` required). Three further guards, each found by the gate: (i) the BT94 fits are evaluated at $\min(T, 10^4\,\text{K})$ — outside their fit range the $T^{0.7}$ term of $\epsilon$ diverges; (ii) both grain terms carry an $x_{\text{HI}}$ weight — the BT94 charging parameter knows only the FUV field, so in ionized gas (EUV-charged grains, grain energy already routed to the IR via `Heat_dust`) it would overheat badly; (iii) the zone

needs a neutral-gas coolant, which is the atomic-hydrogen impact term below — the first gate run without it had no coolant between the fine-structure and Ly$\alpha$ regimes, and the $\epsilon(x)$–$n_e$ feedback (collisional ionization raises $n_e$, neutralizing the grains and raising $\epsilon$ fifty-fold) drove it to a false $\sim 10^4$ K equilibrium.

**Atomic-hydrogen impact on the fine-structure lines.** With $n_{\rm HI}/n_e \sim 10^3$–$10^4$ in neutral gas, hydrogen rather than electrons excites the ground-term fine structure of C I, C II and O I, and those lines are the coolant. This channel is not a switch and is not part of `grain_pe`: it is in the `par%use_metals` cooling whatever heats the gas, as the H-impact term before it was. Since 2026-09-24 it is carried by the $n$-level atoms themselves. `nlevel_load` reads, beside `nlevel_<el>_<stage>.txt`, the file `hcoll_<el>_<stage>.txt` when one exists, and given $n_{\rm HI}$ the rate matrix of the statistical equilibrium becomes

$$R = n_e\, q^e + n_{\rm HI}\, q^{\rm H} + A, \tag{56}$$

one solve in which the two colliders saturate the populations together. The rates, written by `tools/fitting/make_hcoll_fine_structure.py`, are:

- C I $^3P_{0,1,2}$ (`hcoll_c_1.txt`) and O I $^3P_{2,1,0}$ (`hcoll_o_1.txt`): Abrahamsson, Krems & Dalgarno (2007, ApJ 654, 1171) Eq. 1, $\ln(k_{\rm exc}/10^{-11}\,{\rm cm^3\,s^{-1}}) = \sum_{i=0}^{N} a_i\, T^{-ib}$, Table 2 (C + H, 5–1000 K) and Table 4 (O + H, 30–1000 K), all three transitions of each term. The negative signs lost from the PDF text layer were restored from the printed glyphs, and the generator checks itself against the paper's Tables 1 and 3 to $\leq 1.0\%$; the one larger deviation, O I $k_1$ at 50 K (+9.1%), is the paper's own fit at the end of its range.
- C II $^2P_{1/2}$–$^2P_{3/2}$ (`hcoll_c_2.txt`): the de-excitation rate $7.6\times10^{-10}(T/100)^{0.14}\,{\rm cm^3\,s^{-1}}$ of Barinovs et al. (2005, ApJ 620, 537).

The file row is `lo hi form Tmin Tmax c_1..c_10`: form 1 is the Abrahamsson excitation fit ($c_1 = b$, $c_2, \ldots = a_i$), form 2 the de-excitation power law. The de-excitation rate is evaluated at $T$ held inside $[T_{\rm min}, T_{\rm max}]$ — the range of the calculation, no extrapolation — and the excitation rate follows by detailed balance at the actual $T$ with the level energies and statistical weights of the atom file.

Both consumers read the same atom. The emitted lines (`lines_mod`, `nlevel_emissivities(atom, T, ne, emis, nHI)`) now include hydrogen impact, and the thermal balance adds, for every stage with hydrogen rates, $n_{\rm ion}$ times `nlevel_hcoll_cooling` $= P(e{+}{\rm H}) - P(e)$ — the radiated power of the atom with both colliders minus that with electrons alone — to the electron share taken from the $(T, n_e)$ table (`species_mod::metal_cooling_H`). So every ground-term transition is included ([O I] 63/145 $\mu$m, [C I] 609/370 $\mu$m, [C II] 158 $\mu$m), and the cooling and the emitted line are one physics.

The term it replaces was physically wrong in four ways. Only [C II] 158 and [O I] 63 $\mu$m had it, as a separate two-level formula; C I had no hydrogen impact at all. The [O I] coefficient $9.2\times10^{-11}(T/100)^{0.67}$ carried no source and is 0.20–0.46 of Abrahamsson's over 50–1000 K. The line output was electron impact only, so the same line was cooled by one physics and emitted by another. And the electron and hydrogen saturation denominators were separate. Against it, in the $n_e \to 0$ limit the earlier O I hydrogen cooling is 0.22–0.35 of the new at $n_{\rm HI} = 10^2$ (100–1000 K) and 0.65–1.02 at saturation ($n_{\rm HI} = 10^6$); C II moves by only 1.001–1.014, the difference being the $A$-value (the fixed $2.321\times10^{-6}$ against the atom file's CHIANTI $2.29\times10^{-6}\,{\rm s^{-1}}$). There is no switch because the channel was always on; this is the

same channel carried correctly. Gate `tests/hcoll/` (`check_hcoll.f90`, `run_check.sh`): at $n_e \to 0$, $n_{\rm HI} = 10^{-4}$ the line photon rates reproduce the paper tables (worst $3.0\times10^{-3}$); at $n_{\rm HI} = 10^{14}$ the populations are Boltzmann, including 5000 K above the 1000 K fit edge (worst $1.1\times10^{-9}$); and the thermal increment equals the emissivity difference ($4.4\times10^{-16}$). The earlier gate `tests/two_level/`, which checked the deleted two-level formula, is removed. Measured on `tests/pdr/pdr_L5.in` (pre-change binary against the current one, every item of the 2026-09-24 update included): the neutral shell ($x_{\rm HI} > 0.9$) cools from a median 259 K to 200 K (the volume mean from 550 to 485 K), with the shell-averaged profile running from ∼270 K behind the front to ∼170 K at the outer edge (was ∼400 to 208 K); the line list, which now carries the hydrogen collisions, gives [O I] 63 $\mu$m ×39.5, 145 $\mu$m ×27.5, [C II] 158 $\mu$m ×1.98 and [C I] 609/370 $\mu$m ×1.38/×1.09, while the ionized interior moves by −22 K in $\langle T[N_pN_e]\rangle$.

Scope note: this is the atomic PDR skin — no $H_2$/CO chemistry, no line transfer, no dust–gas collisional coupling — so PDR temperatures are first-order estimates, adequate for the ion structure and rough [C II] emissivities.

## 4.19 *n*-level solver and line luminosities

`nlevel_mod` evaluates exactly the recipe of Section 3.10 (Clenshaw evaluation of the Chebyshev series) and solves the statistical equilibrium $\dot{n} = 0$, $\sum_i n_i = 1$ (partial-pivot elimination; de-excitation $q_{ul} = 8.629\times10^{-6}\,\Upsilon/(g_u\sqrt{T})$, excitation by detailed balance). `lines_mod` integrates $L_{\rm line} = \sum_{\rm leaf} j\, n_{\rm ion} V$ on the converged state and normalizes to the SH95 H$\beta$. The Fortran chain was verified against the Python reference solver on an identical state: 26 line luminosities agree to four significant digits. With `par%emis_output` the same solve also writes the emissivities leaf by leaf (`_emis`: an `emis_<el>_<stage>` block for each ion and `emis_h` for the SH95 lines, [erg s$^{-1}$ cm$^{-3}$] with $\sum \mathrm{emis}\cdot V = L_{\rm line}$, verified to $4\times10^{-4}$%), together with `n_H`, $n_e$, $T_e$, the H/He fractions, and the metal stage fractions — a self-contained input for line maps and position–velocity work.

## 4.20 The two-photon continuum: emission, cooling, transport

The metastable $2s$ ($2^1$S in He I) levels decay by simultaneous emission of two photons sharing the transition energy. That one channel enters the code in three separate places, and the three treatments are deliberately *not* the same; they are collected here because the differences are easy to mistake for inconsistencies.

**Emission** (`nebcont_mod`, Section 3.11). The emission coefficient is written per $n_e n_{\rm ion}$ and is fed by the *recombination* cascade, one term for each of the three ions:

| ion | weight | $\nu_0$ [Ryd] | rate into the emitting level |
|---|---|---|---|
| H I | $n_e n_{\rm HII}$ | 0.7496 | $\alpha_B(\mathrm{H\,II})\, f_{2s}(1, T, \mathrm{case})$ |
| He I | $n_e n_{\rm HeII}$ | 1.514 | $\alpha_B(\mathrm{He\,II})\left[f_{1S} + (1-p_{584})\, f_{1P}\right]$ (+ $2^3$S transfers) |
| He II | $n_e n_{\rm HeIII}$ | 3.00 | $\alpha_B(\mathrm{He\,III})\, f_{2s}(2, T, \mathrm{case})$ |

with the Nussbaumer & Schmutz (1984) shape $A(y)$, $y = \nu/\nu_0$, for H I and He II (`hydrogenic_twophoton_mod`), and the Drake, Victor & Dalgarno (1969) tabulated shape, `data/HeI2phot.dat`, for He I (`hei_twophoton_mod`).

For the two hydrogenic ions the rate is MoCHII's own $\alpha_B$, the one the ionization balance uses, times the share of the

captures into $n \geq 2$ that end in $2s$,

$$f_{2s}(Z, T, \text{case}) = \frac{\alpha_{\text{eff}}(2s, \text{case})}{\alpha_B}, \tag{57}$$

numerator and denominator from *one nl*-resolved radiative cascade in the zero-density limit (`hydrogenic_recombination_mod`, whose data `src/hydrogenic_recombination_data.inc` are written by `tools/fitting/make_hydrogenic_recombination.py`; the same cascade gives $\alpha_B$ of both ions, Section 3.2). The cascade takes the hydrogenic A-values and photoionization cross sections of Storey & Hummer (1991, the CPC routines `ACOFZ1` and `PIXZ1`, unmodified), forms $\alpha_{nl}(T)$ by the Milne relation, and passes each level's effective coefficient down in proportion to its A-values from $n = 499$, the largest $n$ the routines allow, with the captures above it added as a power-law tail (0.026% of $\alpha_B$ at $10^4$ K, 1.2% at 30 K, the low end of the table, $\log T = 1.5$–$7.0$). Case A keeps every decay; case B deletes $np \to 1s$. The case is that of the recombination lines (`par%lyman_lines`: case A when the Lyman lines escape), so a run writes Balmer lines and their two-photon partner in the same case. The fraction is relative to $\alpha_B$ in both cases, since a capture into $1s$ never feeds $2s$. Checks: the case-B H I $\alpha_{\text{eff}}(2s)$ is within 0.3% of Osterbrock & Ferland (2006) Table 4.11 over $10^4$–$2\times10^4$ K; the same cascade's $\alpha_{\text{eff}}(\text{H}\beta)$ reproduces the SH95 case-A emissivity at $n_e = 100$ to 0.06% over 7500–30000 K (the case-A branching, tested directly) and case B to 0.04–1.0% over 30000–3000 K; the He II case-B $\alpha_{\text{eff}}(2^2\text{S})$ is 0.92% (5000 K) to 0.11% (3e4 K) below SH95's at $n_e = 100$, the collisional $l$-mixing SH95 include and this limit leaves out. O&F's 5000 K entry, $1.38\times10^{-13}$, is 4% above the cascade's $1.326\times10^{-13}$ — and above SH95's He II value at $2\times10^4$ K taken to H I by hydrogenic scaling, $\alpha_{\text{H}}(T) = \alpha_{\text{HeII}}(4T)/2 = 1.324\times10^{-13}$. At $10^4$ K, $f_{2s} = 0.3234$ (B) and 0.1231 (A) for H I, 0.2661 and 0.0909 for He II; case A sends 62% fewer recombinations through H I $2s$, since a $np$ level that may decay to $1s$ mostly does. The same cascade gives the exit shares of the He III cascade in the diffuse field, $f_{2s}$ and the Lyman lines, in either case (Section 4.10.1).

Before 2026-09-26 the H I term was the case-B Pengelly (1964) power law $0.8368\times10^{-13}(T_4)^{-0.723}$ whatever the lines did, so a case-A run wrote a two-photon continuum 2.6 times its own case; and the He II term and the channel-5 branching were SH95's $\alpha_{\text{eff}}(2^2\text{S})$ at $n_e = 100$ on four anchors, 5000–30000 K, held at the end values outside them (24% low at 3000 K, 22% high at $4\times10^4$ K and 62% at $6\times10^4$ K), case B only. In case B the written H I term, now the cascade's own $\alpha_{\text{eff}}(2s)$ since $\alpha_B$ is the cascade's too, moves from the power law by –0.16% at $10^4$ K and –4.0% at 5000 K.

Each shape is normalized on *its own* integral, so that $\int_0^1 n(y)\,\text{d}y = 2$ exactly: a decay emits a pair, and the pair count is a property of the transition rather than a constant supplied alongside the shape. The port instead divided each shape by a separately quoted $A_{2q}$ — $8.2249\,\text{s}^{-1}$ for H I, $8.226\times64$ for He II, 51.3 for He I — which reproduces two photons per decay only to –0.014%, +0.05% and +0.28% respectively. The He I case was the one that mattered, because `hei_twophoton_mod` had always normalized the *same* table on its own integral for the transported packets: the tabulation carries $A = 50.93\,\text{s}^{-1}$, not 51.3, so the written continuum radiated 1.9945 photons per decay while the packets transported 2. The value the table carries is confirmed externally — CHIANTI's `heseq_2photon.dat` is the same Drake et al. distribution divided by $A = 50.94\,\text{s}^{-1}$, and MoCHII's table reproduces it times that $A$ to the rounding of the published three decimals. Comparing the two shape determinations directly, MoCHII's H-sequence fit and CHIANTI's Goldman & Drake (1981) $\psi$ tables agree to 0.42% at worst (at $y = 0.98$, where the tabulation ends) and to 0.2% over the bulk, with identical $\langle y \rangle = 0.5$; CHIANTI's own $Z = 1$ and $Z = 2$ tables differ by less than 0.05%, which is what licenses using one shape for both ions.

The He I entry has a history worth recording. The port took MOCASSIN's $6.23\times10^{-14}(T_4)^{-0.827}$, which is the recombination coefficient to *all* singlets — the exact sum $2.06\times10^{-14}+4.17\times10^{-14}$ of the $2^1$S and $2^1$P coefficients of Eq. (49), refitted as a single power law, so the exponent −0.827 is a fit to the sum and not a sum of exponents. Using it is MOCASSIN's stated assumption that every He I singlet ends in $2^1$S, i.e. the $p_{584}\to0$ limit of Eq. (51), and it disagreed with the $p_{584}\to1$ limit `diffuse_mod` was transporting. That constant is no longer used. The emitting $2^1$S population is the one the cascade itself feeds (`hei_recombination_photons_mod::hei_twophoton_decays`), per unit volume

$$r_{2q}=n_e\,n(\mathrm{He^+})\,\alpha_B(\mathrm{He\,II})\left[f_{1S}+(1-p_{584})\,f_{1P}\right], \tag{58}$$

with the branch fractions and the local $p_{584}$ of the transported channel and the on-the-spot sources, and, with the $2^3$S population coupled, the collisional transfers into the singlets added (Section 3.13), so the written continuum, the transported field and the on-the-spot sources describe one nebula. Until 2026-09-26 it took $2.06\times10^{-14}(T_4)^{-0.451}+[1-p_{584}]\,4.17\times10^{-14}(T_4)^{-0.695}$ in absolute units, while the cascade normalizes the same fits by their sum and scales by $\alpha_B$: the two differed by that sum, $0.91\,\alpha_B$ at 8000 K and 0.985 at $10^4$ K, and the coupled transfers were missing. On a 1e5 K blackbody in the HII40 geometry the He two-photon continuum near 1 Ryd moved by +5.3%, the total two-photon continuum by +0.45%. Only H I carries a collisional suppression. A $2s$ atom decays by two photons at $A_{2s}$ or is moved to $2p$, and so to Ly$\alpha$, by a passing charged particle; $2s$ and $2p$ are separated only by the Lamb shift ($0.0353\,\mathrm{cm^{-1}}$ to $2p_{1/2}$) and the fine structure ($0.330\,\mathrm{cm^{-1}}$ to $2p_{3/2}$). With `par%twophoton_quenching = 'colliders'` (the default) every partner enters with its own rate,

$$\gamma_{2q}\to\gamma_{2q}\,\frac{A_{2s}}{A_{2s}+n_eq_\mathrm{e}+n_\mathrm{p}q_\mathrm{p}+n(\mathrm{He^+})\,q_{\mathrm{He^+}}+n(\mathrm{He^{2+}})\,q_{\mathrm{He^{2+}}}},\qquad A_{2s}=8.2246\ \mathrm{s^{-1}}, \tag{59}$$

$A_{2s}$ from Drake (1986) eq. 27 for $Z=1$ with the proton reduced-mass factor. The rates follow Pengelly & Seaton (1964): a particle of charge $Z$ at impact parameter $R$ drives the transition with probability $P=4Z^2e^2S/(3\hbar^2R^2v^2)$, $P=1/2$ inside $R_1$, and the integral is cut at the smallest of $1.12\hbar v/\Delta E$, $0.72\,v\tau$ and the Debye radius, giving $Q=\pi R_1^2[\frac{1}{2}+\ln(R_c/R_1)]$ (their eqs. 34–37, 42). `tools/fitting/make_h2s_collisional_mixing.py` integrates that $Q$ over the Maxwellian of the relative velocity for each final level ($2p_{1/2}$ with 1/3 of the line strength, $2p_{3/2}$ with 2/3) and each collider, with the limit $Q=\pi R_c^2/2$ where $R_c<R_1$ that their closed form (eq. 43) does not carry; the $\Delta E$ cutoff is the smallest wherever $n_e\lesssim2\times10^{12}\,\mathrm{cm^{-3}}$, so the table is the low-density limit. Osterbrock & Ferland (2006) Table 4.10, which "comes from" Pengelly & Seaton, prints the proton and electron rates for both final levels at $10^4$ and $2\times10^4$ K; the quadrature reproduces their level split and temperature dependence but lies above them by 13% ($10^4$ K) and 9.5% ($2\times10^4$ K) for protons and 4.7% and 2.8% for electrons, so the printed values are taken and the quadrature supplies the temperature dependence outside that interval and the $He^+$ and $He^{2+}$ rates, which the table does not give, as their ratio to the proton rate. At $10^4$ K: $q_\mathrm{e}=5.70\times10^{-5}$, $q_\mathrm{p}=4.74\times10^{-4}$, $q_{\mathrm{He^+}}=5.08\times10^{-4}$, $q_{\mathrm{He^{2+}}}=1.52\times10^{-3}\,\mathrm{cm^3\,s^{-1}}$ (`data/atomic/h_2s_2p_mixing.txt`). Protons do 8.3 times what electrons do.

`'lumped_ne'` keeps the form carried over from MOCASSIN, $1+n_eq/8.23$ with $q=5.31\times10^{-4}$ ($T\le10^4$ K) to $4.71\times10^{-4}$ ($T\ge2\times10^4$ K). Those two numbers are the Table 4.10 proton-plus-electron sums, multiplied by $n_e$ alone: where hydrogen is ionized and helium neutral that is nearly right ($n_\mathrm{p}=n_e$), where $He^+$ also collides it is right to 0.6–0.8% by accident (He/H = 0.1 all $He^+$, $10^4$–$2\times10^4$ K), and where the electrons come from metals ($n_\mathrm{p}\ll n_e$) it over-suppresses. It is retained to reproduce the earlier gates. On the converged HII20 state the default moves the two-photon continuum by −0.043%. At the $3000\,\mathrm{cm^{-3}}$ of the PN150 standard the continuum surviving is 0.84.

A factor-six disagreement once recorded here against COLT is a defect of that implementation: COLT takes the Pengelly & Seaton cutoff $1.12\hbar\nu/\Delta E$ with `nu_12_HI = c * 82258.9544` (`source/nebcont.cc`:295), the Ly$\alpha$ frequency in place of the $2s$–$2p$ separation, so the logarithm of that cutoff goes negative, `minpos` (`:301`) drops it and the Debye cutoff alone is left ($q_e + q_p = 3.24\times10^{-3}$ at $10^4$ K, $n_e = 100$). With the physical separations its expression gives $5.79\times10^{-4}$.
No collisional factor is applied to the helium two-photon terms: $A_{2q}$ is $50.9\,s^{-1}$ for He I $2^1S$ and $526\,s^{-1}$ for He II $2s$. For He I the critical density is some four orders of magnitude above hydrogen's: the fastest collisional exit of $2^1S$ is to $2^3S$, $2.1\times10^{-7}$ cm$^3$ s$^{-1}$ at $10^4$ K by detailed balance of the $q_{31a}$ row of `hei_metastable.txt`, so $n_{\rm crit} \approx 2\times10^8$ cm$^{-3}$. For He II $2s$ the $\sim 10^6$ quoted in `nebcont_mod`'s header is not rechecked here. The three terms are summed into the total $L_\nu$ of `_nebcont.txt` and also written separately as its last column. Because the population is the recombination cascade alone, collisional excitation of $2s$ contributes nothing to the emitted continuum, which is therefore a lower bound in the partly neutral gas of an ionization front.
**Cooling** (`cooling_mod`, Section 3.9). The H I line-cooling fit is the $n_e \to 0$ statistical-equilibrium limit of the 25-level CHIANTI atom, and level 2 of that atom *is* $2s$ ($82258.956\,cm^{-1}$, $A = 8.229\,s^{-1}$, the two-photon rate) with the $1s \to 2s$ collision strength present. Collisional excitation of $2s$ therefore cools with its full 10.2 eV, and the leading term of the fit, $A_1 = 4.4262\times10^{-17}$ at $T_1 = 1.1844\times10^5$ K, is exactly that excitation temperature.
The cooling must *not* carry the density quenching the continuum does, and does not: proton and electron $l$-mixing converts $2s$ to $2p$ and the two-photon continuum into Ly$\alpha$, but the gas loses the same 10.2 eV on either route. (CHIANTI `h_1` tabulates no $2s \leftrightarrow 2p$ collision strength, so the equilibrium solve leaves $2s$ unquenched — harmless for the same reason. Nor does the level produce a spurious line: `lines_mod` extracts only the H$\alpha$/H$\beta$/H$\gamma$ windows of the collisional component, so nothing is double counted against the continuum.) Helium has no collisional-excitation line cooling at all — `cooling_total` carries recombination, free-free, collisional ionization and the H I line term only — so collisional two-photon cooling of He I $2^1S$ and He II $2s$ is absent, as it is in MOCASSIN, TORUS, Wood et al., and CMACIONIZE.
**Transport** (`diffuse_mod`, Section 4.10). Only helium two-photon photons can ionize anything: the H I continuum stops at 10.2 eV, so omitting it costs nothing. The He I $2^1S$ branch is one of the diffuse channels and carries its H-ionizing fraction 0.55644 from the Drake table (Section 4.10; the flat surrogate's 0.56 is no longer used). The He II pair, which spans 40.813 eV and so ionizes both H and He I, is the two-photon term of the fifth channel, `par%heii_diffuse` (Section 4.10.1). It is not transported alone: it is one of three paths a He III case-B recombination takes, and the smallest of the three that matters.

# 5. Validation gates and results

## 5.1 Rate integrals vs. analytic attenuation

Uniform sphere ($n_{\rm H} = 1$, $R = 1$ pc, $x_{\rm HI} = 0.1$, $y_{\rm He} = 0.1$), central $10^5$ K source, 16 bins, $10^8$ packets. Against the cell-volume-averaged analytic attenuation with identical bins: $\Gamma_{\rm HI}$ mean bias +0.004%, median 0.38% (pure Monte Carlo noise), max 2.6%; $\Gamma_{\rm HeI}$ bias +0.011% (Figure 2).

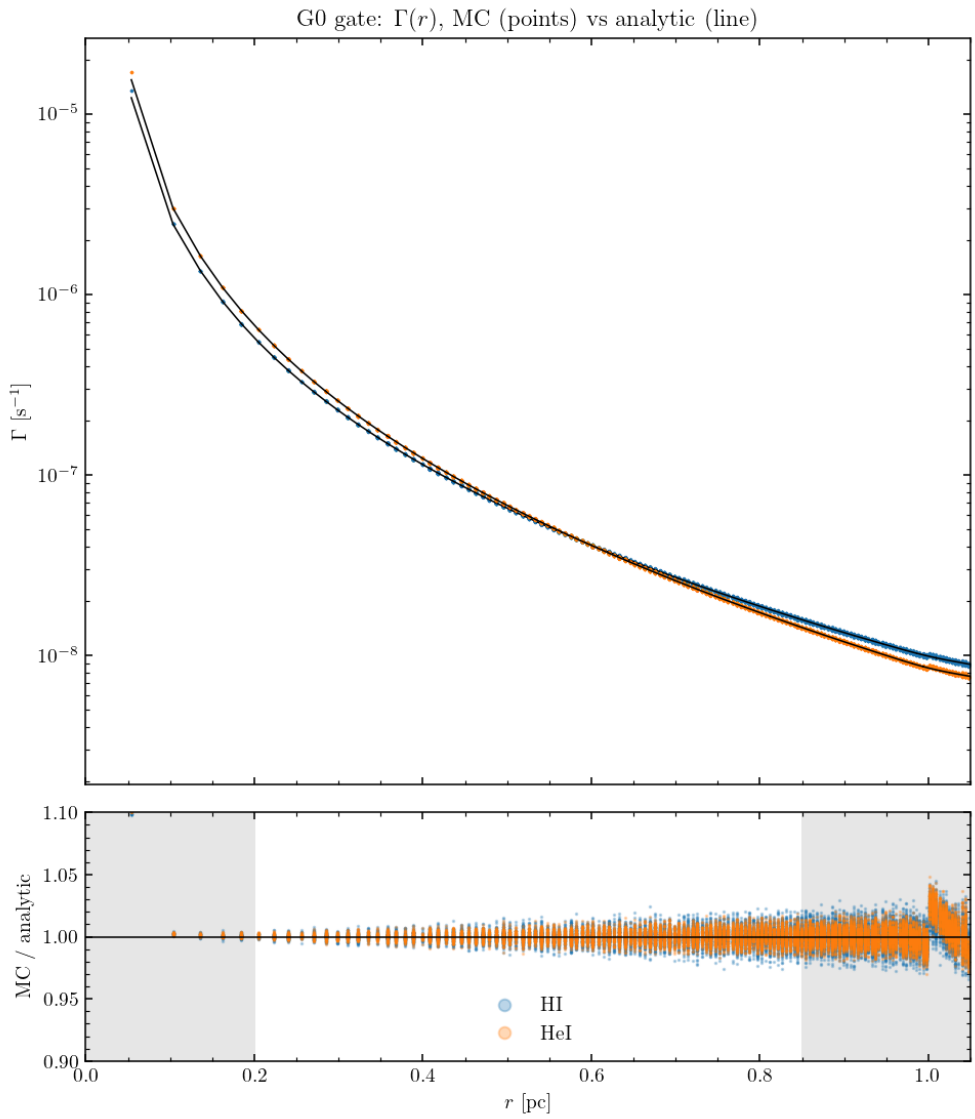



Figure 2: Rate-integral gate: $\Gamma(r)$ from the Monte Carlo tally (points) against the analytic point-source attenuation (lines); the lower panel shows the ratio (shaded regions excluded: cell-size effects at the center, the voxelized sphere edge outside).

## 5.2 Strömgren sphere

Uniform $n_{\rm H} = 100$ sphere, $Q_{\rm H} = 10^{49}\,{\rm s}^{-1}$ ($T_\star = 4\times10^4$ K, 32 bins), case B, $T_e = 10^4$ K. The reference is an exact 1D radial marching with identical bins and rates — necessarily *Gauss–Seidel* (each shell solved with the already-updated interior opacity): a Jacobi sweep advances the front only one absorption length per pass and never converges. Sharp-edge analytic $R_S = 3.157$ pc; the 1D reference gives $R_{\rm eff} = (3V_{\rm ion}/4\pi)^{1/3} = 3.0355$ pc (the $\sim$4% gap is physical: He consumes photons and the model re-emits none).

| run | leaves | $R_{\rm eff}$ [pc] | vs. 1D |
|---|---|---|---|
| uniform level 6 | 262 144 | 3.0675 | +1.05% |
| front-refined levels 4–6 | 74 320 | 3.0676 | +1.06% |
| center-refined levels 4–6 | 7 568 | 3.1205 | +2.80% |

The front-refined octree reproduces the uniform grid to four decimals with 3.5$\times$ fewer cells; the center-refined control (coarsest cells *at* the front) shows why the I-front is the one place needing resolution (Figure 3).

## 5.3 Thermal H/He nebula

Same configuration with the thermal balance on (Figure 4): $T_e(r)$ matches the 1D thermal reference to median 0.12%, mean bias –0.006%, max 3.4% over $0.3 < r < 2.8$ pc (`results/continuous_energy/m10_drift_test/g2a_thermal`, the current defaults and the hydrogenic $\alpha_B$, against the reference of `MoCHII_paper/python/make_g2a_thermal.py`, which takes its rates from `tools/python/mochii_rates.py` and carries the hydrogen share of the He I and He II on-the-spot photons, the $2^3$S coupling and the band from 4.78 eV on 256 stellar-band bins; the 0.61% recorded before was the m4 run against a 32-bin reference without those terms, whose bin quadrature alone put $T_e$ low by 0.42%) — including the front drop. Inside $r = 0.24$ pc run and reference both sit at `te_max`: the He II cascade absorbed on the spot heats gas optically thin to it (§4.9). Without metals the nebula runs at 19–24 kK, the known pure-H/He overshoot that motivates the metal frame.

## 5.4 The MOCASSIN Lexington benchmarks

*Converged runs (2026-09-25, current defaults 2026-09-26).* The numbers of this section are from `results/continuous_energy/m5_con` and, where the current code is quoted, from `results/continuous_energy/m10_drift_test`: those states restarted with every default of 2026-09-25 in force (charge-exchange heating, the $2^3$S coupling with the band from 4.78 eV, the on-the-spot helium photons; `m8_defaults`), with the hydrogenic $\alpha_B$ of §3.2 (`m9_hydrogenic_alpha`), and continued to the stopping test as revised on 2026-09-26 (52 and 59 further iterations in all): the production inputs run to the drift stopping test (§4.2; 64 iterations at HII40, 55 at HII20), with the HII40 gas extended to 5.2 pc because the converged hydrogen front sits on the $1.46\times10^{19}$ cm outer radius of the MOCASSIN benchmark input. The earlier stopping test, on the $x_{\rm HII}$ step alone, ended the runs while the structure was still moving outward: the same HII40 input stopped at iteration 37 with $\langle T[N_pN_e]\rangle = 8175.3$ K, the He I front at 4.103 pc (CLOUDY 4.305), $\langle {\rm He}^+\rangle/\langle {\rm H}^+\rangle = 0.700$ and $L({\rm H}\beta) = 1.927\times10^{37}$; converged these are 8206.7 K, 4.271 pc (a residual of 0.034 pc), 0.758 and $2.000\times10^{37}$. Numbers quoted below as "early-stopped" come from the earlier rule.

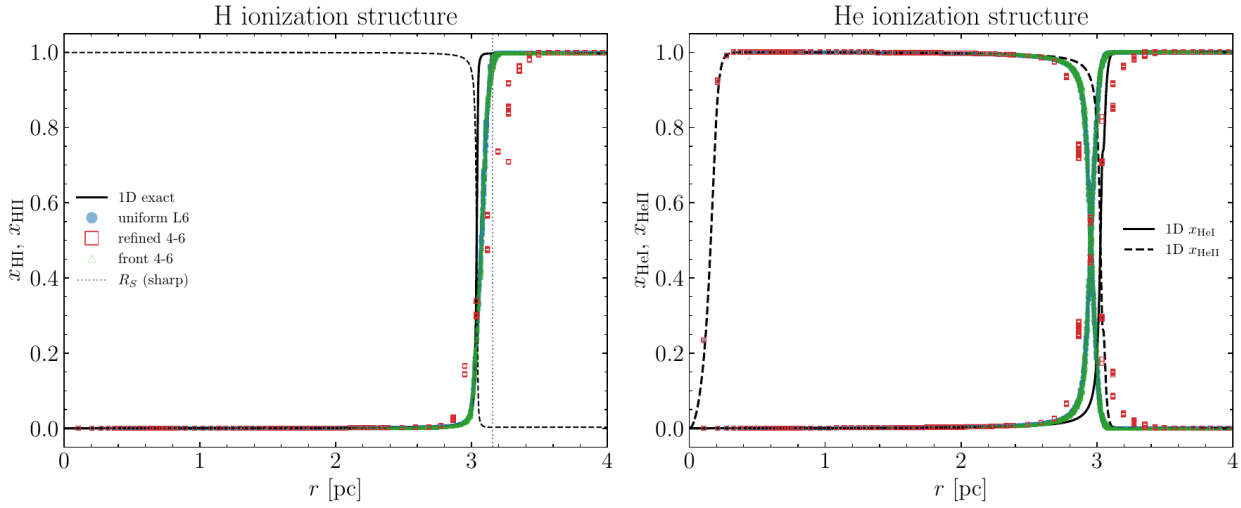



Figure 3: Strömgren gate: H and He ionization structure against the 1D exact reference (black); uniform level-6 (blue), center-refined (orange), front-refined (green, overlying blue).

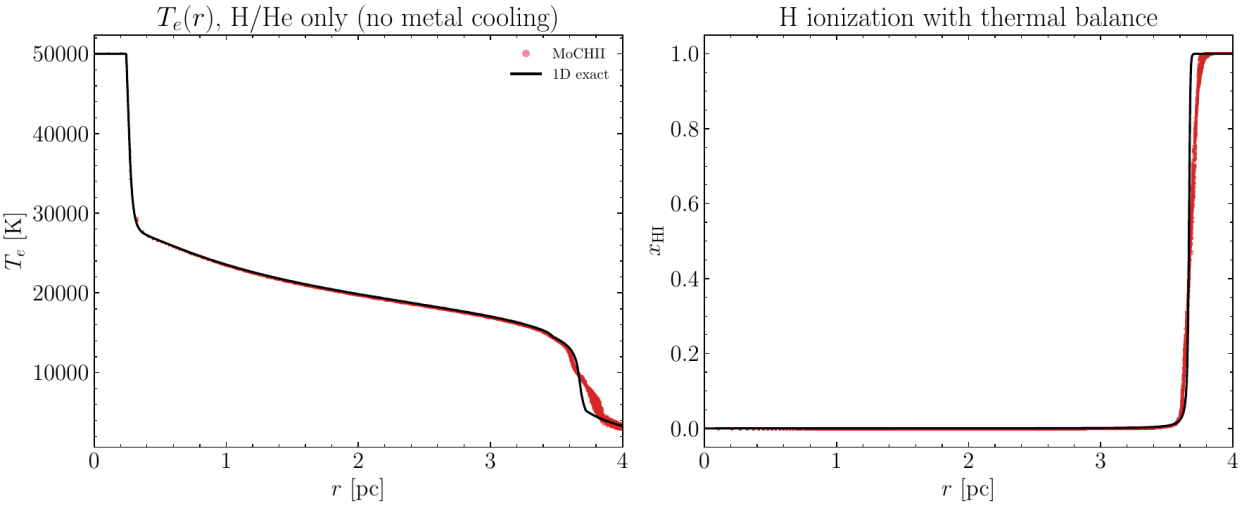



Figure 4: Thermal gate: $T_e(r)$ and $x_{\rm HI}(r)$ with the thermal balance, against the 1D exact reference.

**Temperatures.** $T_e$ weighted by $n_{\rm HII} n_e$, MoCHII with its current defaults (threshold-aligned grid, corrected charge exchange of Section 3.8, and the density-suppressed cooling of Section 3.9); the benchmark inputs carry the ionizing-band luminosity that reproduces the specified $Q_{\rm H}$, transport the diffuse field, and use case-B recombination lines:

| | MoCHII (`local_ne`) | MoCHII (`low_density`) | CLOUDY c25.00 | Lexington-2000 (Ercolano+03) | MOCASSIN 40-it |
|---|---|---|---|---|---|
| HII20 | 6940 K | 6282 K | 6910 K | 6402–6749 K | — |
| HII40 | 8180 K | 7877 K | 8210 K | 7720–8199 K | 8176 K |

With the density-suppressed cooling default (`par%cooling_model = local_ne`) MoCHII gives $T_e$ = 8180 K (HII40) and 6940 K (HII20), matching a current CLOUDY c25.00 run (8210 and 6910 K) to −0.37% and +0.44% (`results/continuous_energy/m10_drift_test`, 2026-09-26; the hydrogenic $\alpha_B$, 1.1% below the Badnell-minus-Milne value it replaced, lowered them by 27 and 18 K from the 8207 and 6958 K of `m8_defaults`, and the continuation to the revised stopping test moved them by +0.2 and +0.0 K while the HII40 hydrogen front moved out by a further 0.012 pc; `m5_converged`, before the defaults of 2026-09-25, gave 8207 and 6946 K and the `low_density` limit 7907 and 6295 K; the `low_density` column is the same input with `cooling_model = 'low_density'`). Of the +12 K at HII20, the band below the hydrogen edge carries about all: the other switches moved it by −0.7 K (`m6_new_physics`). Both modern codes sit *above* the medians of the 2000-era published ranges of Ercolano et al. (2003) — $\langle T[N_p N_e] \rangle$ = 7720–8199 K (median 8026) at HII40 and 6402–6749 K (median 6662) at HII20 — whose tabulated columns carry older atomic data: at HII40 at the top of the range (MoCHII inside it, CLOUDY 11 K above), at HII20 160–190 K above it; the two modern codes now agree with each other far more closely than either does with the historical spread.

The ∼9% HII40 excess reported in earlier revisions (a superseded $T_e$ = 8913 K) is *resolved*: it was two defects in the Tier-1 cooling fits (Section 3.9), compounded by the then-missing density suppression. **(D1)** The ground-term fine-structure populations were distributed by Boltzmann factors — the *high*-density limit — instead of the $n_e \to 0$ limit, which suppressed the far-infrared fine-structure cooling ([O III] 52/88 $\mu$m, [N III] 57 $\mu$m, [C II] 158 $\mu$m) by factors 1.6–2.9. **(D2)** The `.scups` Burgess–Tully *scaling* energy was used as the physical transition energy in both the Boltzmann factor and the radiated photon, in place of the observed `.elvlc` level difference (e.g. [O II] cooling low by 0.58× at 8 kK). Both were located by a fixed-state cooling-budget comparison against CLOUDY c25.00 (its `save cooling each` output): D1 was 23.9% and D2 10.6% of the HII40 cooling budget, while missing channels ([S IV] 10.5 $\mu$m at 1.2%, now added; all others $< 0.2\%$) and trace-species coupling ($n_e$ plus metal photoheating, together +0.2%) were excluded. Regenerating the fits from the $n_e \to 0$ statistical-equilibrium limit with observed energies gives HII40 7894 K and HII20 6376 K (the `low_density` option); restoring the density suppression at the local $n_e$ (the `local_ne` default) raises them to 8211 and 7025 K, and the subsequent corrections to the diffuse field — the Milne ground continua, the He I cascade and the $\alpha_1$ correction of §4.10 — settle them at 8169 and 6925 K; the corrected oxygen charge exchange of Section 3.8 then adds +2 K (8170 and 6926 K), and the 2026-09-24 atomic-data update (inner-shell metal cross sections, Barragan $O^{2+}/N^{2+}$ charge exchange, He–H charge exchange, hydrogen-impact fine structure, recombination cooling from Eq. 11 and the metal ionization-energy loss) gave 8177 and 6936 K: at HII40 the first five items moved it by −9.4 K and the two cooling terms by +16.4 K. All of these steps were measured on early-stopped runs, and step differences of a few K

are within that rule's resolution; run to the drift test the code gave 8207 and 6946 K, and the `low_density` limit 7907 and 6295 K (it was 7894 and 6376 K when the fits were regenerated); the defaults of 2026-09-25 make these 8207 and 6958 K and 7907 and 6304 K, and the hydrogenic $\alpha_B$ 8180 and 6940 K and 7877 and 6282 K. The former "cooling-side residual" was thus the low-density over-cooling of the low-$n_{\rm crit}$ fine-structure lines, and there is no open temperature discrepancy with the current reference code.

**Ionization structure** ($n_e$-weighted means, HII40). The He and metal ionization is set correctly once the ionizing grid resolves the He I edge. With `par%ion_align_edges` on (the default) or nnu_ion $\geq 128$ the He-ionizing photon budget returns to the exact blackbody value ($Q_{\rm HeI}/Q_{\rm HI} = 0.108$ against 0.127 for a 32-bin log grid), He I rises from 0.058 to 0.163 (MOCASSIN 0.207, closing most of the gap), and the metal 3+ fractions fall correspondingly. A separate in-code swap of MOCASSIN-style recombination (Pequignot 1991 RR + Nussbaumer–Storey 1983 DR) showed the recombination-coefficient vintage to be a minor lever: MoCHII's Badnell 2023 $\alpha$(X III $\rightarrow$ X II) runs $\sim$0.8$\times$ MOCASSIN's, but the swap moved the fractions by only $\sim$0.026 and closed $\sim$17% of the ionization gap. The table below is the old 32-bin log grid, the reference point for the alignment discussion (corrected CX vs. the converged 40-iteration MOCASSIN):

| ion | MoCHII (log) | MOCASSIN | ion | MoCHII (log) | MOCASSIN |
|---|---|---|---|---|---|
| C II | 0.188 | 0.355 | O II | 0.344 | 0.491 |
| C III | 0.805 | 0.640 | O III | 0.653 | 0.496 |
| N II | 0.180 | 0.386 | Ne II | 0.524 | 0.587 |
| N III | 0.818 | 0.600 | Ne III | 0.474 | 0.401 |
| S II | 0.194 | 0.066 | S III | 0.750 | 0.786 |

Switching on the aligned-grid default brings the He and metal fractions substantially closer to MOCASSIN — He I 0.058 $\rightarrow$ 0.163, O III 0.653 $\rightarrow$ 0.601, C III 0.805 $\rightarrow$ 0.694 — so most of the apparent over-ionization in the table is the log-grid discretization at the He I edge, not a physics error. At HII20 the CX correction is negligible (low excitation): the structure matches everywhere except Ne I/II (the 21.6-eV threshold at a 20-kK source, a known code-spread hotspot) and the S II/III split.

**Line fluxes** (ratios to H$\beta$). With the corrected cooling the optical collisional lines — formerly systematically high because $T_e$ ran $\sim$9% hot — now fall inside the published spread: [O III] 5007 drops from $\sim$4.3 to $\sim$2 against the published 1.64–3.27, and the [N II], [S II] and [Ne III] ratios move into range. Absolute $L(\mathrm{H}\beta) \approx 1.9 \times 10^{37}$ (HII40) and $4.7 \times 10^{36}\,\mathrm{erg\,s^{-1}}$ (HII20) sat $\sim$3% below the published $2.01$–$2.10 \times 10^{37}$ and $4.83$–$5.04 \times 10^{36}$ in the early-stopped runs; converged they are $2.00 \times 10^{37}$ and $4.79 \times 10^{36}$, 0.5% and 0.8% below the lowest published values, with the case-B recombinations at 0.98 and $1.00\,Q_{\rm H}$. The emissivity chain itself is exact (Section 4.19).

## 5.5 Re-refinement gate and the TNG demonstration

The re-refinement of Section 4.11 on the Strömgren configuration (level-5 start, rebuild to level 7 at iteration 15, 32 768 $\rightarrow$ 467 272 leaves; runs of 2026-09-26 with the hydrogenic $\alpha_B$, `results/continuous_energy/m10_drift_test/g1_g4/`):

| run | leaves | $R_{\rm eff}$ [pc] | vs. 1D |
|---|---|---|---|
| re-refined 5→7 | 467 272 | 3.0579 | +0.74% |
| native uniform level 7 | 2 097 152 | 3.0577 | +0.73% |

The two agree to +0.006% with 4.5× fewer leaves, and both improve on level 6 (+1.05%) — resolution-convergent, as the resolution-convergence requirement expects. The G1 input on the same uniform level-7 grid gives 3.0578 pc (+0.73%). The front profiles overlie exactly.

The post-processing demonstration runs MoCHII on an Illustris-TNG 60-kpc cutout (the MoCafe converter output, $n_{\rm H}$ and $x_{\rm HI}$ columns from the simulation) with a $Q_{\rm H} = 10^{53}\,{\rm s}^{-1}$, 40-kK source at the density peak, thermal balance + metals + re-refinement (Figure 5): dense clumps cast sharp neutral shadows, the circumgalactic gas sits at $x_{\rm HI} \sim 10^{-4}$ with $n_e$-weighted mean $T_e = 7502$ K, and the refinement (32 768 → 617 135 leaves) tracks the *distributed* ionization topology of a turbulent medium — clump skins and partially ionized gas, not a single thin shell. The run also demonstrates why the max-cell convergence test stalls in such media (clump-surface jitter); a volume-integrated criterion is queued.

## 5.6 Dusty Strömgren gate

With the MW D03 dust of Section 4.12 added to the Strömgren configuration ($\tau_{\rm d,abs}(13.6\,{\rm eV}) = 1.7$ over the dust-free radius), against the 1D reference carrying the identical dust absorption in every bin:

| case | $R_{\rm eff}$(1D) [pc] | $R_{\rm eff}$(MC) [pc] | dev |
|---|---|---|---|
| dust-free | 3.0355 | 3.0675 | +1.05% |
| full MW dust | 2.1966 | 2.2120 | +0.70% |
| `laursen09_live` | 3.0206 | 3.0518 | +1.03% |

(All runs of 2026-09-26, level 6.) The gate is on what the dust adds: each dusty deviation less the dust-free one on the same grid, −0.35% and −0.02%, against a tolerance of 1%; the dust-free +1.05% is the finite width of the front on the 0.125 pc cells, which the level-7 grid reduces to +0.73%. The gate used to test the deviation alone, which measured that resolution bias along with the dust and failed the nearly dust-free `laursen09_live` run at +1.03% once the hydrogenic $\alpha_B$ moved the fronts out. The classic dusty reduction ($R/R_0 = 0.724$ in 1D, 0.721 between the MC runs) is reproduced; the live ionization-dependent dust drives itself to near dust-free ($R/R_0 = 0.995$: interior dust vanishes with the computed $x_{\rm HII}$), confirming the dust–ionization feedback loop converges self-consistently. Energy budget: dust absorbs 61% of the band luminosity in the full-dust case ($L_{\rm abs} = 1.935 \times 10^{38}$ of $3.178 \times 10^{38}$ erg s$^{-1}$).

**Scattering bracket.** Turning the scattering on must land the sphere between two analytically defined bounds: treating the dust extinction entirely as absorption (scattered photons destroyed; 1D $R_{\rm eff} = 2.057$ pc) and the absorption-only treatment (scattered photons never interact; 2.197 pc, MC counterpart 2.212 pc). The full-scattering run gives $R_{\rm eff} = 2.203$ pc — inside the bracket (a criterion of `tests/d_dusty/check_dusty.py` since 2026-09-26) and close to the absorption-only bound, exactly as the strong forward scattering ($g = 0.67$–$0.99$) predicts: forward-scattered photons continue nearly radially, and only the small diffused fraction is lost from the radial budget. The dust-absorbed luminosity rises by 0.7% (rescattered photons reabsorbed) — energy bookkeeping consistent.

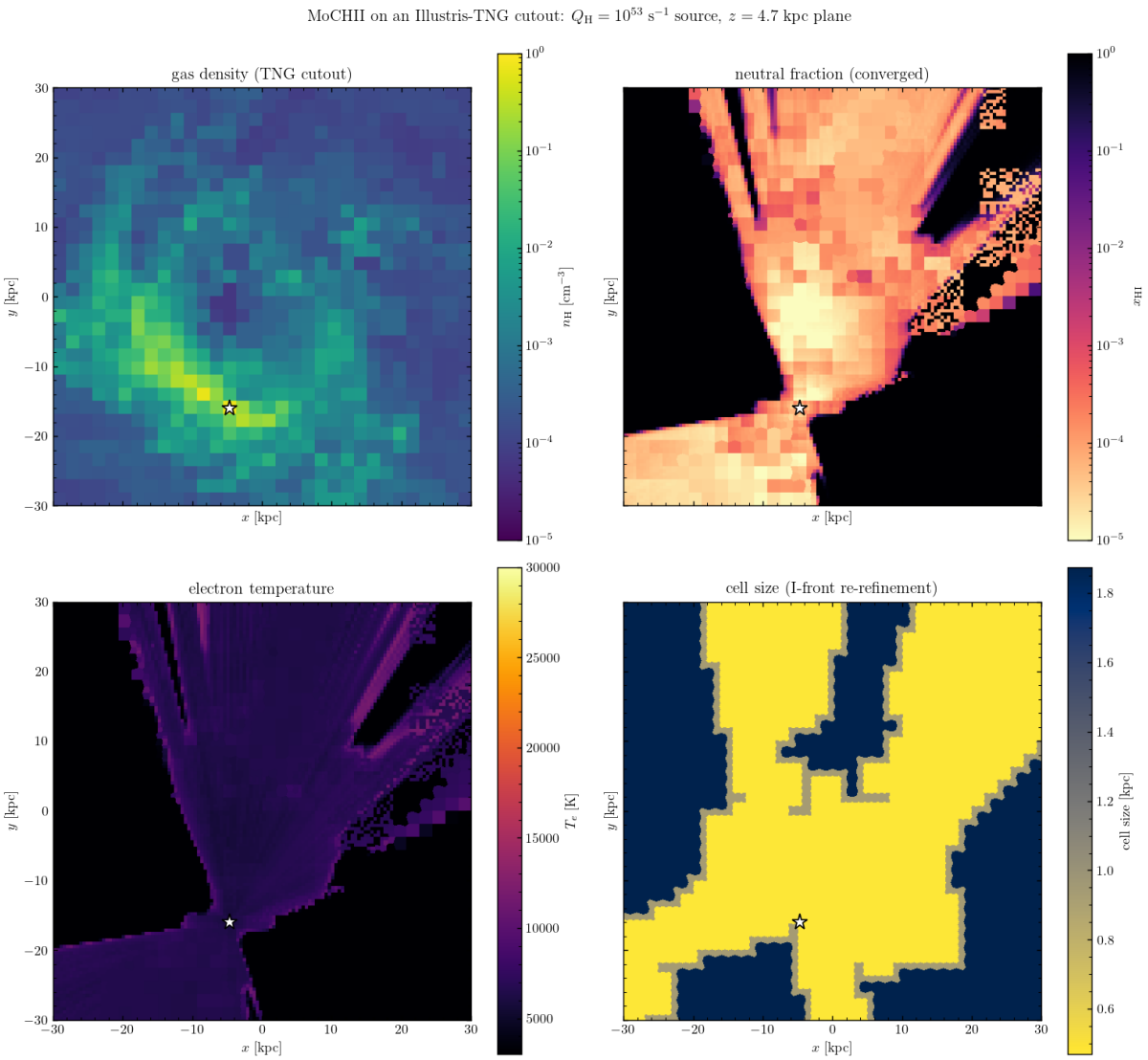



Figure 5: MoCHII on an Illustris-TNG cutout: gas density, converged neutral fraction (note the clump shadows), electron temperature, and the re-refined cell size, in the source plane.

**Dust temperature and IR.** With the band widened to 6 eV (the band luminosity rescaled by the exact blackbody ratio 2.0969 so $Q_{\rm H}$ is preserved) and scattering on, the equilibrium mixture temperature runs from $T_{\rm d} = 87$ K at 0.3 pc to 28 K at the sphere edge — the canonical HII-region gradient — and the IR spectrum integral closes on the absorbed power to $L_{\rm IR}/L_{\rm abs} = 1.000$. (This demo predates the two-segment `add_fuv` grid and rescaled the single-segment band by hand; the +1.5% bin-coarsening shift it measured is what the two-segment grid now removes.)

## 5.7 FUV extension gate: PDR-side C II and Mg II

The `add_fuv` smoke test (dusty Strömgren, 32 ionizing + 16 FUV bins, thermal + metals with Mg/Fe, dust scattering on; `tests/d_dusty/d_fuv_opt.in`) verifies the band mechanics and the physics it was built for. Mechanics: the two-segment grid reports $L_{\rm FUV}/L_{\rm ion} = 1.0969$ for the $4\times10^4$ K Planck source at a 6 eV edge — exactly the blackbody band ratio applied by hand in the earlier demo — with $Q_{\rm H}$ preserved by construction. Physics: inside the front (2.2 pc with this dust and cooling) the metals sit at C III/Mg III as before; *beyond* it, where H is neutral and the ionizing bins are exhausted, the FUV bins propagate under dust extinction alone and hold the low-threshold stages fully singly ionized — radially averaged C II = 1.000 and Mg II = 1.000 from the front out to the sphere edge, the PDR-side structure that the ionizing-only band cannot produce (there the same zone recombines to C I/Mg I). Two documented limitations in that zone: electrons from metal ionization are not fed back into $n_e$ (the trace-species design), and with photoheating $\approx 0$ the thermal bisection pins $T_e$ at `te_min` — PDR temperatures require photoelectric heating, outside the present scope. The front-zone cells with vanishing heating also dominate the $\max|\Delta T_e|$ convergence test (another argument for the queued volume-integrated criterion); the H II-interior structure and $T_e$ (8.1–10.5 kK) converge normally.

## 5.8 External field and multiple sources

The source-component model of Section 2.2 is gated in the `ext_field`, `multi_src`, and `peel_ext` directories under `tests/`:

- **External field.** In an optically thin box ($\tau \sim 7\times10^{-3}$) the interior band-integrated mean intensity recovers $J_{\rm ext}$ to 0.9981 (`'rec'`) and 0.9991 (`'sph'`), uniform and isotropic (axis asymmetry $< 0.1\%$); the entering power equals $\pi J A_{\rm surf}$ [Eq. (2)] exactly.
- **Superposition.** Two point sources against the analytic sum $\sum_i L_i/(16\pi^2 r_i^2)$: median deviation 0.76%, mean ratio 0.9989; a mixed point + external run, median 0.75%. Two-temperature additivity — a two-source run against the sum of the two single-source runs — agrees to a bin-wise median of 0.12% and to 1.0001 in the band total.
- **Spectra.** A multi-column `src_spectrum_file` tabulating the same Planck curves as its `src_tstar` twin reproduces it to 0.010% in the worst bin; `ext_spectrum` against its `ext_tstar` twin to 0.019% (this check also caught and fixed the single-component path ignoring `ext_tstar`/`ext_spectrum`).
- **External peel.** The thin-box image-integrated direct flux against $F = J A_{\rm proj}/d^2$: –0.22% (`'rec'`); the `direc0` flux +0.15% at chord $\tau \sim 2$; the measured central-pixel $\tau_b = -\ln(\text{direct}/\text{direc0})$ reproduces $n_{\rm H}\,\sigma_{\rm HI}(E_b)\,\ell$ bin by bin to four digits with the exact hydrogenic cross section (deviating from the $E^{-3}$ power law by up to $\sim$10%, as $\sigma_{\rm exact}$ does).

- **Physical-unit spectra.** A `'per_ev'` source against its `'shape_ev'` twin agrees to 0.004% in the worst bin (the derived `src_lum` is exact); the `'per_ang'` and `'per_um'` wavelength conversions reproduce it to 0.013%; rescaling a physical file to a set scale matches its shape twin to 0.0000%. A physical `ext_spectrum` $J_E$ file recovers the interior $\langle J\rangle/J = 0.998$, and doubling `ext_intensity` doubles the field exactly (2.0000).
- **ISRF presets.** The measured FUV energy density matches the analytic value to $u_{\rm FUV,meas}/u_{\rm FUV} = 1.0002$–$1.0003$: 8.94 (Draine), 5.23 (Habing), $6.01\times10^{-14}$ erg cm$^{-3}$ (Mathis), with exactly zero in the ionizing bins.
- **Backward compatibility.** Single-source runs are bit-compatible with the pre-change binary (worst difference 1 ULP, $9.9\times10^{-16}$ relative).

## 5.9 Plane-parallel slab

The slab of Section 4.17 is gated under `tests/slab/` against the analytic plane-parallel transport, with the `xy_periodic` wrap exercised throughout:

- **xy-periodic wrap.** A tall thin periodic column ionizes fully ($\langle x_{\rm HII}\rangle = 0.9999$) where the isolated (non-periodic) control reaches only 0.7072 — the wrap re-injects the lateral flux; the octree (`car_walk = 'amr'` / `grid_type = 'amr'`) and the DDA agree to $8.5\times10^{-17}$ for `nx`$\geq$2, and after the exact-face wrap fix (Section 4.17) also at `nx = 1` ($1.3\times10^{-18}$). The AMR leaf-list and car density-cube file inputs both run correctly under `xy_periodic`.
- **Beam attenuation.** A frozen neutral slab illuminated by a collimated beam gives a photoionization-rate profile $\Gamma_{\rm HI}(z)$ that is a *perfect* single exponential ($R^2 = 1.00000$); at $\theta = 60^\circ$ ($\mu = 0.5$) the decay slope is exactly $\mathrm{slope}_0/\mu$ (ratio 1.0000) — the $e^{-\tau/\mu}$ law and the $\cos\theta$ convention are correct. The isotropic mode gives $F_z = \pi I$ exactly and a shallower monotone profile.
- **Emergent $I(\mu)$ and energy budget.** A normal beam transmits $e^{-\tau_\perp} = 0.8874$ exactly (budget closes, $0.88739 + 0.11261 = 1.0$), with $I(\mu)$ a single spike at $\mu = 1$ and zero reflection (no scattering); a $\theta = 60^\circ$ beam transmits $e^{-\tau/0.5} = 0.7874$ exactly; isotropic illumination gives a broad $I(\mu)$ over all bins (mean $\mu = 0.583$, forward-shifted by the $e^{-\tau/\mu}$ weighting). Two-face illumination is a symmetric superposition.
- **Photoionization slab.** A full ionization solve (case B OTS) builds a plane-parallel ionization front at the Strömgren depth $d = \Phi/(n_{\rm H}^2\alpha_B)$; the front position matches to within $\sim$5–7% (recombination-coefficient and finite-front-width uncertainty) and scales with the incident flux ($d \propto F_z$, ratio 1.94 for a factor-of-two change, the small deficit being the roughly constant front width).

## 5.10 Argon as a data operation

Adding argon touched no algorithm: two cooling fits (Ar III, Ar IV), two $n$-level files, one element file (VFKY96 rows $Z$=18, Voronov, Badnell RR + SVS82 DR2, KF96 CX), one abundance parameter. PyNeb cross-check: [Ar III] 7136/7751 within 0.7%; [Ar IV] 4740/4711 within 4.0% (a density pair — the same database-spread class as [O II] and [S II]; PyNeb uses RB97/RGJ19 there). End-to-end smoke test: [Ar III] 8.99 $\mu$m, 7136, 7751 appear in the line table with 7136/7751 = 4.17, the $A$-ratio, as expected.

## 5.11 Magnesium and iron

Mg and Fe repeat the recipe with three new data wrinkles, all absorbed by the readers: the Fe II power-law RR (`RR2`), the SVS82-form DR (`DR2`), and the Mg/Fe charge-exchange forms and stage attribution (Section 3.8). Products: cooling fits Mg II (0.05% max error), Fe II (0.11%), Fe III (0.42%); $n$-level files Mg II (3 levels), Fe II (13 levels, the IR/optical forbidden lines), Fe III (14 levels); element files with VFKY96 $Z$=12, 26 rows (since 2026-09-24 also the `PHINNER`/`PHSHELL` rows of §3.1: Mg I/II and Fe I–III have their $2p/3p$ edges inside the band, so the smoke numbers below predate a cross-section change of up to two orders of magnitude above 55 eV for Mg; on the 11-element smoke run in the HII20 geometry the Mg, Fe and Ca stage fractions do not move at the $10^{-4}$ level, since a 20 kK blackbody puts almost no photons above the 34–88 eV inner edges). PyNeb cross-check: the [Fe III] 4658/4702 ratio agrees to −6% to −16% — the largest gate deviation so far, and a known property of iron: PyNeb's default Fe III collision strengths (BB14) differ from CHIANTI v11's by tens of percent (the iron collision-strength spread in the literature), compounded by the 14-level truncation of a 34-level model. The number is recorded as a data-vintage bracket, not a code error — the same chain reproduces [O III] to 0.6%. For iron-line studies `par%fe_levels_full` loads expanded 16/34-level Fe II/III models (`nlevel_fe_{2,3}_full.txt`, 298 [Fe III] lines vs 38, e.g. 4658/4702 +3%) in place of the compact defaults. End-to-end smoke test (HII20 sphere with $n_{\rm Mg}/n_{\rm H} = 3\times10^{-6}$, $n_{\rm Fe}/n_{\rm H} = 3\times10^{-7}$, depleted gas-phase values): Mg loads 3 stages, Fe 4 stages; the line table gains the Mg II 2796/2803 doublet (ratio 1.98, the $A$-ratio limit) and the [Fe II] 1.257 $\mu$m, 5.34 $\mu$m, and 26 $\mu$m lines.

## 5.12 Silicon, chlorine, calcium (five stages)

Si, Cl, and Ca bring the registry to eleven elements, each tracked through five stages. The one framework extension was chlorine's cross section (the subshell sum of §3.1 at every energy — `phfit2` sets $E_{\rm inn} = 0$ for Cl, which is absent from the VFKY96 outer-shell table; until 2026-09-24 only the outer $3p$ subshell fit was carried, and the $3s$ subshell is now added); Si carries the KF96 charge exchange from the MOCASSIN `chex` rows (recombination direction), Cl and Ca have no transcription available. $n$-level files (all $\leq 0.65\%$ worst transition): Si II/III/IV, Cl II/III/IV, Ca II, Ca V; Cl V and Ca III/IV have no CHIANTI v11 level data and are tracked without lines (the Ar II pattern). Cooling fits: Si II 1.1%, Si IV 0.2%, Ca II 0.4%; Si III 12% and the Cl ions 8–9% resist the multi-exponential form — accepted, since at $\sim3\times10^{-7}$ abundances their share of the thermal balance is $\sim10^{-4}$ and the diagnostics run through the exact $n$-level solve. PyNeb gates: [Cl II] 8579/9124 to 1.5%; Si III] 1892/1883 to 6.9% and [Cl III] 5518/5538 to 6.6% (PyNeb's DK94/BZ89 compilations against CHIANTI v11 — the established database-spread class). End-to-end smoke: [Si II] 34.8 $\mu$m enters the line table at 4.4% of H$\beta$ (the strongest metal line after [C II] — the ionization-front/PDR coolant it is), with Si III] 1883/1892, Si IV 1394/1403, [Cl II] 8581/9126, and the [Cl III] pair at $5518/5538 = 1.402$, the low-density limit.

## 5.13 Line and continuum outputs

The converged-state outputs now include: the SH95 H I recombination lines (smoke-state check: H$\alpha$/H$\beta$ = 2.94 at $\sim$7 kK); the $n$-level collisional lines for the 24 registry ions (including [C II] 158 $\mu$m, C III] 1909, [N III] 57 $\mu$m, Mg II 2796/2803, and the [Fe II] IR lines); and the grid-integrated nebular continuum $L_\nu$ (free-bound + free-free from the verified tables and the Milne extension above the edges, plus the three two-photon continua), written on a 400-point

log-$\nu$ grid from 11.4 $\mu$m to 217 Å.

## 5.14 He I $2^3$S metastable and 10833 Å absorption

The $2^3$S population (Section 3.13) is gated in `tests/hei_meta/` on three runs (low/high density and an `add_fuv` run) and its coupling in `tests/hei_metastable_coupling/`. **Current (2026-09-26, Bray et al. rows, $\alpha_3 = f_{3S}\alpha_B$; the three inputs, which pin the coupling off, rerun at 16 ranks in `results/hei_metastable_gate_2026-09-26/`):** all `check_hei_meta.py` gates pass. $\alpha_3(10^4\,\mathrm{K}) = 2.1311\times10^{-13}$ ($1.0148\times$ OH18's $2.10\times10^{-13}$), the low-density ratio $\alpha_3 n_e/A_{31} = 1.6754\times10^{-7}$ at $n_e = 100$, the plateau $5.8669\times10^{-6}$, $n_{\rm crit} = 3502$ cm$^{-3}$ (the $n_e$-scan half-plateau crossing at 3498); the module $n_3$ matches the Python evaluation of the data file on each run's own state to $\leq 1.0\times10^{-15}$; the low-density core sits within median 3.12% / max 4.26% of the radiative limit and the high-density core at 0.701 of the plateau; $\Phi_3$ matches the bin-center quadrature to $1.5\times10^{-2}$ (bound $4.5\times10^{-2}$, max $\Phi_3 = 9.04\times10^{-5}$ s$^{-1}$); the Stage 2 results below are unchanged (high-density column conservation $6.9\times10^{-15}$, doublet/$J'$=0 ratio 8.003). The coupling gate: Bray Tables 2/3 reproduced at their nodes to $7.8\times10^{-16}$, the rate sums and $q_{31g}$ detailed balance to $7.1\times10^{-15}$, $\alpha_3 = f_{3S}\alpha_B$ exactly, the closed-form $n_3$ against a direct $2\times2$ solve within $0.75\,n_3/n_{\rm HeI}$ (the size of the $n_1 = n_{\rm HeI}$ approximation), the energy ledger to $3.8\times10^{-16}$, the diffuse low-density limit to $2.5\times10^{-13}$, and `solve_ion_cell` balances with the $2^3$S terms to $1.1\times10^{-10}$. The earlier record follows; its $\alpha_3$, plateau and $n_{\rm crit}$ are those of the OH18 coefficient set and are superseded. From the 2026-09-24 evening binary ($q_{31g}$, the García Muñoz $Q_{31}$, the path estimator for $\Phi_3$, the exact collision constant and $C_3$), run on those three inputs in `results/hei_metastable/` with the `check_hei_meta.py` of that day; the outputs stored under `tests/hei_meta/` are dated 2026-07-24 and predate all of it. Against the 12:40 binary of the same day (which had $q_{31g}$, $Q_{31}$ and the path estimator but not the constant or $C_3$) the total $2^3$S column changes by –0.136% (`hei_lowd`, `hei_fuv`) and –3.37% (`hei_highd`); against the balance before the morning update it had already moved by –0.16%, –3.85% and –0.16% ($\Phi_3$ by $\leq 0.2\%$). The evening binary also carried, for a few hours, the Kimura He+H$^+$ rate (since withdrawn, §3.8); it moved only the neutral leaves ($x_{\rm HI} > 0.99$), which carry $< 3\times10^{-7}$ of the column, so the column changes above are those of the constant and $C_3$. **Stage 1**: the data-file analytic limits reproduce the low-density ratio $\alpha_3 n_e/A_{31} = 1.6509\times10^{-7}$ (spec $1.65\times10^{-7}$ at $n_e = 100$, $10^4$ K), the plateau $5.887\times10^{-6}$, and $n_{\rm crit} = 3566$ cm$^{-3}$ (the $n_e$-scan half-plateau crossing lands at 3601); the Fortran module $n_3$ matches the independent Python closed form on each run's own state to worst relative $3.4\times10^{-15}$ (machine precision); the low-density core sits within median 3.07% / max 4.21% of the radiative limit and the high-density run ($n_e = 1.1\times10^4 > n_{\rm crit}$) at 0.698 of the plateau; $\Phi_3$ from the path estimator matches a bin-center $J_\nu$ quadrature to $1.4\times10^{-2}$, inside twice the quadrature's own bound ($4.5\times10^{-2}$; max $\Phi_3 = 9.05\times10^{-5}$ s$^{-1}$); and $\alpha_3(10^4\,\mathrm{K}) = 0.998\times0.75\,\alpha_B$(He II), the diffuse `HEI_F3S` consistency (as recorded 2026-07-24; the current checker does not evaluate it, and neither input of it changed). **Stage 2 (10833 absorption)**: the LINE10833 constants carry the $f$ ratio 1:3:5 and $A = 1.0216\times10^7$ s$^{-1}$ each (NIST/p-winds, not the CHIANTI $1.15\times10^7$); a synthetic uniform cube reproduces the column conservation $\sum NA_{\rm pix} = \sum nV$ and the interior column $n_3\,\ell$ exactly, and the line-center $\tau_0 = \int\kappa_0\,dl$ per component and for the doublet matches the hand integral $\kappa_0^{\rm hand}\,\ell$ to $< 2\times10^{-16}$; the optically-thin case raises no warning while a $\times10^5$ scaled case pushes $\tau_0 > 1$ and fires it. On the high-density run the column conserves to $8\times10^{-15}$ and the doublet/$J'$=0 $\tau_0$ ratio is 8.003, equal to the $(f_1{+}f_2)/f_0 = 8.002$ oscillator-strength ratio of the blended lines (that run's metastable column is optically thick, $\tau_0 \sim 10^4$ — flagged by the warning, as designed).

# 6. Known limitations and next steps

- By default He recombination emits only the three ground continua; the He I excited-channel ionizing photons ($\sim$60% of He recombinations) are re-emitted only when `par%hei_diffuse` is set — their effect on the He I/II split against MOCASSIN is small (the split is set by the >24.6-eV budget). The He III case-B cascade — the He II Ly$\alpha$ (40.8 eV), the $2^2$S two-photon continuum and the free-bound continuum into $n \geq 2$, all of which ionize H — is transported only when `par%heii_diffuse` is set (default off; Section 4.10.1). Immaterial wherever $He^{2+}$ is absent, as in both Lexington benchmarks; a hard spectrum must have it on.

- With the default astrodust or DL07 SEDust optics, the EUV part of the leaf field lies on the 1762-point EUV Q-table grid and is not folded into a shorter-wavelength bin. A custom Q/kext table that does not cover the transported band is rejected at setup rather than silently clamped.

- The thermal-loop line cooling is suppressed to the local $n_e$ by default (`cooling_model = local_ne`) through the $(T, n_e)$ table of Section 3.9, and the atomic-hydrogen impact on the C I, C II and O I ground terms (`metal_cooling_H`, §4.18) is solved in the same statistical equilibrium as the electrons, so the two colliders saturate together. What remains is the *cold* limit of the electron share: the cooling fits hold over $10^3$–$10^5$ K and collapse below that floor (their lowest excitation temperature is 2054 K for C I, far above the 23.6 K of [C I] 609.1 $\mu$m), so in gas colder than $\sim 10^3$ K only the hydrogen increment is evaluated from the atoms directly; the hydrogen rates themselves are held at their fit edges outside 5–1000 K (C I), 30–1000 K (O I) and 20–2000 K (C II). Ar II carries no cooling or lines (no CHIANTI v11 level data).

- PDR zones: with the Section 4.18 switches off (the default), the FUV extension gives the PDR-side *ion structure* only; with them on, $n_e$, metal photoheating, and photoelectric heating make the PDR temperatures first-order estimates — $H_2$/CO chemistry and line-transfer cooling remain out of scope.

- The cell-max convergence test stalls at the front-cell Monte Carlo noise floor (and at clump-surface jitter in turbulent media); the volume-integrated criterion (`conv_crit = 'vol'`) resolves this and is the recommended production setting.

- The MOCASSIN Lexington benchmark temperatures are resolved (Section 5.4): with the density-suppressed cooling default MoCHII (HII40 8180 K, HII20 6940 K, run to the drift stopping test) matches a current CLOUDY c25.00 run to −0.37% and +0.44%, the earlier excess having been traced to two Tier-1 cooling defects (D1/D2) and the missing density suppression, all fixed. The one remaining note is that both modern codes sit above the medians of the 2000-era published Lexington temperatures (at the top of the HII40 range, 160–190 K above the HII20 one) — a data-vintage difference, not a code defect.

- The HII40 He I front sits 0.034 pc (0.8% in radius) inside the CLOUDY c25.00 one. The 0.18 pc once reported here as unexplained was, to 0.14 pc, premature stopping of the gas iteration (§5.4); the single-parameter exclusions of Sections 4.10, 3.4 and 4.7 were measured on early-stopped runs and are records of that search, not current bounds.

- The registry-wide recombination audit (Section 3.7) found no defect — no transcription error, and no form a better source could replace — but it leaves one question open: the rates that form *neutral* atoms differ between Badnell

and Verner & Ferland (1996) by 10–16%, and for the metals no third determination is available to say which is right. Adjudicating it needs an independent determination for each ion (R-matrix C I/O I/Ne I recombination, or CHIANTI totals not derived from Badnell). For He I, where such a determination does exist, MoCHII is within 0.3% of it.

- *Auger emission after inner-shell photoionization is not carried* (flagged 2026-09-24). Above their first inner edges Mg I ($2p$, 54.9 eV), Ca I ($3p$, 34.4 eV) and Fe I–III ($3p$, 66.0–87.1 eV) absorb with the full subshell cross section (§3.1), but the code moves the ion up one stage and sends the hole energy out as fluorescence, whereas the Kaastra & Mewe (1993) yields give two electrons with probability $\geq 0.995$ for these vacancies. The stage fractions and the heating of Mg, Ca and Fe in gas that sees 35–100 eV photons are affected; the fix needs the stage matrix of HS6.

- *Charge-exchange heating rests on unidentified products for six reactions* ($C^{2+}$, $C^{3+}$, $S^{3+}$, $Ar^{3+}$, $Si^{3+}$, $Fe^{3+}$ + H), which take the KF96 electronic defect as distributed with CLOUDY (§3.8, Table 2). Two products were settled from the papers on 2026-09-25: Mg + $H^+$ branches into $Mg^+$ 3p and the endothermic $Mg^+$ 4s channel (Allan et al. 1988 Table 1; 66% 4s at $10^4$ K, so the reaction cools by 1.26 eV per event; below the table's 5000 K the 4s share is carried down by its threshold Boltzmann factor), and $He^{2+}$ + H is radiative at low energy (West, Lane & Cohen 1982). Its heat is set to zero, which is an approximation marked at the code site: the products $He^+$ + $H^+$ repel, and the gas keeps $\Delta E(\infty) - \Delta E(R)$ of the emission distance, ∼3–10 eV from their Fig. 2, but they give no emission spectrum to average over. The same holds for the radiative $X^{2+}$ + H rows of the metals.

- Further metal ions as science needs them.